WHICH PROJECTION ALGEBRAS EMBED IN A PARTITION MONOID? WIRING OBSTRUCTIONS AND THE FIRST NON-EMBEDDABLE EXAMPLES

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Abstract. The Wagner–Preston theorem embeds every inverse semigroup in a symmetric inverse monoid. For regular $*$-semigroups the natural analogue of the symmetric inverse monoid is the partition monoid $\mathcal{P}_X$, and we ask which regular $*$-semigroups, and which projection algebras, embed in one. A necessary condition is the projection condition: projections with a common lower bound commute. We show that it is far from sufficient, and that the further obstructions are of a single kind, which we call wiring: a matched path $a \bowtie c \bowtie b$ of projections (meaning $aca = a$, $cac = c$, and likewise for b, c) which the geometry of $\mathcal{P}_X$ forces to close, $a \bowtie b$, as soon as auxiliary projections commute with, lie below, or "see alike" suitable vertices of the path. We prove eight such quasi-identities valid in every $\mathcal{P}_X$, and an infinite family, the chain conditions, which shows that the quasivariety generated by the projection algebras of partition monoids is not finitely based. The eight conditions together with the projection condition characterise embeddability for projection algebras of order at most 8: there are exactly nine minimal non-embeddable projection algebras of order at most 8 satisfying the projection condition, and we describe them. On the other side we show that the local structure imposes no restriction at all, every finite completely simple regular $*$-semigroup being embeddable, and that for graph semigroups the projection condition and a single wiring condition are also sufficient.

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1. Introduction

[Introduction: add the related work (East–Gray–Azeef Muhammed–Ruškuc, East–Azeef Muhammed) and a paragraph on the structure of the paper.]

The Wagner–Preston theorem represents every inverse semigroup faithfully by partial bijections: S embeds in the symmetric inverse monoid $\mathcal{I}_X$ on its own underlying set. Inverse semigroups are exactly the regular $*$-semigroups in which the projections commute, the symmetric inverse monoid is the generic example of one, and the theorem says that it is also the universal one: the inverse semigroups are precisely the $*$-subsemigroups of the $\mathcal{I}_X$. For regular $*$-semigroups in general – semigroups with an involution $x \mapsto x^*$ satisfying $xx^*x = x$, introduced by Nordahl and Scheiblich [7] – the generic examples are the diagram monoids: the partition monoid $\mathcal{P}_X$ and its submonoids, the Brauer, Temperley–Lieb, Motzkin and partial Brauer monoids, in all of which the involution turns a diagram upside down. The partition monoid is the largest of them; it consists of the set partitions of $X \sqcup X'$, where X' is a disjoint copy of X, under the product by concatenation, and it contains the symmetric group, the symmetric inverse monoid and the other diagram monoids as $*$-subsemigroups. In the recent structure theory of regular $*$-semigroups [2, 3] it plays, as a source of examples and of intuition, the role that $\mathcal{I}_X$ plays for inverse semigroups. It is therefore natural to ask for the analogue of the Wagner–Preston theorem: *does every regular $*$-semigroup embed in some partition monoid, and if not, which ones do?*

The answer to the first question is no, and the second question is the subject of this paper. The obstructions live in the *projection algebra*. Recall that the projections of a regular $*$-semigroup are the elements $p = p^2 = p^*$, that they form a projection algebra under the binary operation $\theta_p(q) = pqp$, that every abstract projection algebra, in the axiomatic sense of Jones, is the projection algebra of some regular $*$-semigroup (Imaoka), and that the projection algebra determines much of the structure of S [5, 6, 2, 3]. For an inverse semigroup the projection algebra is the semilattice of idempotents, and every semilattice embeds in some $\mathsf{P}(\mathcal{I}_X)$; so in the inverse case the projection algebra imposes nothing, and Wagner–Preston is the statement that nothing else does either. For regular $*$-semigroups the projection algebra imposes a great deal. A $*$-embedding $S \to \mathcal{P}_X$ restricts to an embedding of projection algebras $\mathsf{P}(S) \to \mathsf{P}(\mathcal{P}_X)$, so every quasi-identity satisfied by the projection algebras $\mathsf{P}(\mathcal{P}_X)$ is a necessary condition for embeddability, and all the obstructions we find are of this kind; whether an embedding of $\mathsf{P}(S)$ can always be extended to one of S is an open question (Section 9). We therefore study the projection algebras: *which projection algebras embed in some $\mathsf{P}(\mathcal{P}_X)$?* By a standard argument (Proposition 8.1) a finite projection algebra embeds iff it satisfies all quasi-identities valid in the

$\mathsf{P}(\mathcal{P}_X)$, so the question is one of finding these quasi-identities, and of deciding whether finitely many suffice. We show that they do not, and we find the first nine of them, with a complete classification of the obstructions up to order 8.

The question has a second ancestry. Every $\mathcal{P}_X$ embeds in the involuted semigroup of binary relations on 2^X (send a partition α to the relation $\{(A,B) : A \sqcup B' \text{ is a union of blocks of } \alpha\}$), so $*$-subsemigroups of partition monoids are representable involuted semigroups in the sense of Schein [8], and Schein's quasi-identities for representability are necessary conditions; they carry no information here (every regular $*$-semigroup of order ≤ 5 is representable by relations, including ones that embed in no $\mathcal{P}_X$), because what distinguishes the relations coming from $\mathcal{P}_X$ is that each is a Boolean subalgebra of $2^X \times 2^X$. The analogy holds in the shape of the answer: Schein's characterisation of representability is an infinite system of quasi-identities, Bredikhin [1] showed that no finite system of elementary axioms can replace it, and the embeddable projection algebras, too, are the models of an infinite system of quasi-identities that cannot be replaced by a finite one. (Representable involuted semigroups in which x^* is an inverse of x do not seem to have been considered separately.)

One necessary condition has been known: in $\mathcal{P}_X$, two projections with a common lower bound in the natural order commute, i.e. the quasi-identity

$$\text{(Q1)} \qquad r = prp,\ r = qrq \implies pq = qp$$

holds in $\mathsf{P}(\mathcal{P}_X)$ [attribution of (Q1): found by East, Gould and Stokes (unpublished); the reformulation (2) of Proposition 2.4 is due to the author; to be settled]. Our results are the following.

The projection condition is far from sufficient, and the further obstructions are wiring conditions. All the conditions we find have the same shape. Call projections p, q *matched*, $p \bowtie q$, if $pqp = p$ and $qpq = q$; this is the *friendly* relation of [EGMR], and in $\mathcal{P}_X$ it means that p and q have the same rank and their transversal blocks are paired off by the product. A matched path $a \bowtie c \bowtie b$ need not close ($a \bowtie b$ need not hold), but in $\mathcal{P}_X$ it is forced to close by any of a number of side conditions on auxiliary projections: that an auxiliary commutes with two vertices and sees the third as it sees one of them, that two vertices see an auxiliary alike, that an auxiliary lies above a vertex, and so on. Each such side condition is a quasi-identity in the language of projection algebras, and we prove eight of them in every $\mathcal{P}_X$ (Theorem 1.3), by arguments that analyse the blocks of the projections involved. The simplest, (S), has four variables and explains the smallest non-embeddable example, a projection algebra of order 6; the others have up to seven variables.

The conditions are not finitely many. The wiring conditions extend to an infinite family, the *chain conditions* (W_n), in which the auxiliary is replaced by a chain $c \sim x_1 \sim \cdots \sim x_n \sim b$ of pairwise commuting projections each of which identifies a with c; we prove all of them in every $\mathcal{P}_X$ by a counting invariant (Theorem 1.4), and we exhibit for each n a projection algebra of order $2n+5$ that satisfies (W_{n-1}) and all the other conditions of this paper but violates (W_n). Consequently the quasivariety generated by the projection algebras of partition monoids is not finitely based, and the class of finite embeddable projection algebras is not the class of finite models of any finite set of quasi-identities (Theorem 1.5).

The small examples are classified. Up to order 8 the obstructions are completely understood: a projection algebra of order at most 8 embeds in some $\mathsf{P}(\mathcal{P}_X)$ if and only if it satisfies (Q1) together with the wiring conditions of Theorem 1.3, and there are exactly nine minimal non-embeddable projection algebras of order at most 8 satisfying (Q1), one of order 6, one of order 7 and seven of order 8, each the witness of one condition (Theorem 1.6). The computations behind this (an enumeration of projection algebras and of regular $*$-semigroups up to order 9, and embedding searches into $\mathsf{P}(\mathcal{P}_m)$ for $m \leq 8$) are described in Section 8.

The local structure is no obstruction, and for graph semigroups the conditions are sufficient. Every finite completely simple regular $*$-semigroup embeds in a finite partition monoid (Theorem 1.1), so no condition on the principal factors can be necessary; and for the graph semigroups $Q(G, \perp)$ of [ref], and for a layered generalisation of them, (Q1) and (S) are also sufficient (Theorem 1.7).

The digraph conditions. Besides the wiring conditions there is a family of conditions of a different shape, indexed by digraphs, whose simplest member is the *triangle condition* (Tri): three pairwise commuting projections with common product w, three with common product z, matched crosswise, force $w = z$. In every projection algebra the triangle condition implies all the others of its family; it holds in every $\mathcal{P}_X$, and it fails in a 61-element quotient of a $*$-subsemigroup of $\mathcal{P}_3$ that satisfies (Q1) (Theorem 1.2); the hexagon condition, which was its original form, is equivalent to it on finite projection algebras but not in general.

The methods are elementary but the proofs are long, because the only tool is the block calculus of $\mathcal{P}_X$: a projection of $\mathcal{P}_X$ is a partition of X with some of its classes marked, a product of projections is described by the join of their partitions, and a quasi-identity is proved by following chains of blocks. Two general devices organise these arguments: a *counting invariant* attached to pairs of projections, which is constant along chains of commuting identifiers (Section 6), and a *finite descent* principle (Lemma 2.10), which replaces the rank of $\mathcal{P}_X$ by the size of a principal down-set in an arbitrary finite projection algebra and lets several arguments be finished algebraically.

Main theorems.

Theorem 1.1 (local structure; Section 3)**.** *Every finite completely simple regular $*$-semigroup embeds in a finite partition monoid. $\mathcal{P}_4$ contains non-orthodox completely simple $*$-subsemigroups.*

Theorem 1.2 (digraph conditions; Section 4)**.** *For every set X and every digraph E on at least three vertices, (D_E) holds in $\mathcal{P}_X$ if and only if E is not a star; in every projection algebra the triangle condition* (Tri) *implies (D_E) for every non-star E, and on finite projection algebras it is equivalent to the hexagon condition. The triangle condition fails in a 61-element regular $*$-semigroup satisfying* (Q1)*, a quotient of a $*$-subsemigroup of $\mathcal{P}_3$, whose projection algebra $\mathsf{P}(Q_0)$ has order 11, and in no projection algebra of order at most 10 satisfying* (Q1)*.*

Theorem 1.3 (wiring conditions; Section 5)**.** *The conditions* (S), (R), (V), (R′), (P_1), (Q), (K4) *and* (K0) *of Section 5 hold in every $\mathcal{P}_X$. In every projection algebra* (Q) *implies* (S) *and* (R′) *implies* (R)*; apart from this the conditions are independent: each of* (V), (R′), (P_1), (Q), (K4), (K0) *is violated by an explicit projection algebra of order 8 which satisfies* (Q1), (Tri)*, all chain conditions and the other five, and* (S), (R) *have witnesses of orders 6 and 7.*

Theorem 1.4 (chain conditions; Section 6)**.** *For every $n \geq 1$ the chain condition (W_n) holds in every $\mathcal{P}_X$. For every n there is a projection algebra $\mathfrak{P}_n^{\mathrm{ch}}$ of order $2n+5$ which satisfies (W_{n-1}),* (Q1), (Tri) *and the eight wiring conditions, and violates (W_n).*

Theorem 1.5 (no finite basis)**.** *The quasivariety generated by the projection algebras $\mathsf{P}(\mathcal{P}_X)$, X finite, is not finitely based; the class of finite projection algebras embeddable in some $\mathsf{P}(\mathcal{P}_X)$ is not the class of finite models of any finite set of quasi-identities.*

Theorem 1.6 (orders at most 8; Sections 5 and 8)**.** *A projection algebra of order at most 8 embeds in some $\mathsf{P}(\mathcal{P}_X)$ if and only if it satisfies* (Q1) *and the eight wiring conditions, and then it embeds in $\mathsf{P}(\mathcal{P}_7)$. The minimal non-embeddable projection algebras of order at most 8 satisfying* (Q1) *are exactly the nine algebras $\mathfrak{P}$ (order 6), $\mathfrak{P}_7$ (order 7) and $\mathfrak{P}_8^1, \ldots, \mathfrak{P}_8^7$ (order 8) of Sections 5 and 7.*

Theorem 1.7 (graph semigroups; Section 7)**.** *A graph semigroup $Q(G,\perp)$, and more generally a layered graph semigroup, embeds in a partition monoid if and only if it satisfies* (Q1) *and* (S)*.*

Evidence levels. Throughout we distinguish three kinds of statement: a quasi-identity is *proved* when it is shown to hold in every $\mathcal{P}_X$; it is *verified for $m \le k$* when a SAT solver has shown it to hold in $\mathsf{P}(\mathcal{P}_m)$ for all $m \le k$ (with variables allowed to coincide); and a finite algebra is *non-embeddable for $m \le k$* when a search has found no embedding into $\mathsf{P}(\mathcal{P}_m)$ for $m \le k$. Only the first is a theorem; the others are reported as such, and the main results use only proved statements.

2. Preliminaries

2.1. **Regular $*$-semigroups and projection algebras.** A *regular $*$-semigroup* is a semigroup S with a unary operation $x \mapsto x^*$ satisfying $x^{**} = x$, $(xy)^* = y^*x^*$ and $xx^*x = x$. Its *projections* are the elements $p = p^2 = p^*$; we write $\mathsf{P}(S)$ for the set of projections. For projections p, q the element $\theta_p(q) = pqp$ is again a projection, and $\mathsf{P}(S)$ with the binary operation $(p,q) \mapsto \theta_p(q)$ is the *projection algebra* of S. In the notation $p \cdot q := \theta_q(p)$ (the value lies below the second argument) the identities satisfied by all projection algebras of regular $*$-semigroups are

$$p \cdot p = p, \qquad (q \cdot p) \cdot p = q \cdot p, \qquad (p \cdot q) \cdot p = q \cdot p, \tag{1}$$

$$((r \cdot p) \cdot q) \cdot p = r \cdot (q \cdot p), \qquad (((r \cdot p) \cdot q) \cdot p) \cdot q = (r \cdot p) \cdot q, \tag{2}$$

and conversely an algebra $(P, \cdot)$ satisfying (1)–(2) is the projection algebra of some regular $*$-semigroup [5, 6] [check that (1)–(2) is the axiom system of [6] (or of [2]) as stated there]. In θ-notation the axioms read $\theta_p(p) = p$, $\theta_p\theta_p = \theta_p$, $\theta_p(\theta_q(p)) = \theta_p(q)$, $\theta_p\theta_q\theta_p = \theta_{\theta_p(q)}$ and $\theta_q\theta_p\theta_q\theta_p = \theta_q\theta_p$. We call an abstract algebra satisfying them a projection algebra; everything below about projection algebras uses only these axioms, except where finiteness is invoked explicitly.

The *natural order* on projections is $q \le p$ iff $q = pq = qp$, equivalently $q = \theta_p(q)$. We write $\downarrow p = \{q : q \le p\}$. Two projections *commute*, $p \sim q$, if $pq = qp$; then pq is a projection, $pq \le p, q$, and $pq = \theta_p(q) = \theta_q(p)$. In the projection algebra alone commutation is expressed by $\theta_p(q) = \theta_q(p)$:

Lemma 2.1. *For projections p, q of a regular $*$-semigroup, $pq = qp$ iff $\theta_p(q) = \theta_q(p)$.*

Proof. If $pq = qp$ then $pqp = pq = qpq$. Conversely let $w = pqp = qpq$. Then $(pq)(pq)^* = pqqp = w$, so by regularity $pq = (pq)(pq)^*(pq) = w\,pq = pqp\,pq = (pq)^2 = p(qpq) = pw = p\,pqp = w$; symmetrically $qp = w$. □

We note in passing that the product of two projections is always idempotent, since $pq = (pq)(pq)^*(pq) = pqqppq = pqpq$; in general it is not a projection.

Definition 2.2 (matched pairs)**.** Projections p, q are *matched*, $p \bowtie q$, if $\theta_p(q) = p$ and $\theta_q(p) = q$.

This is the *friendly* relation $p\,\mathcal{F}\,q$ of [3]; we keep a separate symbol because in $\mathcal{P}_X$ it comes with a bijection of transversal blocks (Lemma 2.6(c)) and the word "matched" refers to that bijection. The relation is reflexive and symmetric, not transitive, and matched pairs can be manufactured at will:

Lemma 2.3. *For all projections p, q: $\theta_p(q) \bowtie \theta_q(p)$, and $\theta_{\theta_p(q)}(q) = \theta_p(q)$.*

Proof. Both statements follow from the axioms in θ-notation. Using $\theta_p\theta_q\theta_p = \theta_{\theta_p(q)}$ and then $\theta_p(\theta_q(p)) = \theta_p(q)$ twice,

$$\theta_{\theta_p(q)}(\theta_q(p)) = \theta_p\theta_q\theta_p(\theta_q(p)) = \theta_p\theta_q(\theta_p(q)) = \theta_p(\theta_q(p)) = \theta_p(q),$$

and symmetrically $\theta_{\theta_q(p)}(\theta_p(q)) = \theta_q(p)$; so $\theta_p(q) \bowtie \theta_q(p)$. Likewise $\theta_{\theta_p(q)}(q) = \theta_p\theta_q\theta_p(q) = \theta_p(\theta_q(p)) = \theta_p(q)$. □

2.2. **The projection condition.** In $\mathcal{P}_X$, as we recall in Section 2.3, two projections with a common lower bound commute. The corresponding quasi-identity

$$\theta_p(r) = r = \theta_q(r) \implies \theta_p(q) = \theta_q(p) \tag{Q1}$$

is therefore necessary for embeddability. It has several equivalent forms.

Proposition 2.4. *For a regular $*$-semigroup S the following are equivalent.*

1. $\mathsf{P}(S)$ *satisfies* (Q1).
2. *Every idempotent of S lying above a projection in the natural order is a projection.*
3. *For every projection p, the set $S_p = \{x \in S : px = xp = p\}$ is an inverse semigroup.*
4. *Two projections commute if and only if they have a common lower bound.*

Proof. S_p is closed under multiplication and under $*$ (if $px = xp = p$ then $px^* = (xp)^* = p$), so it is a regular $*$-semigroup with zero p; it is inverse iff all its idempotents are projections, and its idempotents are exactly the idempotents $e \geq p$. This gives (2)⇔(3).

(2)⇒(1). Let $r \leq p, q$. The element $e = pq$ is idempotent, and $er = pqr = pr = r = re$, so e lies above the projection r; by (2) it is a projection, and $pq = e = e^* = qp$.

(1)⇒(2). Let $e = e^2$ and let r be a projection with $er = re = r$. Then $r = r^* = (re)^* = e^*r$ and likewise $r = re^*$, so the projections $f = ee^*$ and $g = e^*e$ both lie above r: $fr = ee^*r = er = r$, $rf = re\,e^* = re^* = r$, and similarly for g. By (1), $fg = gf$, that is, $ee^*e^*e = e^*ee^*$. Multiplying this on the left by e gives $fg = e\,e^*e^*e = (ee^*)(ee^*) = f$, and multiplying it on the right by e gives $fg = f(ge) = (fg)e = (gf)e = e^*e\,ee^*\,e = (e^*e)(e^*e) = g$ (using $ge = e^*ee = g$). Hence $ee^* = e^*e$, and $e = ee^*e = e^*ee = e^*e$ is a projection.

(1)⇔(4). If $pq = qp$ then pq is a projection below p and q; the converse is (Q1) with Lemma 2.1. □

Remark 2.5. In $\mathcal{P}_X$, an element x lies above a projection p iff x maps every $\mathsf{T}(p)$-block onto itself by a block bijection and permutes the $\mathsf{N}(p)$-blocks, so that $(\mathcal{P}_X)_p \cong \prod_{A \in \mathsf{T}(p)} \mathcal{I}_A^* \times \mathcal{J}_p$ with $\mathcal{I}_A^*$ the dual symmetric inverse monoid on A and $\mathcal{J}_p$ an inverse monoid of partial block bijections; for the projection p with all blocks singletons and nontransversal, $(\mathcal{P}_X)_p$ is the symmetric inverse monoid $\mathcal{I}_X$. So (Q1) says that these "unitary" subsemigroups are inverse.

Two consequences of (Q1) are used throughout: the projections above a fixed projection pairwise commute; and two distinct matched projections have no common lower bound (they would commute, and then $p = pqp = pq = q$).

2.3. **Partition monoids and the block calculus.** For a set X let $X' = \{x' : x \in X\}$ be a disjoint copy. The partition monoid $\mathcal{P}_X$ consists of the set partitions of $X \sqcup X'$; the product $\alpha\beta$ is obtained by identifying the lower points of α with the upper points of β, taking connected components, and forgetting the middle layer, and α^* interchanges X and X'. A block is *transversal* if it meets both X and X'; the *rank* of α is the number of transversal blocks.

A projection p of $\mathcal{P}_X$ is determined by a pair $(\varepsilon_p, \mathsf{T}(p))$, where ε_p is a partition of X and $\mathsf{T}(p)$ is a set of ε_p-classes: the blocks of p are $A \cup A'$ for $A \in \mathsf{T}(p)$ (the T-*blocks*) and B, B' for the remaining classes B (the N-*blocks*, forming $\mathsf{N}(p)$). We put $\operatorname{supp} p = \bigcup \mathsf{T}(p)$, so that $\operatorname{rank} p = |\mathsf{T}(p)|$. We say "block of p" for a class of ε_p, with the label T or N understood. For projections p, q the *join* $\varepsilon_p \vee \varepsilon_q$ is the join of partitions of X; its classes are the connected components of the graph on the blocks of p and q in which two blocks are adjacent when they meet. A *chain* in $\varepsilon_p \vee \varepsilon_q$ is a sequence of blocks, alternately of p and of q, in which consecutive blocks meet; two blocks lie in a common class iff they are joined by a chain. For a class K we

write U_K for the union of the $\mathsf{T}(p)$-blocks contained in K and L_K for the union of the $\mathsf{T}(q)$-blocks in K. The following lemma is the whole of our toolkit.

Lemma 2.6. *Let p, q be projections of $\mathcal{P}_X$.*

(a) *The blocks of $\theta_p(q) = pqp$ are the $\mathsf{N}(p)$-blocks and the sets U_K for the classes K of $\varepsilon_p \vee \varepsilon_q$ meeting $\operatorname{supp} p$; U_K is a T-block of $\theta_p(q)$ iff K meets $\operatorname{supp} q$. In particular the blocks of $\theta_p(q)$ not contained in $\operatorname{supp} p$ are exactly the $\mathsf{N}(p)$-blocks; consequently, if $e \leq p$ then $\operatorname{supp} e \subseteq \operatorname{supp} p$, every $\mathsf{T}(e)$-block is a union of $\mathsf{T}(p)$-blocks, and every block of e not contained in $\operatorname{supp} p$ is an $\mathsf{N}(p)$-block.*

(b) *Suppose $pq = qp$. Then every class K of $\varepsilon_p \vee \varepsilon_q$ is of one of four types:* (i) *$K = U_K = L_K$;* (ii) *$K = U_K$ is a single $\mathsf{N}(q)$-block;* (iii) *$K = L_K$ is a single $\mathsf{N}(p)$-block;* (iv) *K is a single block of both p and q. The T-blocks of pq are the classes of type* (i)*; hence $\operatorname{supp}(pq) = \operatorname{supp} p \cap \operatorname{supp} q$, and every $\mathsf{N}(q)$-block contained in $\operatorname{supp} p$ is a union of $\mathsf{T}(p)$-blocks.*

(c) *Suppose $p \bowtie q$. Then every class of $\varepsilon_p \vee \varepsilon_q$ meeting $\operatorname{supp} p \cup \operatorname{supp} q$ contains exactly one $\mathsf{T}(p)$-block and exactly one $\mathsf{T}(q)$-block. In particular $\operatorname{rank} p = \operatorname{rank} q$, no $\mathsf{T}(p)$-block is an $\mathsf{N}(q)$-block, and the classes of $\varepsilon_p \vee \varepsilon_q$ induce a bijection $\mathsf{T}(p) \to \mathsf{T}(q)$. Conversely, if every class meeting $\operatorname{supp} p \cup \operatorname{supp} q$ contains exactly one $\mathsf{T}(p)$-block and one $\mathsf{T}(q)$-block then $p \bowtie q$.*

Proof. (a) In the product pq the middle layer is partitioned by $\varepsilon_p \vee \varepsilon_q$, since every block of p has a lower part and every block of q an upper part in the middle; the transversal blocks of pq are $U_K \cup L'_K$ for the classes K with $U_K \neq \emptyset \neq L_K$, the upper nontransversal blocks are the $\mathsf{N}(p)$-blocks and the sets U_K with $L_K = \emptyset$. Now $\theta_p(q) = (pq)(pq)^*$ is the projection whose T-blocks are the upper parts of the transversal blocks of pq and whose N-blocks are the upper nontransversal blocks of pq. For the last sentence use $e = \theta_p(e)$.

(b) The projection pq is symmetric, so $U_K \cup L'_K$ is a block only if $U_K = L_K$ (when both are nonempty), and the family of upper nontransversal blocks $\mathsf{N}(p) \cup \{U_K : L_K = \emptyset \neq U_K\}$ equals the family of lower ones $\mathsf{N}(q) \cup \{L_K : U_K = \emptyset \neq L_K\}$. Let K be a class. If $U_K \neq \emptyset \neq L_K$, an $\mathsf{N}(p)$-block inside K would be an N-block of pq, hence (as $U_K \neq \emptyset$) an $\mathsf{N}(q)$-block; but a set which is a block of both p and q is a class by itself; so K contains no $\mathsf{N}(p)$-block and dually no $\mathsf{N}(q)$-block, and $K = U_K = L_K$, type (i). If $U_K \neq \emptyset = L_K$ then U_K is an $\mathsf{N}(q)$-block; K is a union of q-blocks containing the q-block U_K and p-blocks; as before the remaining p-blocks in K would be common blocks, so $K = U_K$, type (ii). Type (iii) is dual, and if $U_K = L_K = \emptyset$ then K consists of $\mathsf{N}(p)$- and $\mathsf{N}(q)$-blocks, which must coincide as above, type (iv). Finally an $\mathsf{N}(q)$-block $B \subseteq \operatorname{supp} p$ lies in a class K which is not of type (i) or (iii) (these consist of $\mathsf{T}(q)$-blocks only) nor of type (iv) (B meets $\operatorname{supp} p$), so $K = B$ is of type (ii).

(c) $pqp = p$ says that the upper projection of pq is p: its T-blocks $\{U_K : U_K \neq \emptyset \neq L_K\}$ are exactly the $\mathsf{T}(p)$-blocks, and its N-blocks $\mathsf{N}(p) \cup \{U_K : U_K \neq \emptyset = L_K\}$ are exactly the $\mathsf{N}(p)$-blocks. So every class meeting $\operatorname{supp} p$ contains exactly one $\mathsf{T}(p)$-block and at least one $\mathsf{T}(q)$-block; by symmetry, exactly one of each. A $\mathsf{T}(p)$-block which is an $\mathsf{N}(q)$-block would be a class by itself, containing no $\mathsf{T}(q)$-block. The converse is read off from (a). □

We shall use (a) constantly in the following form: for projections p, q, *p sees q as $\theta_p(q)$*, and $\theta_p(q)$ is determined by the partition of $\operatorname{supp} p$ into the sets U_K together with the information which of them are *bright* (meet $\operatorname{supp} q$) and which are *dark*. Thus "p sees q and r alike", $\theta_p(q) = \theta_p(r)$, means that $\varepsilon_p \vee \varepsilon_q$ and $\varepsilon_p \vee \varepsilon_r$ induce the same partition of $\operatorname{supp} p$ with the same bright parts; we then say that p *identifies* q with r and write $p \in \mathrm{Id}(q, r)$. Two orders of magnitude of this statement will matter: $\theta_p(q) = p$ (every class meeting $\operatorname{supp} p$ contains one $\mathsf{T}(p)$-block and is bright: p is *half-matched* with q) and $\theta_p(q) = q$ ($q \leq p$).

Lemma 2.7. *Let p, q be projections of $\mathcal{P}_X$.*

(a) *$q \leq p$ iff every block of p lies in a block of q and every $\mathsf{N}(p)$-block is a block of q. In that case every block of q is an $\mathsf{N}(p)$-block or a union of $\mathsf{T}(p)$-blocks, and every $\mathsf{N}(p)$-block is an $\mathsf{N}(q)$-block.*

(b) *$\theta_p(q) = p$ iff every class of $\varepsilon_p \vee \varepsilon_q$ meeting $\operatorname{supp} p$ contains exactly one $\mathsf{T}(p)$-block and meets $\operatorname{supp} q$. In particular every block of q then meets at most one $\mathsf{T}(p)$-block.*

Proof. (a) If $q = \theta_p(q)$, Lemma 2.6(a) describes the blocks of q as stated. Conversely, if the two conditions hold then $\varepsilon_p \vee \varepsilon_q = \varepsilon_q$, every class of ε_q meeting $\operatorname{supp} p$ is a union of $\mathsf{T}(p)$-blocks, and it is a $\mathsf{T}(q)$-block iff it meets $\operatorname{supp} q$; so $\theta_p(q) = q$ by Lemma 2.6(a). (b) is Lemma 2.6(a) read for $\theta_p(q) = p$. □

For a matched pair $p \bowtie q$, Lemma 2.6(c) gives a bijection $\varphi\colon \mathsf{T}(p) \to \mathsf{T}(q)$, sending a $\mathsf{T}(p)$-block to the $\mathsf{T}(q)$-block in its class of $\varepsilon_p \vee \varepsilon_q$. Restrictions of q are transported to restrictions of p along φ:

Lemma 2.8 (transport)**.** *Let $p \bowtie q$ in $\mathcal{P}_X$ with bijection $\varphi\colon \mathsf{T}(p) \to \mathsf{T}(q)$, and let $x \leq q$. For a set Y which is a union of $\mathsf{T}(q)$-blocks put $\varphi^{-1}[Y] = \bigcup\{\varphi^{-1}(C) : C \in \mathsf{T}(q),\ C \subseteq Y\}$. Then the blocks of $\theta_p(x)$ are the $\mathsf{N}(p)$-blocks and the sets $\varphi^{-1}[Y]$ for the blocks $Y \subseteq \operatorname{supp} q$ of x, with the label of Y; and the class of $\varepsilon_p \vee \varepsilon_x$ containing $\varphi^{-1}[Y]$ is the union of the classes of $\varepsilon_p \vee \varepsilon_q$ containing the $\mathsf{T}(p)$-blocks in $\varphi^{-1}[Y]$, which meets $\operatorname{supp} p$ in $\varphi^{-1}[Y]$ and $\operatorname{supp} q$ in Y. These are all the classes of $\varepsilon_p \vee \varepsilon_x$ meeting $\operatorname{supp} p$.*

Proof. By Lemma 2.7(a) the blocks of x outside $\operatorname{supp} q$ are $\mathsf{N}(q)$-blocks and those inside are unions of $\mathsf{T}(q)$-blocks. The classes of $\varepsilon_p \vee \varepsilon_q$ meeting $\operatorname{supp} p$ each contain exactly one $\mathsf{T}(p)$-block A and exactly one $\mathsf{T}(q)$-block $\varphi(A)$, and all other blocks in them are N-blocks. Passing to $\varepsilon_p \vee \varepsilon_x \geq \varepsilon_p \vee \varepsilon_q$, a block $Y \subseteq \operatorname{supp} q$ of x merges the classes of the $\mathsf{T}(q)$-blocks it contains, and the other blocks of x lie inside single classes; since each class contains one $\mathsf{T}(q)$-block, which lies in one Y, the classes of $\varepsilon_p \vee \varepsilon_x$ meeting $\operatorname{supp} p$ are the unions described. Lemma 2.6(a) then gives the blocks $\varphi^{-1}[Y]$ of $\theta_p(x)$, transversal iff the class meets $\operatorname{supp} x$, i.e. (as the class meets $\operatorname{supp} q$ exactly in Y) iff Y is a T-block of x. □

Finally, a counting invariant. For equivalence relations π on X write $|\pi|$ for the number of classes of π and $|\pi|_Y$ for the number of classes meeting a set $Y \subseteq X$. For projections p, q, x put

$$\delta_{pq}(x) = |\varepsilon_x \vee \varepsilon_p| - |\varepsilon_x \vee \varepsilon_q|, \qquad \delta'_{pq}(x) = |\varepsilon_x \vee \varepsilon_p|_{\operatorname{supp} p} - |\varepsilon_x \vee \varepsilon_q|_{\operatorname{supp} q};$$

both depend on x only through ε_x.

Lemma 2.9 (counting invariant)**.** *Let p, q be projections of $\mathcal{P}_X$, let w identify p with q ($\theta_w(p) = \theta_w(q)$) and let $z \leq w$. Then $\delta_{pq}(z) = \delta_{pq}(w)$ and $\delta'_{pq}(z) = \delta'_{pq}(w)$. Consequently, if $x \sim y$ and $y \in \operatorname{Id}(p, q)$, then $\delta_{pq}(xy) = \delta_{pq}(y)$ and $\delta'_{pq}(xy) = \delta'_{pq}(y)$.*

Proof. Put $S = \operatorname{supp} w$, $\rho_p = \varepsilon_w \vee \varepsilon_p$ and $\rho_q = \varepsilon_w \vee \varepsilon_q$. By Lemma 2.6(a) the blocks of $\theta_w(p)$ inside S are the sets $K \cap S$ for the classes K of ρ_p meeting S, transversal iff K meets $\operatorname{supp} p$; so $\theta_w(p) = \theta_w(q)$ says that ρ_p and ρ_q induce the same partition σ of S, and that for every part G of σ the ρ_p-class containing G meets $\operatorname{supp} p$ iff the ρ_q-class containing G meets $\operatorname{supp} q$. By Lemma 2.7(a), every block of $z \leq w$ is an $\mathsf{N}(w)$-block or a union of $\mathsf{T}(w)$-blocks, so $\varepsilon_z = \varepsilon_w \vee \mu$ for an equivalence relation μ all of whose nontrivial classes lie inside S. Now $\varepsilon_z \vee \varepsilon_p = \rho_p \vee \mu$, and its classes are of two kinds. A class of ρ_p disjoint from S is untouched by μ and remains a class. The classes of ρ_p meeting S are merged according to $\sigma \vee \mu|_S$: a path alternating ρ_p-steps and nontrivial μ-steps has all its μ-steps inside S, so between two consecutive μ-steps it joins a point of S to a point of S within one ρ_p-class, i.e. within one part of σ; hence two ρ_p-classes meeting S

lie in one class of $\rho_p \vee \mu$ iff their parts of σ lie in one class of $\sigma \vee \mu|_S$. So the classes of the second kind correspond bijectively to the classes of $\sigma \vee \mu|_S$, and such a class meets $\operatorname{supp} p$ iff one of its constituent ρ_p-classes does. The same description holds for q, with the same σ and the same $\mu|_S$, and corresponding constituents (same part G) meet $\operatorname{supp} p$, resp. $\operatorname{supp} q$, simultaneously. Therefore

$$|\varepsilon_z \vee \varepsilon_p| - |\varepsilon_z \vee \varepsilon_q| = \big(|\rho_p| - |\sigma|\big) - \big(|\rho_q| - |\sigma|\big) = \delta_{pq}(w),$$

and the same computation restricted to the classes meeting $\operatorname{supp} p$, resp. $\operatorname{supp} q$ (the classes of the first kind meeting $\operatorname{supp} p$ are the ρ_p-classes meeting $\operatorname{supp} p$ but not S, whose number is $|\rho_p|_{\operatorname{supp} p}$ minus the number of parts G whose ρ_p-class meets $\operatorname{supp} p$, and the latter number is the same for q) gives $\delta'_{pq}(z) = \delta'_{pq}(w)$. For the last assertion, $xy = yx$ is a projection with $xy \le y$, so the values are transported from y to xy. □

Lemma 2.9 is the engine of the chain conditions in Section 6; it also enters the proof of (K4).

For a matched pair $a \bowtie c$ we write $A_1, \dots, A_m$ for the $\mathsf{T}(a)$-blocks, C_i for the $\mathsf{T}(c)$-block in the class of A_i, and Γ_i for that class of $\varepsilon_a \vee \varepsilon_c$; so $\Gamma_i \cap \operatorname{supp} a = A_i$, $\Gamma_i \cap \operatorname{supp} c = C_i$, and the other classes of $\varepsilon_a \vee \varepsilon_c$ meet neither support. For a matched path $a \bowtie c \bowtie b$ we index the $\mathsf{T}(b)$-blocks through c: B_i is the $\mathsf{T}(b)$-block in the class Δ_i of $\varepsilon_b \vee \varepsilon_c$ containing C_i. We also put

$$u = \theta_a(b), \qquad v = \theta_b(a),$$

so that $u \le a$, $v \le b$, $u \bowtie v$ (Lemma 2.3), and $a \bowtie b$ iff $u = a$ and $v = b$. By Lemma 2.6(a) the T-blocks of u are the sets $D \cap \operatorname{supp} a$ for the classes D of $\varepsilon_a \vee \varepsilon_b$ meeting both supports, and $a \bowtie b$ iff every class of $\varepsilon_a \vee \varepsilon_b$ meeting $\operatorname{supp} a \cup \operatorname{supp} b$ contains exactly one A and one B.

2.4. **Finite descent.** In $\mathcal{P}_X$ many arguments end with a rank count: a projection below a projection of the same rank equals it. The following lemma provides the same conclusion in every finite projection algebra, with the size of a principal down-set in place of the rank.

Lemma 2.10 (finite descent)**.** *Let P be a projection algebra.*

(a) *If $p \bowtie q$ then θ_q maps $\downarrow p$ into $\downarrow q$, θ_p maps $\downarrow q$ into $\downarrow p$, and the two maps are mutually inverse bijections.*
(b) *If P is finite, $q \le p$, and p, q lie in one component of the graph of $\bowtie$, then $q = p$.*
(c) *If P is finite, $\theta_p(a) = p$, and a, p lie in one component of the graph of $\bowtie$, then $a \bowtie p$.*
(d) *If P is finite and $a \bowtie c \bowtie b$, then $a \bowtie b$ iff $\theta_a(b) = a$ iff $\theta_b(a) = b$.*

Proof. (a) $\theta_q(e) \le q$ for every e. For $e \le p$ the axiom $\theta_p\theta_q\theta_p = \theta_{\theta_p(q)}$ gives $\theta_p(\theta_q(e)) = \theta_p\theta_q\theta_p(e) = \theta_{\theta_p(q)}(e) = \theta_p(e) = e$, and symmetrically. (b) By (a) and induction along a chain $q = x_0 \bowtie x_1 \bowtie \dots \bowtie x_k = p$, $|\downarrow q| = |\downarrow p|$; as $\downarrow q \subseteq \downarrow p$, the two sets coincide, so $p \le q$ and $q = p$. (c) Put $a' = \theta_a(p) \le a$. By Lemma 2.3, $a' \bowtie \theta_p(a) = p$, so a' lies in the component of p, hence of a, and (b) gives $a' = a$, i.e. $\theta_a(p) = a$. (d) If $u = \theta_a(b) = a$ then $v = \theta_b(a) \le b$ and $v \bowtie u = a \bowtie c \bowtie b$, so $v = b$ by (b); symmetrically. □

In $\mathcal{P}_X$ the down-set $\downarrow p$ depends only on $\operatorname{rank} p$, so (b) is the rank count; but the lemma holds in every finite projection algebra, and this has consequences for the logical relations among our conditions: a quasi-identity whose proof in $\mathcal{P}_X$ uses only the block calculus and rank counts along matched chains often has a finite-algebraic twin which is not derivable equationally. We shall meet one instance in Section 5.8.

3. The local structure is unconstrained

A natural first guess for a condition beyond (Q1) is a restriction on the "local" pieces of S, that is, on its principal factors, which are completely 0-simple regular $*$-semigroups. Some care is needed in saying what such a restriction could be. The zero of a principal factor is

not in general an element of S, and when it is – when S contains a completely 0-simple $*$-subsemigroup together with its zero – (Q1) already forces that subsemigroup to be a Brandt semigroup, since two distinct matched projections have no common lower bound (Section 2.2). So the zero pattern of a principal factor is not local information: it is decided by how the $\mathcal{D}$-class sits inside S, and it interacts with the rest of S, as the triangle condition of Section 4 shows. What remains genuinely local is the nonzero part of a principal factor: a set of projections of a $\mathcal{D}$-class with pairwise nonzero products, together with the sandwich entries and the maximal subgroup. These are exactly the completely simple $*$-subsemigroups of S, and a condition on them would be a condition on finite completely simple regular $*$-semigroups. There is none.

Recall that a completely simple regular $*$-semigroup is a Rees matrix semigroup $\mathcal{M}[G; I, I; P]$ with involution $(i, g, j)^* = (j, g^{-1}, i)$, and that the sandwich matrix may be normalised so that $p_{ii} = 1$ and $p_{ji} = p_{ij}^{-1}$ for all $i, j \in I$ [7]. Its projections are the elements $(i, 1, i)$.

Theorem 3.1. *Every finite completely simple regular $*$-semigroup embeds in a finite partition monoid. Explicitly, let $S = \mathcal{M}[G; I, I; P]$ be in normal form, let $G \leq \mathrm{Sym}(k)$ act on $[k] = \{1, \dots, k\}$ on the right, and let*

$$X = \{a_{i,s} : i \in I,\ s \in [k]\} \ \sqcup\ \{c_{ij,s} : i \neq j \in I,\ s \in [k]\}, \qquad |X| = k|I|^2.$$

For $i \in I$ let π_i be the partition of X into the singletons $\{a_{i,s}\}$, the triples

$$T_{i;l,s} = \{c_{il,s},\ a_{l,sp_{il}},\ c_{li,sp_{il}}\} \qquad (l \in I \setminus \{i\},\ s \in [k])$$

and the singletons $\{c_{lm,s}\}$ with $i \notin \{l, m\}$. Let $\varphi(i, g, j)$ be the partition of $X \sqcup X'$ with transversal blocks $\{a_{i,s}, a'_{j,sg}\}$ ($s \in [k]$), upper nontransversal blocks the blocks of π_i other than the singletons $\{a_{i,s}\}$, and lower nontransversal blocks the primed copies of the blocks of π_j other than the singletons $\{a_{j,s}\}$. Then $\varphi\colon S \to \mathcal{P}_X$ is an injective $*$-homomorphism.

Proof. That π_i is a partition of X is checked by locating each point: $a_{l,s}$ with $l \neq i$ lies in $T_{i;l,sp_{li}}$ only, $c_{il,s}$ in $T_{i;l,s}$ only, $c_{li,s}$ in $T_{i;l,sp_{li}}$ only, and the other points are singletons. The blocks of π_i other than the $\{a_{i,s}\}$ cover $X \setminus \{a_{i,s} : s \in [k]\}$, so $\varphi(i, g, j)$ is a partition of $X \sqcup X'$, of rank k. The description of $\varphi(i, g, j)^*$ is obtained by interchanging i with j and g with g^{-1}, so φ respects the involution; and φ is injective, since the transversal blocks determine i, j and the permutation $s \mapsto sg$, hence g.

For the product $\varphi(i, g, j)\varphi(l, h, m)$ consider the middle layer, which carries the partition π_j (from the lower blocks of the first factor, with the points $a_{j,t}$ attached to the transversal blocks $\{a_{i,s}, a'_{j,sg}\}$) and the partition π_l (from the upper blocks of the second factor, with the points $a_{l,t}$ attached to the transversal blocks $\{a_{l,t}, a'_{m,th}\}$). If $j = l$ the two partitions coincide, so the components of the middle layer are the blocks of π_j; the point $a_{j,sg}$ is a singleton of π_j and is attached to $a_{i,s}$ above and $a'_{m,sgh}$ below, and no other middle block meets a transversal block. The product therefore has transversal blocks $\{a_{i,s}, a'_{m,sgh}\}$, upper nontransversal blocks those of the first factor and lower ones those of the second: it is $\varphi(i, gh, m) = \varphi((i, g, j)(j, h, m))$, as $p_{jj} = 1$. If $j \neq l$, put $t = sg\,p_{jl}$ and consider the set

$$M_s = \{a_{j,sg},\ c_{jl,sg},\ c_{lj,t},\ a_{l,t}\} = T_{l;j,t} \cup T_{j;l,sg} \cup \{a_{j,sg}\} \cup \{a_{l,t}\},$$

where $T_{l;j,t} = \{c_{lj,t}, a_{j,tp_{lj}}, c_{jl,tp_{lj}}\} = \{c_{lj,t}, a_{j,sg}, c_{jl,sg}\}$ is a block of π_l and $T_{j;l,sg} = \{c_{jl,sg}, a_{l,t}, c_{lj,t}\}$ is a block of π_j. Each point of M_s lies, in each of π_j and π_l, in a block contained in M_s (the points $a_{j,sg}$ and $a_{l,t}$ are singletons of π_j and π_l respectively), so M_s is a component of the middle layer; it contains exactly one transversal end $a_{j,sg}$ of the first factor, attached to $a_{i,s}$, and exactly one transversal start $a_{l,t}$ of the second, attached to $a'_{m,th}$. Hence the product has the transversal blocks $\{a_{i,s}, a'_{m,sgp_{jl}h}\}$, which are those of $\varphi(i, gp_{jl}h, m)$. The upper nontransversal blocks of the first factor have no points in the middle layer, so they are

blocks of the product; every other upper point lies in a transversal block; so the upper non-transversal blocks of the product are those of $\varphi(i,\cdot,\cdot)$, and dually for the lower ones. Thus $\varphi(i,g,j)\varphi(l,h,m) = \varphi(i,gp_{jl}h,m) = \varphi((i,g,j)(l,h,m))$. □

Since every finite group embeds in some $\mathrm{Sym}(k)$, every finite completely simple regular $*$-semigroup embeds in some $\mathcal{P}_n$. Two remarks on what this rules out.

Remark 3.2. Orthodoxy of the completely simple $*$-subsemigroups is not necessary either, although $\mathcal{P}_n$ for small n looks "flat": the projections

$$p : \{0\},\{1\} \text{ transversal, } \{2,3\} \text{ nontransversal;}$$
$$q : \{1\},\{3\} \text{ transversal, } \{0,2\} \text{ nontransversal;}$$
$$r : \{0\},\{3\} \text{ transversal, } \{1,2\} \text{ nontransversal}$$

of $\mathcal{P}_4$ satisfy $p \bowtie q \bowtie r \bowtie p$, but $pqr \neq pr$: the sandwich matrix of the rank-2 $\mathcal{D}$-class of $\mathcal{P}_4$ has a non-flat triangle, and the $*$-subsemigroup generated by p,q,r is a non-orthodox completely simple regular $*$-semigroup $\mathcal{M}[\mathbb{Z}_2;3,3;P]$ with $p_{12} \neq 1$.

Remark 3.3 (normal $*$-bands)**.** For a strong semilattice Y of square $*$-bands $A_\alpha \times A_\alpha$ with structure maps $f_{\alpha\beta}\colon A_\alpha \to A_\beta$ ($\alpha \geq \beta$), (Q1) holds iff all $f_{\alpha\beta}$ are injective: a projection (c,c) of a lower component lies below (a,b) iff $f(a) = f(b) = c$. If Y has a least element 0, injectivity lets us regard every A_α as a subset of $A = A_0$, and the band becomes the $*$-subsemigroup $\{(\alpha,a,b) : a,b \in A_\alpha\}$ of $Y \times (A \times A)$. Since semilattices and square bands embed in partition monoids and $\mathcal{P}_X \times \mathcal{P}_Y \leq \mathcal{P}_{X \sqcup Y}$, a finite normal $*$-band embeds in a partition monoid iff it satisfies (Q1).

4. The digraph conditions and the triangle condition

Before the wiring conditions we present a family of necessary conditions of a different shape, indexed by digraphs: k pairwise commuting projections with common product w, k pairwise commuting projections with common product z, matched crosswise according to the digraph, force $w = z$. The family has a single strongest member, the triangle condition, which implies all the others in every projection algebra, and its witness is a quotient of $\mathcal{P}_3$.

4.1. **The conditions and their validity.** Let $k \geq 3$ and let E be a set of ordered pairs (i,j) of distinct elements of $[k] = \{1,\dots,k\}$, a loopless digraph. For projections $a_1,\dots,a_k,b_1,\dots,b_k$ of a regular $*$-semigroup consider the hypotheses

(H1) $a_ia_j = a_ja_i =: w$ and $b_ib_j = b_jb_i =: z$ for all $i \neq j$, the products being independent of the pair $\{i,j\}$;
(H2) $a_ib_i = b_ia_i =: m_i$ for all i;
(H3) $a_i \bowtie b_j$ for all $(i,j) \in E$,

and let (D_E) be the quasi-identity "(H1)–(H3) imply $w = z$". By Lemma 2.1 all hypotheses are equations of the projection algebra, so (D_E) is a quasi-identity in $2k$ variables. (In (H1) it suffices to assume the "cyclic" equations $\theta_{a_2}(a_1) = \theta_{a_3}(a_2) = \theta_{a_1}(a_3)$ and so on: if w denotes the common value, then $\theta_{a_1}(a_2) = a_1a_2a_1 = (a_1a_2a_1)^2 = a_1(a_2a_1a_2)a_1 = a_1wa_1 = a_1(a_3a_1a_3)a_1 = (a_1a_3a_1)^2 = w$, and the a_i pairwise commute with product w by Lemma 2.1.) Two instances are of particular interest: $k = 3$ with E all six pairs, the *hexagon condition*, in which the six matched pairs form the hexagon $a_1\,b_2\,a_3\,b_1\,a_2\,b_3$; and $k = 3$ with $E = \{(1,2),(2,3),(3,1)\}$ a directed 3-cycle, the *triangle condition* (Tri) (Figure 1), which assumes only $a_1 \bowtie b_2$, $a_2 \bowtie b_3$, $a_3 \bowtie b_1$ and therefore implies the hexagon condition trivially. Call E a *star* if it is empty, or all its edges have the same tail, or all have the same head; E is not a star iff it has two distinct

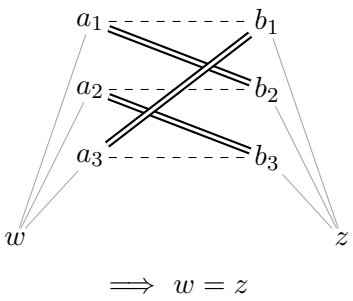


FIGURE 1. The triangle condition (Tri): the matched pairs $a_i \bowtie b_j$ follow the directed 3-cycle $1 \to 2 \to 3 \to 1$ of indices. Double lines are matched pairs, dashed lines commuting pairs; $w = a_i a_j$ and $z = b_i b_j$ for $i \neq j$.

tails and two distinct heads. The hypothesis $k \geq 3$ cannot be dropped: for two matched pairs $a_1 \bowtie b_2$, $a_2 \bowtie b_1$ the statement fails already in $\mathcal{P}_2$.

The matchings contribute through the following *transport relations*.

Lemma 4.1. *Under* (H1)–(H3)*, for every edge* $(i,j) \in E$*:*

$$a_i m_j a_i = w, \qquad b_j m_i b_j = z, \qquad a_i z a_i = m_i, \qquad b_j w b_j = m_j. \tag{3}$$

Proof. $a_i m_j a_i = a_i a_j b_j a_i = a_j(a_i b_j a_i) = a_j a_i = w$; $b_j m_i b_j = b_j b_i a_i b_j = b_i(b_j a_i b_j) = b_i b_j = z$; $a_i z a_i = a_i b_j b_i a_i = (a_i b_j a_i) b_i = m_i$; and $b_j w b_j = b_j a_i a_j b_j = (b_j a_i b_j) a_j = m_j$. □

Call E *admissible* if every vertex has an out-edge and an in-edge, and no vertex is incident to all edges. The triangle is admissible; so are the directed cycles C_ℓ, $\ell \geq 3$, and the union of two disjoint 2-cycles on four vertices.

Theorem 4.2. *Let* E *be an admissible digraph on* $[k]$*. Then* (D_E) *holds in every partition monoid* $\mathcal{P}_X$*. Moreover, under* (H1)–(H3)*,* $a_i = b_j$ *for all* $(i,j) \in E$*, and if the bipartite graph* $\{a_i — b_j : (i,j) \in E\}$ *is connected then all* $2k$ *projections coincide.*

Proof. Write $A_i = \operatorname{supp} a_i$, $B_i = \operatorname{supp} b_i$, $W = \operatorname{supp} w$, $Z = \operatorname{supp} z$. By admissibility every i has an out-edge and an in-edge, so by Lemma 4.1 for every i there are j, j' with $a_i m_j a_i = w$, $a_i z a_i = m_i$, $b_i m_{j'} b_i = z$ and $b_i w b_i = m_i$.

Step 0: supports. By (H1) and Lemma 2.6(b), $A_i \cap A_j = W$ and $B_i \cap B_j = Z$ for $i \neq j$; by (H2), $A_i \cap B_i = \operatorname{supp} m_i$. In particular a point of X lies in at most one A_i unless it lies in W (and then in all of them), and similarly for the B_i and Z. Since $k \geq 3$,

$$\text{every } x \notin W \cup Z \text{ lies outside } A_i \cup B_i \text{ for some } i. \tag{4}$$

Step 1: ε_w and ε_z agree outside $W \cup Z$. By (3) and Lemma 2.6(a), applied to $z = \theta_{b_i}(m_{j'})$, $w = \theta_{a_i}(m_j)$, $m_i = \theta_{a_i}(z)$ and $m_i = \theta_{b_i}(w)$, for every i: the blocks of z not contained in B_i are exactly the $\mathsf{N}(b_i)$-blocks; the blocks of w not contained in A_i are exactly the $\mathsf{N}(a_i)$-blocks; and the blocks of m_i not contained in A_i (resp. B_i) are exactly the $\mathsf{N}(a_i)$-blocks (resp. $\mathsf{N}(b_i)$-blocks). Consequently, for $x \notin A_i \cup B_i$, the block of z containing x is the $\mathsf{N}(b_i)$-block of x, which is the block of m_i containing x, which is the $\mathsf{N}(a_i)$-block of x, which is the block of w containing x. By (4), ε_w and ε_z have the same block at every point of $X \setminus (W \cup Z)$; moreover such a common block is an $\mathsf{N}(a_i)$-block and an $\mathsf{N}(b_i)$-block, hence contained in $X \setminus (A_i \cup B_i) \subseteq X \setminus (W \cup Z)$. It follows that a block of w or of z is either a common block contained in $X \setminus (W \cup Z)$, or is contained in $W \cup Z$.

Step 2: the $\mathsf{T}(w)$- and $\mathsf{T}(z)$-blocks correspond. Fix an edge $(i,j) \in E$. From (H1) and (3), $wzw = a_i(a_j z a_j)a_i = a_i m_j a_i = w$ and $zwz = b_j(b_i w b_i)b_j = b_j m_i b_j = z$, so $w \bowtie z$. By

Lemma 2.6(c), every class of $\varepsilon_w \vee \varepsilon_z$ meeting $W \cup Z$ contains exactly one $\mathsf{T}(w)$-block and exactly one $\mathsf{T}(z)$-block. Common blocks are classes by themselves, so the class of a $\mathsf{T}(w)$-block Ω consists of blocks contained in $W \cup Z$: these are $\mathsf{T}(w)$-blocks, $\mathsf{N}(w)$-blocks contained in $Z \setminus W$, $\mathsf{T}(z)$-blocks and $\mathsf{N}(z)$-blocks contained in $W \setminus Z$. An $\mathsf{N}(z)$-block contained in W meets only $\mathsf{T}(w)$-blocks, so it meets only one of them (two would lie in a common class); hence every $\mathsf{N}(z)$-block contained in W lies inside a single $\mathsf{T}(w)$-block, and dually. Therefore the class of Ω consists of Ω, the $\mathsf{N}(z)$-blocks inside Ω, the $\mathsf{T}(z)$-blocks meeting Ω and the $\mathsf{N}(w)$-blocks inside those, and since it contains exactly one $\mathsf{T}(z)$-block, *there is exactly one* $\mathsf{T}(z)$*-block* Z_Ω *meeting* Ω, and dually exactly one $\mathsf{T}(w)$-block W_Θ meeting a given $\mathsf{T}(z)$-block Θ.

Step 3: $\Omega \subseteq Z_\Omega$. Suppose $y \in \Omega \setminus Z_\Omega$. Then $y \notin Z$ (a $\mathsf{T}(z)$-block containing y would meet Ω, hence be Z_Ω), so the block N of z containing y is an $\mathsf{N}(z)$-block, and by Step 2, $N \subseteq \Omega \subseteq W \subseteq A_1 \cap \cdots \cap A_k$. For each i, N is a block of $z = \theta_{b_i}(m_{j'})$, so by Lemma 2.6(a) either $N \subseteq B_i$ or N is an $\mathsf{N}(b_i)$-block. As $N \cap Z = \emptyset$ and $B_i \cap B_l = Z$, $N \subseteq B_i$ holds for at most one index $i = m$; by admissibility there is an edge $(i,j) \in E$ with $m \notin \{i,j\}$, so that N is both an $\mathsf{N}(b_i)$-block and an $\mathsf{N}(b_j)$-block. Since $N \subseteq A_i$ and $a_i b_i = b_i a_i$, Lemma 2.6(b) shows that N is a union of $\mathsf{T}(a_i)$-blocks; let τ be one of them. Now $a_i \bowtie b_j$ by (H3). The class of $\varepsilon_{a_i} \vee \varepsilon_{b_j}$ containing τ contains the b_j-block $N \supseteq \tau$, hence all $\mathsf{T}(a_i)$-blocks inside N; by Lemma 2.6(c) there is only one, so $N = \tau$. But then N is simultaneously a $\mathsf{T}(a_i)$-block and an $\mathsf{N}(b_j)$-block, which is impossible for a matched pair by Lemma 2.6(c). Hence $\Omega \subseteq Z_\Omega$.

Step 4: conclusion. Dually $Z_\Omega \subseteq W_{Z_\Omega}$ (with the roles of a and b exchanged: an $\mathsf{N}(w)$-block $N \subseteq Z \subseteq B_j$ is a union of $\mathsf{T}(b_j)$-blocks because b_j commutes with a_j, and $b_j \bowtie a_i$); since Ω meets Z_Ω and W_{Z_Ω} is the only $\mathsf{T}(w)$-block meeting Z_Ω, we get $W_{Z_\Omega} = \Omega$ and so $\Omega = Z_\Omega$. Thus $\mathsf{T}(w) = \mathsf{T}(z)$ and $W = Z$. Inside $W = Z$ both ε_w and ε_z consist of these T-blocks, and outside $W \cup Z$ they agree by Step 1. Hence $w = z$.

Finally, $w = z$ is a common lower bound of a_i and b_j, so by (Q1) $a_i b_j = b_j a_i$ for all i, j; for $(i,j) \in E$ this gives $a_i = a_i b_j a_i = a_i b_j$, i.e. $a_i \leq b_j$, and dually $b_j \leq a_i$, so $a_i = b_j$. If the bipartite graph of E is connected, all $2k$ projections coincide. □

4.2. **The triangle condition is a master condition.** In this subsection $a_1, \ldots, a_k, b_1, \ldots, b_k$ are projections of a regular $*$-semigroup satisfying (H1), (H2) for some $k \geq 2$, and (H3) for a digraph E when so stated. We show that the matchings of E serve only to establish transport relations, and that once these are available, matchings can be manufactured at will, so that the triangle condition implies (D_E) for every non-star E. We shall use repeatedly that θ_p is monotone in every projection algebra: $x \leq y$ implies $\theta_p(x) \leq \theta_p(y)$, since in the binary notation the axioms give $(x * y) * (z * y) = x * (z * y)$ and $(x * y) * (y * z) = (x * y) * z$, whence $x * y = x$ implies $(x * p) * (y * p) = x * (y * p) = (x * y) * (y * p) = (x * y) * p = x * p$.

Lemma 4.3. *If $z a_i z = z$ then $b_j m_i b_j = z$ for all $j \neq i$; if $w b_i w = w$ then $a_j m_i a_j = w$ for all $j \neq i$.*

Proof. Put $e = b_j m_i b_j$. Writing $m_i = b_i a_i$ and $b_j b_i = z$ gives $e = z a_i b_j$; writing $m_i = a_i b_i$ gives $e = b_j a_i z$. Hence $e = ze = ez$, so $e \leq z$. Conversely $zez = z m_i z = z a_i z = z$; so $z \leq e$ and $e = z$. Dually for w. □

Lemma 4.4 (bounded elements)**.** *Let $e \leq a_i$, $e' \leq a_j$ with $i \neq j$, and $w \leq e, e'$. Then $ee' = e'e = w$. Dually two elements under distinct b's and above z commute with product z. If $e \leq a_i$, $f \leq b_i$ and $m_i \leq e, f$, then $ef = fe = m_i$.*

Proof. $ee'e = e(a_i a_j)e'(a_j a_i)e = ewe'we = ewe = w$, and symmetrically $e'ee' = w$; so $ee' = e'e = w$ by Lemma 2.1. Likewise $efe = e(a_i b_i)f(b_i a_i)e = e m_i f m_i e = m_i = fef$. □

Call the family *transported* if the four relations

$$a_i z a_i = m_i, \qquad b_i w b_i = m_i, \qquad z a_i z = z, \qquad w b_i w = w \tag{5}$$

hold at every index i. In a transported family $w \bowtie z$ (as $wzw = a_i(a_j z a_j)a_i = a_i m_j a_i = w$ by Lemma 4.3, and dually), and $b_j m_i b_j = z$, $a_j m_i a_j = w$ for all $i \neq j$.

Theorem 4.5 (non-star patterns)**.** (a) *If E is not a star, then* (H1)–(H3) *imply, in every projection algebra, that* (5) *holds at every non-isolated index i of E; moreover each isolated index i can be replaced by the pair $a_i' := \theta_{a_i}(b_{j_0})$, $b_i' := \theta_{b_i}(a_{j_0})$, where j_0 is any non-isolated index, after which* (H1)–(H3) *still hold with the same w, z and the family is transported.*

(b) *If E is a star, then $\mathcal{P}_2$ contains projections satisfying* (H1)–(H3) *for E with $w \neq z$.*

Proof. (a) For an index i with an out-edge (i,l) we have $a_i z a_i = a_i b_l b_i a_i = (a_i b_l a_i) b_i = m_i$ and $z a_i z = z b_l a_i b_l z = z$; dually an in-edge (l,i) gives $b_i w b_i = m_i$ and $w b_i w = w$.

Step 1: $b_i w b_i = m_i$ at every tail i. Let $i \neq i'$ be tails. Since $w \leq a_i$, $w b_i w = w a_i b_i a_i w = w m_i w = w(a_i z a_i)w = wzw$, and likewise $w b_{i'} w = wzw$. Hence, $b_i w b_i$ being a projection,

$$b_i w b_i = b_i(w b_i w) b_i = b_i(w b_{i'} w) b_i = b_i a_i a_{i'} b_{i'} a_{i'} a_i b_i = m_i\, m_{i'}\, m_i = a_i b_i\, (b_{i'} a_{i'} b_{i'})\, b_i a_i = a_i z a_{i'} z a_i = a_i z a_i = m_i,$$

using $b_i b_{i'} = z$, $z a_{i'} z = z$ and $a_i z a_i = m_i$. Dually $a_j z a_j = m_j$ at every head j, since there are two distinct heads.

Step 2: $w \bowtie z$. Choose a tail i and a head j with $i \neq j$. Then $wzw = a_i a_j z a_j a_i = a_i m_j a_i = a_i a_j b_j a_j a_i = w b_j w = w$ by Step 1 and the in-edge at j, and $zwz = b_j b_i w b_i b_j = b_j m_i b_j = b_j b_i a_i b_i b_j = z a_i z = z$ by Step 1 and the out-edge at i.

Step 3: the remaining relations. At a tail i we now have $a_i z a_i = m_i$, $z a_i z = z$ and $b_i w b_i = m_i$, and $w b_i w = w$ follows from $w \bowtie z$ by monotonicity: $z \leq b_i$ gives $w = wzw \leq w b_i w \leq w$. Dually at a head. So (5) holds at every non-isolated index.

Step 4: isolated indices. Let i be isolated and j_0 non-isolated. Since $w \leq a_i$, monotonicity gives $z = zwz \leq z a_i z \leq z$, so $z a_i z = z$, and dually $w b_i w = w$; by Lemma 4.3, $b_l m_i b_l = z$ and $a_l m_i a_l = w$ for $l \neq i$. The other two relations of (5) may fail at i, but the two sandwiches are equal: $u_i := a_i z a_i = b_i w b_i = m_i m_{j_0} m_i$. Indeed $z = b_i m_{j_0} b_i$ (Lemma 4.3 at j_0), and writing $m_{j_0} = a_{j_0} b_{j_0}$, $b_{j_0} b_i = z$ in one copy and $m_{j_0} = b_{j_0} a_{j_0}$ in the other, $z = b_i a_{j_0} z = z a_{j_0} b_i$; hence $a_i z a_i = a_i(b_i a_{j_0} z)(z a_{j_0} b_i) a_i = m_i(a_{j_0} z a_{j_0}) m_i = m_i m_{j_0} m_i$, and dually $b_i w b_i = m_i(b_{j_0} w b_{j_0}) m_i = m_i m_{j_0} m_i$. Thus $u_i = \theta_{m_i}(m_{j_0}) \leq m_i$, and for $l \neq i$ we get $b_l u_i b_l = z$: it is $\leq b_l m_i b_l = z$ by monotonicity, and $z(b_l u_i b_l) z = z u_i z = z a_i z a_i z = z$; dually $a_l u_i a_l = w$.

Put $a_i' = a_i b_{j_0} a_i$ and $b_i' = b_i a_{j_0} b_i$. Then $a_i' \leq a_i$ and $w \leq a_i'$, since $a_i' w a_i' = a_i(b_{j_0} w b_{j_0}) a_i = a_i m_{j_0} a_i = w$; dually $z \leq b_i' \leq b_i$. By Lemma 4.4, a_i' commutes with every a_l ($l \neq i$), and with a_l' if l is also isolated, with product w, and dually for b_i': (H1) holds for the new family, with the same w and z. For (H2),

$$a_i' b_i' a_i' = a_i b_{j_0}(m_i a_{j_0} m_i) b_{j_0} a_i = a_i b_{j_0}\big(b_i(a_i a_{j_0} a_i) b_i\big) b_{j_0} a_i = a_i(b_{j_0} u_i b_{j_0}) a_i = a_i z a_i = u_i,$$

and dually $b_i' a_i' b_i' = b_i w b_i = u_i$; so $a_i' b_i' = b_i' a_i' = u_i =: m_i'$ by Lemma 2.1. (H3) is unaffected, as i lies on no edge. Finally (5) holds at i for the new pair: $a_i' z a_i' = a_i b_{j_0}(a_i z a_i) b_{j_0} a_i = a_i(b_{j_0} u_i b_{j_0}) a_i = u_i = m_i'$, dually $b_i' w b_i' = m_i'$, and $z a_i' z = z$, $w b_i' w = w$ as before since $w \leq a_i'$, $z \leq b_i'$. The relations at the other indices only involve a_l, b_l, m_l, w, z for $l \neq i$ and are unchanged.

(b) In $\mathcal{P}_2$ let F be the projection with the single block $\{1, 2, 1', 2'\}$ (rank 1) and e the projection with blocks $\{1,2\}$, $\{1',2'\}$ (rank 0); then $e \leq F$ and $F \bowtie F$. For an out-star with tail v (or for $E = \emptyset$) take $a_v = F$, $a_i = e$ for $i \neq v$, and $b_j = F$ for all j: then $w = e$, $z = F$, all pairs commute, $a_v \bowtie b_j$ for all j, and $w \neq z$. For an in-star with head v take $a_i = F$ for all i, $b_v = F$ and $b_j = e$ for $j \neq v$. ∎

Lemma 4.6 (edge creation)**.** *Let the family be transported, $k \geq 2$, and $i \neq j$. Put $a_i' = \theta_{a_i}(b_j)$, $b_j' = \theta_{b_j}(a_i)$, and replace a_i by a_i' and b_j by b_j'. The new family is again transported, with the same w, z and $m_1, \ldots, m_k$; $a_i' \bowtie b_j'$; and every pair a_l, b_m with $l \neq i$, $m \neq j$ is unchanged.*

Proof. $a_i' \leq a_i$, and $w, m_i \leq a_i'$: indeed $a_i' w a_i' = a_i(b_j w b_j)a_i = a_i m_j a_i = w$ and $a_i' m_i a_i' = a_i(b_j m_i b_j)a_i = a_i z a_i = m_i$. Dually $b_j' \leq b_j$ and $z, m_j \leq b_j'$. By Lemma 4.4, a_i' commutes with every a_l ($l \neq i$) with product w, and with b_i with product m_i; dually for b_j'; so (H1), (H2) hold with the same w, z, m_l. The transport relations at i: $a_i' z a_i' = a_i(b_j m_i b_j)a_i = m_i$, $b_i w b_i = m_i$ is unchanged, $z a_i' z = z$ since $z = zwz \leq z a_i' z \leq z$ by monotonicity ($w \leq a_i'$), and $w b_i w = w$ is unchanged; dually at j; the other indices are untouched. Finally $a_i' \bowtie b_j'$ by Lemma 2.3. □

Theorem 4.7 (master condition)**.** (a) *Every transported family on $k \geq 3$ indices yields, on any three of its indices, a triangle configuration with the same w and z. Consequently, in every projection algebra, the triangle condition implies* (D_E) *for every non-star digraph E.*

(b) *For every set X with $|X| \geq 2$,* (D_E) *holds in $\mathcal{P}_X$ iff E is not a star.*

Proof. (a) Apply Lemma 4.6 three times on the indices $1, 2, 3$: create $(1, 2)$, replacing a_1, b_2; then $(2, 3)$, replacing a_2, b_3; then $(3, 1)$, replacing a_3, b_1. Each step leaves the pairs created before untouched, since they involve different elements: after the three steps $a_1' \bowtie b_2'$, $a_2' \bowtie b_3'$, $a_3' \bowtie b_1'$, and (H1), (H2) hold with the original w, z. For the consequence, take a configuration for a non-star E, replace its isolated indices (Theorem 4.5(a)), obtaining a transported family with the same w, z, and build the triangle configuration; the triangle condition gives $w = z$. (b) (Tri) holds in $\mathcal{P}_X$ by Theorem 4.2, so the non-star conditions hold by (a); the stars fail already in $\mathcal{P}_2 \leq \mathcal{P}_X$ by Theorem 4.5(b). □

Thus only the triangle needs the combinatorial proof of Theorem 4.2; every other non-star instance follows from it algebraically. The hexagon condition, which was the original form of the theorem, is equivalent to the triangle condition on finite projection algebras, but for a subtle reason: no fixed sextuple of terms in $a_1, \ldots, b_3$ satisfies the hexagon hypotheses in every triangle configuration (in the initial model of the Horn theory "axioms plus triangle hypotheses" neither $w = z$ nor any such hypothesis holds), and the six projections to which the hexagon condition is applied are found by an iteration.

Proposition 4.8. *Let P be a finite projection algebra satisfying the hexagon condition, and let $a_1, \ldots, a_k, b_1, \ldots, b_k$ be a transported family in P, $k \geq 3$. Then $w = z$. In particular, on finite projection algebras the hexagon condition is equivalent to the triangle condition and to* (D_E) *for every non-star E on three vertices.*

Proof. Call a projection x *bounded* if $x \leq a_i$ and $w, m_i \leq x$ for some i, or $x \leq b_j$ and $z, m_j \leq x$ for some j; the $2k$ vertices are bounded. Fix three indices, say $1, 2, 3$, and for $x \leq a_1$ consider the *walk*

$$x \xrightarrow{\theta_{b_3}} x_1 \xrightarrow{\theta_{a_2}} x_2 \xrightarrow{\theta_{b_1}} x_3 \xrightarrow{\theta_{a_3}} x_4 \xrightarrow{\theta_{b_2}} x_5 \xrightarrow{\theta_{a_1}} x_6 =: \Phi(x) \leq a_1,$$

whose elements lie under $b_3, a_2, b_1, a_3, b_2, a_1$ respectively. (i) *The walk preserves boundedness.* Let $x \leq a_i$ be bounded and $y = \theta_b(x)$ with $b = b_{i-1}$ (indices mod 3). Since $z \leq b$, $yzy = b\,x(bzb)x\,b = b(xzx)b$, and $xzx = x(a_i z a_i)x = x m_i x = m_i$, so $yzy = b m_i b = z$, i.e. $z \leq y$; similarly $y m_{i-1} y = b(x m_{i-1} x)b = b(x(a_i m_{i-1} a_i)x)b = b(xwx)b = bwb = m_{i-1}$, using the transport relations $a_i m_{i-1} a_i = w$ (Lemma 4.3) and $b_{i-1} w b_{i-1} = m_{i-1}$. The step $y \leq b_j \mapsto x' = \theta_{a_{j-1}}(y)$ is the same computation with a and b exchanged. (ii) *Stabilisation.* If $x_0 \to x_1 = \theta_{p_1}(x_0) \to \cdots \to x_n = \theta_{p_n}(x_{n-1}) = x_0$ is a closed walk of projections in a regular $*$-semigroup S with finitely many projections, then $x_{j-1} \bowtie x_j$ for all j. Indeed, put $s_j = x_{j-1} p_j$, so that

$s_j^* s_j = x_j$ and $s_j s_j^* = \theta_{x_{j-1}}(p_j) \leq x_{j-1}$, and $s = s_1 s_2 \cdots s_n$. Since $s_j^* s_j s_{j+1} = x_j s_{j+1} = s_{j+1}$, we get $s^* s = x_n = x_0$, and $ss^* \leq s_1 s_1^* = \theta_{x_0}(p_1) \leq x_0$. Now $t \mapsto s^* t s$ is an order isomorphism from the projections below ss^* onto the projections below $s^* s$; as $ss^* \leq s^* s$ and there are finitely many projections, $ss^* = s^* s$. Hence $\theta_{x_0}(p_1) = x_0$, so $x_0 x_1 x_0 = (x_0 p_1 x_0)(p_1 x_0) = x_0$, while $x_1 x_0 x_1 = x_1$ by Lemma 2.3; rotating the cycle gives the other pairs. (iii) By Imaoka's theorem $P = \mathsf{P}(S)$ for a regular $*$-semigroup S. The operator Φ is a composite of maps θ_p, hence monotone, and $\Phi(a_1) \leq a_1$; so $a_1 \geq \Phi(a_1) \geq \Phi^2(a_1) \geq \cdots$ is eventually constant, and there is $x^* = \Phi^n(a_1)$ with $\Phi(x^*) = x^*$. By (i) all elements of the walk from x^* are bounded, and the walk $x^* \to x_1 \to \cdots \to x_6 = x^*$ is closed, so by (ii) consecutive elements are matched. Put $A_3 = x_1 \leq b_3$, $A_1 = x_3 \leq b_1$, $A_2 = x_5 \leq b_2$, $B_2 = x_2 \leq a_2$, $B_3 = x_4 \leq a_3$, $B_1 = x_6 \leq a_1$. By Lemma 4.4 the A_j pairwise commute with product z, the B_i with product w, and $A_i B_i = B_i A_i$; the six consecutive matchings are exactly the matchings $A_i \bowtie B_j$, $i \neq j$. So $(B_1, B_2, B_3 \mid A_1, A_2, A_3)$ satisfies the hexagon hypotheses with products w and z, and the hexagon condition gives $w = z$. The last assertion follows with Theorem 4.5(a), since every E on three vertices has fewer hypotheses than the hexagon. □

Remark 4.9. The finiteness in Proposition 4.8 cannot be dropped. "Unrolling" the hexagon of the witness $\mathsf{P}(Q_0)$ below along its cycle of matchings, replacing each vertex by a descending chain $e_{i,0} > e_{i,1} > \cdots$, $f_{i,0} > f_{i,1} > \cdots$ with all matchings taken level by level except one, which shifts the level by one, produces a countable projection algebra H_∞ satisfying (Q1) and the hexagon condition (and (D_E) for every non-star E on $k \geq 4$ vertices) in which $(e_{1,0}, e_{2,0}, e_{3,0} \mid f_{1,0}, f_{2,0}, f_{3,0})$ is a triangle configuration with $w \neq z$. So the hexagon condition does not imply the triangle condition in the variety of projection algebras, and a regular $*$-semigroup with projection algebra H_∞ satisfies (Q1) and the hexagon condition but embeds in no partition monoid. In H_∞ the walk of the proof closes after one turn but the stabilisation step (ii) fails. The verification of the axioms for H_∞ is a finite computation (the operation is determined by the types and levels of its arguments, with level shifts bounded by the depth of the terms involved); we omit it.

4.3. **The witness.**

Theorem 4.10. *Let μ be the relation on $\mathcal{P}_3$ identifying two elements of rank ≤ 1 iff they have the same kernel and the same cokernel (the partitions of $\{1,2,3\}$ induced by their upper and lower blocks). Then μ is a $*$-congruence, $\mathcal{P}_3/\mu$ has 103 elements, and the $*$-subsemigroup Q_0 of $\mathcal{P}_3/\mu$ generated by the images of the six projections of rank 2 has 61 elements, satisfies (Q1), and embeds in no partition monoid; its projection algebra $\mathsf{P}(Q_0)$ has 11 elements and violates the triangle condition.*

Proof. That μ is a $*$-congruence (it is the congruence that collapses the ideal of rank ≤ 1 onto the 5×5 square band $\mathrm{Part}_3 \times \mathrm{Part}_3$ of pairs of partitions of $\{1,2,3\}$, one of the congruences below the Rees congruence of that ideal in the classification of [4]), that the quotients satisfy (Q1) and the element counts were checked by direct computation. Write a_i for the image of the rank-2 projection of $\mathcal{P}_3$ whose transversal blocks are $\{i\}$ and $\{1,2,3\} \setminus \{i\}$, and b_i for the image of the rank-2 projection whose transversal blocks are the two singletons $\{j\}$, $\{k\}$ ($\{i,j,k\} = \{1,2,3\}$) and whose nontransversal block is $\{i\}$. In $\mathcal{P}_3/\mu$ the hypotheses (H1)–(H3) hold with E the set of all six pairs: w is the class of the partition of $\{1,2,3\}$ into one block, z that of the discrete partition, m_i that of the partition $\{i\} \mid \{j,k\}$, and the products $a_i b_j$ $(i \neq j)$ have rank 2; all this is a direct verification in $\mathcal{P}_3$. But $w \neq z$, since the two partitions are different. By Theorem 4.2 no $*$-homomorphism $Q_0 \to \mathcal{P}_X$ is injective on $\{w, z\}$. The projection algebra $\mathsf{P}(Q_0)$ consists of $a_1, \ldots, b_3$ and the five "low" elements w, z, m_1, m_2, m_3, with $\theta_\ell(x) = \ell$ for every low ℓ, $\theta_{a_i}(w) = w$, $\theta_{a_i}(z) = \theta_{a_i}(m_i) = m_i$, $\theta_{a_i}(m_j) = w$ $(j \neq i)$, and dually for the b_i. □

Concretely, $Q_0 = D \cup (\mathrm{Part}_3 \times \mathrm{Part}_3)$, where D is the rank-2 $\mathcal{D}$-class of $\mathcal{P}_3$ with its $\mathcal{H}$-classes collapsed, a 6×6 trace with trivial groups whose zero pattern is the 6-cycle $a_1b_2a_3b_1a_2b_3$ of matched pairs. The obstruction lives in the projection algebra: $\mathsf{P}(Q_0)$ is a minimal non-embeddable projection algebra (its six maximal proper subalgebras, obtained by omitting one of a_i, b_i, embed in $\mathsf{P}(\mathcal{P}_4)$), and it is the smallest projection algebra satisfying (Q1) and violating (Tri): a Mace4 search shows that no projection algebra of order ≤ 10 does, and that the one of order 11 is $\mathsf{P}(Q_0)$ (Section 8). The analogous quotients of $\mathcal{P}_4$ contain those of $\mathcal{P}_3$ and give nothing new; the further obstructions of the next section were found instead by enumerating small projection algebras.

5. Wiring conditions

5.1. **The shape of the conditions.** All the conditions of this section are statements about a matched path $a \bowtie c \bowtie b$, with the conclusion $a \bowtie b$, under hypotheses on one, two or three auxiliary projections. In $\mathcal{P}_X$ the hypotheses $a \bowtie c$, $c \bowtie b$ pair the T-blocks $A_i \leftrightarrow C_i \leftrightarrow B_i$ through the classes Γ_i and Δ_i, but a $\mathsf{T}(a)$-block and a $\mathsf{T}(b)$-block in corresponding classes need not meet or even be connected through a and b: the matching does not transport. The conditions say that it does transport as soon as the auxiliaries are wired to the path in one of the following ways, which we name for later reference: an auxiliary x *commutes* with a vertex ($xp = px$); it *lies below* or *above* a vertex; it *sees two vertices alike* ($\theta_x(p) = \theta_x(q)$); it *is seen alike* by two vertices ($\theta_p(x) = \theta_q(x)$); or it is *half-matched* with a vertex ($\theta_p(x) = p$). The conditions are the following; (S) and (R) are the first two we found, and are consequences of (Q) and (R′) respectively (Corollaries 5.5 and 5.13), which is the order in which they are proved below.

$$
\begin{aligned}
&\text{(S)}: && a \bowtie c \bowtie b \bowtie d,\ cd = dc,\ \theta_a(cd) = \theta_b(cd) \implies a \bowtie d;\\
&\text{(R)}: && a \bowtie c \bowtie b,\ xa = ax,\ xc = cx,\ \theta_x(b) = \theta_x(c) \implies a \bowtie b;\\
&\text{(V)}: && bcb = b,\ ata \leq b,\ ctc \leq a \implies ata = btb;\\
&\text{(R}'\text{)}: && a \bowtie c \bowtie b,\ xc = cx,\ \theta_x(a) = \theta_x(c),\ ya = ay,\ \theta_x(b) = \theta_y(b) \implies a \bowtie b;\\
&\text{(P}_1\text{)}: && a \bowtie c \bowtie b \bowtie d,\ xa = ax,\ xc = cx,\ \theta_a(x) = \theta_b(x),\ \theta_x(c) = \theta_x(d) \implies a \bowtie d;\\
&\text{(Q)}: && a \bowtie c \bowtie b \bowtie d,\ xa = ax,\ \theta_c(x) = \theta_d(x),\ ya = ay,\ yb = by,\ \theta_y(c) = \theta_y(b) \implies a \bowtie d;\\
&\text{(K4)}: && a \bowtie c \bowtie b,\ aya = a,\ b \leq z,\ yc = cy,\ \theta_z(y) = \theta_b(a),\ \theta_z(a) = \theta_a(b) \implies a \bowtie b;\\
&\text{(K0)}: && a \bowtie c \bowtie b,\ yc = cy,\ \theta_y(b) = \theta_b(a),\ \theta_b(z) = b,\ \theta_z(a) = \theta_z(b),\ \theta_y(z) = \theta_a(b) \implies a \bowtie b.
\end{aligned}
$$

(S) and (V) look different: in (S) the path has length three and the auxiliary is the product cd of two commuting vertices; (V) has an order conclusion and a single half-matching hypothesis. But both are of the same kind. (S) is a consequence of (Q) (Corollary 5.13), and (V) says that, when b is half-matched with c and the restrictions $\theta_a(t) \leq b$ and $\theta_c(t) \leq a$ are wired crosswise, the restrictions $\theta_a(t)$ and $\theta_b(t)$ of t coincide; its proof (Theorem 5.6) reduces it by equational reasoning to the statement $\theta_c(\theta_b(t)) = \theta_c(t)$, that c sees t and $\theta_b(t)$ alike, which is of the same kind as the hypotheses above. The two conditions (E_1) and (E_6) of the census (Section 5.9) are consequences of (K0) and (K4), respectively, together with (Q1) and the chain condition (W_2) (Remark 5.23).

Figure 2 draws the eight conditions in a common vocabulary. Table 1 lists, for each condition, the projection algebra that violates it and satisfies all the others; these are the nine minimal non-embeddable algebras of Theorem 1.6; they are defined in Sections 5.9 and 7.1, and Proposition 5.21 records which conditions each of them satisfies.

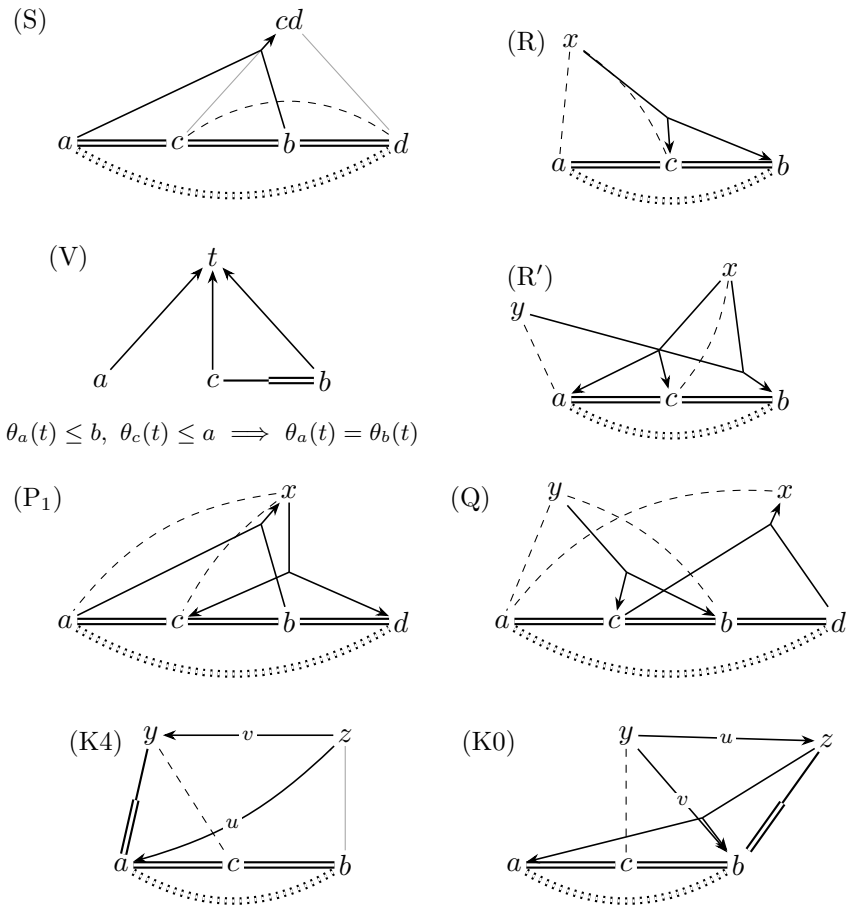


FIGURE 2. The eight wiring conditions. Double lines are matched pairs and the dotted double line is the conclusion; dashed lines are commuting pairs; thin lines are order, the lower element drawn lower; a line that is double at one end only, at p, is a half-matching $\theta_p(x) = p$. An arrow from p to x stands for $\theta_p(x)$: a forked arrow from x to p and q means that x sees p and q alike, $\theta_x(p) = \theta_x(q)$; two arrows merging into x mean that their sources see x alike; a label u or v on an arrow gives its value, $u = \theta_a(b)$, $v = \theta_b(a)$. In (V) the arrows are the three restrictions of t, and the hypotheses on them are written out.

condition	variables	witness	order
(S)	4	$\mathfrak{P}$	6
(R)	$3+1$	$\mathfrak{P}_7$	7
(V)	4	$\mathfrak{P}_8^7$	8
(R′)	$3+2$	$\mathfrak{P}_8^3$	8
(P_1)	$4+1$	$\mathfrak{P}_8^2$	8
(Q)	$4+2$	$\mathfrak{P}_8^5$ (and $\mathfrak{P}_8^4$ via (P_2))	8
(K4)	$3+2$	$\mathfrak{P}_8^6$	8
(K0)	$3+2$	$\mathfrak{P}_8^1$	8

TABLE 1. The wiring conditions and their witnesses.

5.2. **The closing lemma.** Several proofs end in the same way. For a path $a \bowtie c \bowtie b$ in $\mathsf{P}(\mathcal{P}_X)$ consider

(G1) $\varepsilon_c \le \varepsilon_a \vee \varepsilon_b$, (G2) no class of $\varepsilon_a \vee \varepsilon_b \vee \varepsilon_c$ contains two $\mathsf{T}(c)$-blocks.

Lemma 5.1 (closing lemma)**.** *If $a \bowtie c$, $c \bowtie b$,* (G1) *and* (G2) *hold in $\mathcal{P}_X$, then $a \bowtie b$: every class of $\varepsilon_a \vee \varepsilon_b$ meeting* $\operatorname{supp} a \cup \operatorname{supp} b \cup \operatorname{supp} c$ *contains exactly the T-blocks A_i, B_i, C_i for one i.*

Proof. By (G1), $\varepsilon_a \vee \varepsilon_b = \varepsilon_a \vee \varepsilon_b \vee \varepsilon_c \ge \varepsilon_a \vee \varepsilon_c$, $\varepsilon_b \vee \varepsilon_c$. Let D be the class of $\varepsilon_a \vee \varepsilon_b$ containing A_i. Then $\Gamma_i \subseteq D$, so $C_i \subseteq D$, and $\Delta_i \subseteq D$, so $B_i \subseteq D$. If $A_j \subseteq D$ or $B_j \subseteq D$ for some $j \ne i$, then $C_j \subseteq D$ as well, contradicting (G2). The same applies to the class of any B_i or C_i, so the classes meeting $\operatorname{supp} a \cup \operatorname{supp} b$ pair the A's with the B's bijectively, and $a \bowtie b$ by Lemma 2.6(c). □

A second recurring step concerns an auxiliary commuting with both ends of a matched pair: it splits X into a part where the two projections coincide and a part where they are compared through their products with the auxiliary.

Lemma 5.2 (commuting auxiliary)**.** *Let $a \bowtie c$ in $\mathcal{P}_X$ and let x be a projection with $xa = ax =: e$ and $xc = cx =: f$. Put $Y = X \setminus \operatorname{supp} x$. Then every block of a and of c lies inside* $\operatorname{supp} x$ *or is an $\mathsf{N}(x)$-block, $\varepsilon_e = \varepsilon_x \vee \varepsilon_a$ and $\varepsilon_f = \varepsilon_x \vee \varepsilon_c$; and with $U_1, \dots, U_t$ the $\mathsf{T}(e)$-blocks and $V_1, \dots, V_{t'}$ the $\mathsf{T}(f)$-blocks:*

(a) *$A_i \subseteq \operatorname{supp} e$ iff $C_i \subseteq \operatorname{supp} f$, and for the remaining i we have $A_i = C_i \subseteq Y$;*
(b) *an $\mathsf{N}(x)$-block is an $\mathsf{N}(a)$-block iff it is an $\mathsf{N}(c)$-block.*

Thus a and c have the same blocks inside Y, with the same labels.

Proof. By Lemma 2.6(b) the classes of $\varepsilon_x \vee \varepsilon_a$ are of four types: the $\mathsf{T}(e)$-blocks U_l (each a union of A's and a union of $\mathsf{T}(x)$-blocks), the $\mathsf{N}(a)$-blocks inside $\operatorname{supp} x$ (unions of $\mathsf{T}(x)$-blocks), the $\mathsf{N}(x)$-blocks inside $\operatorname{supp} a$ (unions of A's), and the common N-blocks; and $\varepsilon_e = \varepsilon_x \vee \varepsilon_a$. Likewise for c. In particular every block of a and of c lies inside $\operatorname{supp} x$ or inside Y; every A_i lies in some U_l or in an $\mathsf{N}(x)$-block, and every C_i in some V_k or in an $\mathsf{N}(x)$-block. An $\mathsf{N}(x)$-block M containing a $\mathsf{T}(c)$-block C_i is a union of C's; it is not an $\mathsf{N}(a)$-block (an ε_a-block which is a union of ε_c-blocks would be a class of $\varepsilon_a \vee \varepsilon_c$ containing C_i and no A, contradicting $a \bowtie c$), so it is a union of A's as well, and the class of C_i in $\varepsilon_a \vee \varepsilon_c$, being made of the A's and C's inside M, is $A_i = C_i$. Hence if $C_i \not\subseteq \operatorname{supp} f$ then $A_i = C_i \subseteq Y$, so $A_i \not\subseteq \operatorname{supp} e$; by symmetry this gives (a). For (b), an $\mathsf{N}(x)$-block which is an $\mathsf{N}(a)$-block but not an $\mathsf{N}(c)$-block is a union of C's, hence contains some C_i and is then a union of A's, which is absurd. □

5.3. **The condition (R′) and its corollary (R).**

Theorem 5.3. (R′) *holds in every partition monoid: if a, b, c, x, y are projections of $\mathcal{P}_X$ with*

$$a \bowtie c, \quad c \bowtie b, \quad xc = cx, \quad \theta_x(a) = \theta_x(c), \quad ya = ay, \quad \theta_x(b) = \theta_y(b),$$

then $a \bowtie b$.

The proof has two parts. An algebraic part, valid in every regular $*$-semigroup whose projection algebra satisfies (Q1), derives a number of identities from the hypotheses and reduces the theorem to the case $x \le y$; the combinatorial part is a chain argument in $\mathcal{P}_X$ which shows that no class of $\varepsilon_a \vee \varepsilon_b \vee \varepsilon_c$ contains two $\mathsf{T}(c)$-blocks, followed by a count. Most of the algebraic part was found with Prover9, and the combinatorial statements were found by testing candidates against all instances of the hypotheses in $\mathsf{P}(\mathcal{P}_4)$.

Proof. Throughout, $r = \theta_x(b) = \theta_y(b)$, $f = xc = cx$ and $e = ya = ay$.

Step 1: algebraic consequences. The identities of this step hold in every regular $*$-semigroup satisfying (Q1), under the hypotheses of the theorem.

(i) $r \le x$ and $r \le y$, so $xy = yx =: m$ by (Q1), and $r \le m$.

(ii) $\theta_x(c) = xcx = xc = f$, so the hypothesis $\theta_x(a) = \theta_x(c)$ reads $xax = f$. Hence $xa = xa\,ax\,xa = (xax)a = fa$ and, applying $*$, $ax = af$.

(iii) $f \le y$ *and* $cy = yc = f$. First, $\theta_a(r) = a(yby)a = (ay)b(ya) = ebe$. Next, for every element s, $\theta_x(\theta_e(s)) = x\,ya\,s\,ay\,x = m\,asa\,m = \theta_m(\theta_a(s))$, since $xya = ma$ and $ayx = am$. Applied to $s = b$, with $\theta_a(r) = ebe$: $\theta_m(aba) = \theta_x(ebe) = \theta_x(\theta_a(r)) = xa\,xbx\,ax = (xax)b(xax) = fbf$, and $fbf = f$ because $f \le c$ and $cbc = c$ (for $q \le c$, $qbq = qc\,b\,cq = q(cbc)q = q$). So $f = \theta_m(aba) \le m \le y$. Now f is a common lower bound of c and y, so $cy = yc$ by (Q1); the projection $q = cy$ lies below c, so $qbq = q$ by the identity just used, and $qbq = c(yby)c = crc = c(xbx)c = (cx)b(xc) = fbf = f$. Thus $cy = yc = f$.

(iv) $\theta_a(x) = \theta_a(y) = e$ *and* $e \bowtie f$. By (ii) and (iii), $axa = afa = a\,cy\,a = ac\,ya = aca\,y = ay = e$. Using $ay = ya$, $cy = yc$, $aca = a$ and $cac = c$: $efe = ya\,yc\,ya = y\,ac\,ya = y\,aca\,y = e$ and $fef = yc\,ya\,yc = y\,cac\,y = f$.

(v) $yax = ax$, $xya = xa$, $axb = yab$ *and* $xab = xcb$. We have $cbf = cb\,cx = (cbc)x = cx = f$ and, by (ii), $ax = af = acx$; hence $axbf = afbf = a(fbf) = af = ax$. Since $r \le y$, $y\,xbx = yr = r = xbx$; therefore $yax = y\,axbf = y\,a(xbx)c = a\,y(xbx)\,c = a(xbx)c = axbf = ax$. Applying $*$ gives $xay = xa$, i.e. $xya = xa$. Next $xb = (xbx)b = rb = (yby)b = yb$, so $axb = ayb = yab$. Finally $xcb = (xax)b = x(axb) = x(yab) = (xya)b = xab$.

(vi) $y\,ab = y\,acb$. Multiplying $xab = xcb$ on the left by a and using $axa = e$ and $axc = afc = af$ (as $f \le c$): $e\,b = af\,b$; and $y\,acb = (ya)(cb) = a(yc)b = afb$. So $eb = y\,ab = y\,acb = ecb$.

(vii) *Reduction to* $x \le y$. The projection $m = xy$ satisfies the hypotheses in place of x: $mc = xyc = xf = f = fy = cxy = cm$, as $f \le x$ and $f \le y$; $\theta_m(c) = mcm = f$, and $\theta_m(a) = xy\,a\,yx = xex = x(afa)x = (xa)f(ax) = (fa)f(af) = f(afa)f = fef = f$ by (ii) and (iv); and $\theta_m(b) = \theta_m(\theta_x(b)) = \theta_m(r) = r = \theta_y(b)$, since $m \le x$ and $r \le m$. As the conclusion does not involve x, we assume from now on that $x \le y$.

Step 2: the setting in $\mathcal{P}_X$. We use the notation of Section 2.3: A_i, C_i, B_i, Γ_i, Δ_i; for a block E of c we write $\Gamma(E)$ and $\Delta(E)$ for its classes in $\varepsilon_a \vee \varepsilon_c$ and $\varepsilon_b \vee \varepsilon_c$. Put $Y = X \setminus \operatorname{supp} y$. We shall show that

$$\text{no class of } \varepsilon_a \vee \varepsilon_b \vee \varepsilon_c \text{ contains two } \mathsf{T}(c)\text{-blocks}, \tag{6}$$

and then conclude by a count. Since A_i, B_i, C_i lie in one class of $\varepsilon_a \vee \varepsilon_b \vee \varepsilon_c$, (6) implies that every class of $\varepsilon_a \vee \varepsilon_b$ contains at most one $\mathsf{T}(a)$-block and at most one $\mathsf{T}(b)$-block, and that a class containing A_i contains no $\mathsf{T}(b)$-block other than B_i.

By Lemma 2.6(b) applied to the commuting pairs y, a and y, c and x, c: every block of a, of c and of x lies inside $\operatorname{supp} y$ or is an $\mathsf{N}(y)$-block; an $\mathsf{N}(y)$-block is an $\mathsf{N}(a)$-block or a union of $\mathsf{T}(a)$-blocks, and an $\mathsf{N}(c)$-block or a union of $\mathsf{T}(c)$-blocks; every $\mathsf{N}(c)$-block lies inside $\operatorname{supp} x$ (and is then a union of $\mathsf{T}(x)$-blocks) or is an $\mathsf{N}(x)$-block. Since $x \le y$, every block of x is an $\mathsf{N}(y)$-block or a union of $\mathsf{T}(y)$-blocks (Lemma 2.7(a)). The classes of $\varepsilon_y \vee \varepsilon_a$ meeting $\operatorname{supp} y$ are the $\mathsf{T}(e)$-blocks $U_1, \dots, U_t$ (unions of $\mathsf{T}(a)$-blocks and of $\mathsf{T}(y)$-blocks) and the $\mathsf{N}(a)$-blocks inside $\operatorname{supp} y$; the classes of $\varepsilon_y \vee \varepsilon_c$ meeting $\operatorname{supp} y$ are the $\mathsf{T}(f)$-blocks $V_1, \dots, V_{t'}$ and the $\mathsf{N}(c)$-blocks inside $\operatorname{supp} y$.

Step 3: a second reduction. Let M be an $\mathsf{N}(y)$-block which is a union of $\mathsf{T}(a)$-blocks. If M were an $\mathsf{N}(c)$-block, it would be a class of $\varepsilon_a \vee \varepsilon_c$ by itself, containing $\mathsf{T}(a)$-blocks but no $\mathsf{T}(c)$-block, contradicting $a \bowtie c$; so M is a union of $\mathsf{T}(c)$-blocks, the class of any $A_i \subseteq M$ lies inside M and consists of T-blocks only, and $\Gamma_i = A_i = C_i$. Symmetrically, an $\mathsf{N}(y)$-block which is a union of $\mathsf{T}(c)$-blocks is a union of $\mathsf{T}(a)$-blocks. Let x', y' be obtained from x, y by declaring all

these blocks M transversal. We claim that a, b, c, x', y' still satisfy the hypotheses, with $x' \leq y'$; as the conclusion does not change, we may then assume that *every* $\mathsf{N}(y)$*-block is an* $\mathsf{N}(a)$*-block and an* $\mathsf{N}(c)$*-block*, so that all A_i and all C_i lie inside $\operatorname{supp} y$. Indeed, the partitions $\varepsilon_{x'} = \varepsilon_x$, $\varepsilon_{y'} = \varepsilon_y$ are unchanged, and each M is a class by itself in $\varepsilon_x \vee \varepsilon_a$, $\varepsilon_x \vee \varepsilon_c$, $\varepsilon_y \vee \varepsilon_a$ and $\varepsilon_y \vee \varepsilon_c$, being a block of x and of y which is a union of blocks of a and of c. Hence $x'c = cx'$, $y'a = ay'$ and $y'c = cy'$ by the converse of Lemma 2.6(b) (if every class of $\varepsilon_p \vee \varepsilon_q$ is of one of the four types, the description of pq in the proof of Lemma 2.6 shows that pq is symmetric, so $pq = qp$): the class M is now of type (i), the others are unchanged. Further $\theta_{x'}(a) = \theta_{x'}(c)$, because the blocks of these projections are those of $\theta_x(a) = \theta_x(c)$ except that each M becomes a transversal block of both; and $\theta_{x'}(b) = \theta_{y'}(b)$ will follow from Step 4(c) below, whose proof uses only $x \leq y$ and $\theta_x(b) = \theta_y(b)$.

Step 4: the classes of $\varepsilon_y \vee \varepsilon_b$. Let $\sigma = \varepsilon_y \vee \varepsilon_b$.

(a) $\varepsilon_x \vee \varepsilon_b = \sigma$. Let W be a block of x which is a union of $\mathsf{T}(y)$-blocks. If W is a $\mathsf{T}(x)$-block, let $\tilde{K}$ be its class in $\varepsilon_x \vee \varepsilon_b$; then $\tilde{K} \cap \operatorname{supp} x$ is a block of $\theta_x(b) = \theta_y(b)$ contained in $\operatorname{supp} y$, hence equal to $K \cap \operatorname{supp} y$ for a class K of σ, and $W \subseteq K$. If W is an $\mathsf{N}(x)$-block, it is itself a block of $\theta_x(b) = \theta_y(b)$ contained in $\operatorname{supp} y$, so again $W = K \cap \operatorname{supp} y$ for a class K of σ. The remaining blocks of x are $\mathsf{N}(y)$-blocks. So every block of x lies in a single class of σ, i.e. $\varepsilon_x \leq \sigma$, and $\varepsilon_x \vee \varepsilon_b = \sigma$.

(b) *Two kinds of classes.* By (a) and the two descriptions of r, for every class K of σ meeting $\operatorname{supp} y$ either $K \cap \operatorname{supp} y = \tilde{N}$ is an $\mathsf{N}(x)$-block (a union of $\mathsf{T}(y)$-blocks), and then $K \cap \operatorname{supp} b = \emptyset$ since $\tilde{N}$ is a non-transversal block of $\theta_x(b)$; or $K \cap \operatorname{supp} y = K \cap \operatorname{supp} x$. We call the $\mathsf{N}(x)$-blocks inside $\operatorname{supp} y$ the $\tilde{N}$*-blocks*; each $\tilde{N}$-block is $K \cap \operatorname{supp} y$ for a class K of σ with $K \cap \operatorname{supp} b = \emptyset$, and every other class meeting $\operatorname{supp} y$ has $K \cap \operatorname{supp} y \subseteq \operatorname{supp} x$.

(c) In the situation of Step 3, the blocks of $\theta_{x'}(b)$ and $\theta_{y'}(b)$ are indexed by the same classes K of σ (by (a)), and they can differ only for a class $K \supseteq M$, where they are $(K \cap \operatorname{supp} x) \cup M$ and $(K \cap \operatorname{supp} y) \cup M$, both transversal iff K meets $\operatorname{supp} b$. If these differ, then by (b) $K \cap \operatorname{supp} y$ is an $\tilde{N}$-block and $K \cap \operatorname{supp} b = \emptyset$; but then, for $A_i = C_i \subseteq M$, every block of b meeting K lies in K, and every block of c meeting a block Z of b inside K lies in K (a block of c in Y is an $\mathsf{N}(y)$-block, hence lies in a single class of σ; a block of c inside $\operatorname{supp} y$ meets $Z \cap \operatorname{supp} y \subseteq \tilde{N}$, so it is $\tilde{N}$ or contained in it), so a chain from $C_i \subseteq K$ to B_i in Δ_i stays inside K, contradicting $K \cap \operatorname{supp} b = \emptyset$. This completes Step 3.

(d) *Every* $\tilde{N}$*-block is an* $\mathsf{N}(c)$*-block, and its class* $\Delta(\tilde{N})$ *contains no* $\mathsf{T}(b)$*-block and satisfies* $\Delta(\tilde{N}) \cap \operatorname{supp} y = \tilde{N}$. Let K be the class of σ with $K \cap \operatorname{supp} y = \tilde{N}$. As in (c), every block of b or of c reachable by a chain in $\varepsilon_b \vee \varepsilon_c$ from a block of b inside K lies inside K. If $\tilde{N}$ were a union of $\mathsf{T}(c)$-blocks (the alternative, as x and c commute), a block of b meeting some $C_j \subseteq \tilde{N}$ would lie in K, and then $B_j \subseteq \Delta_j \subseteq K$, contradicting $K \cap \operatorname{supp} b = \emptyset$. So $\tilde{N}$ is an $\mathsf{N}(c)$-block, $\Delta(\tilde{N}) \subseteq K$, and the claims follow.

Step 5: the classes of $\varepsilon_x \vee \varepsilon_a$. By Lemma 2.6(a), the blocks of $\theta_x(a)$ inside $\operatorname{supp} x$ are the sets $K \cap \operatorname{supp} x$ for the classes K of $\varepsilon_x \vee \varepsilon_a$ meeting $\operatorname{supp} x$, transversal iff K meets $\operatorname{supp} a$; those of $\theta_x(c)$ are the $\mathsf{T}(f)$-blocks V_k (transversal) and the $\mathsf{N}(c)$-blocks $\tilde{N}'$ inside $\operatorname{supp} x$ (non-transversal; they are the classes of $\varepsilon_x \vee \varepsilon_c$ of type (ii)). Since $\theta_x(a) = \theta_x(c)$: for every k there is a class K_k of $\varepsilon_x \vee \varepsilon_a$ with $K_k \cap \operatorname{supp} x = V_k$, and for every $\mathsf{N}(c)$-block $\tilde{N}' \subseteq \operatorname{supp} x$ a class $K_{\tilde{N}'}$ with $K_{\tilde{N}'} \cap \operatorname{supp} x = \tilde{N}'$ and $K_{\tilde{N}'} \cap \operatorname{supp} a = \emptyset$. Consequently *every block of* c *lies inside a single class of* $\varepsilon_x \vee \varepsilon_a$: a $\mathsf{T}(c)$-block lies in some $V_k \subseteq K_k$; an $\mathsf{N}(c)$-block inside $\operatorname{supp} x$ is some $\tilde{N}' \subseteq K_{\tilde{N}'}$; and the other $\mathsf{N}(c)$-blocks are $\mathsf{N}(x)$-blocks. Hence $\varepsilon_a \vee \varepsilon_c \leq \varepsilon_x \vee \varepsilon_a$, and for an $\mathsf{N}(c)$-block $\tilde{N}' \subseteq \operatorname{supp} x$ we get $\Gamma(\tilde{N}') \subseteq K_{\tilde{N}'}$; so $\Gamma(\tilde{N}')$ *contains no* $\mathsf{T}(a)$*-block, hence no* $\mathsf{T}(c)$*-block, and*

$\tilde{N}'$ is the only block of c inside $\operatorname{supp} x$ *that it contains* (a block of c in $\Gamma(\tilde{N}')$ inside $\operatorname{supp} x$ lies in $K_{\tilde{N}'} \cap \operatorname{supp} x = \tilde{N}'$).

Step 6: the four kinds of blocks of c and their classes. By Steps 2–4, every block of c is of exactly one of the following kinds: a $\mathsf{T}(c)$-block C_i (inside $\operatorname{supp} y$, by Step 3); an $\mathsf{N}(c)$-block inside $\operatorname{supp} x$; an $\tilde{N}$-block; or an $\mathsf{N}(y)$-block inside Y, which is an $\mathsf{N}(a)$-block by Step 3 and is therefore a class of $\varepsilon_a \vee \varepsilon_c$ by itself. Collecting Steps 4(d) and 5:

(1) the blocks of c in Γ_i are C_i and $\tilde{N}$-blocks (an $\mathsf{N}(c)$-block inside $\operatorname{supp} x$ lies in a class without $\mathsf{T}(a)$-blocks, and an $\mathsf{N}(y)$-block in Y is a class by itself);
(2) the blocks of c in Δ_i are C_i, $\mathsf{N}(c)$-blocks inside $\operatorname{supp} x$ and $\mathsf{N}(y)$-blocks in Y (an $\tilde{N}$-block lies in a class without $\mathsf{T}(b)$-blocks);
(3) for an $\mathsf{N}(c)$-block $\tilde{N}' \subseteq \operatorname{supp} x$, the blocks of c in $\Gamma(\tilde{N}')$ are $\tilde{N}'$ and $\tilde{N}$-blocks, and $\Gamma(\tilde{N}')$ contains no $\mathsf{T}(a)$-block;
(4) for an $\tilde{N}$-block $\tilde{N}$, the blocks of c in $\Delta(\tilde{N})$ are $\tilde{N}$ and $\mathsf{N}(y)$-blocks in Y (as $\Delta(\tilde{N}) \cap \operatorname{supp} y = \tilde{N}$), and $\Delta(\tilde{N})$ contains no $\mathsf{T}(b)$-block;
(5) an $\mathsf{N}(y)$-block $E \subseteq Y$ is a class of $\varepsilon_a \vee \varepsilon_c$ by itself.

Step 7: proof of (6). Two blocks E, E' of c lie in one class of $\varepsilon_a \vee \varepsilon_b \vee \varepsilon_c$ iff they are joined by a sequence $E = E_0, E_1, \dots, E_n = E'$ of blocks of c in which consecutive blocks lie in a common class of $\varepsilon_a \vee \varepsilon_c$ or in a common class of $\varepsilon_b \vee \varepsilon_c$: if two blocks W, W' (of a, b or c) meet at a point p, and $E(p)$ is the block of c containing p, then W and $E(p)$ lie in a common class of $\varepsilon_a \vee \varepsilon_c$ or of $\varepsilon_b \vee \varepsilon_c$ (according to whether W is a block of a or of b; if W is a block of c then $W = E(p)$), so a chain of blocks meeting consecutively can be replaced by the sequence of the blocks of c at its meeting points. Now start such a sequence at $E_0 = C_p$ and use (1)–(5) to see which blocks of c can occur. From C_p one can move, within Γ_p, only to $\tilde{N}$-blocks of Γ_p, and within Δ_p only to $\mathsf{N}(c)$-blocks $\tilde{N}' \subseteq \operatorname{supp} x$ of Δ_p and to $\mathsf{N}(y)$-blocks $E \subseteq Y$ of Δ_p. From an $\tilde{N}$-block $\tilde{N}$ of Γ_p one can move within $\Gamma(\tilde{N}) = \Gamma_p$ (back) or within $\Delta(\tilde{N})$ to $\mathsf{N}(y)$-blocks $E \subseteq Y$; from such an E one can move within $\Delta(E) = \Delta(\tilde{N})$ only, and $\Gamma(E) = \{E\}$. From $\tilde{N}' \subseteq \operatorname{supp} x$ in Δ_p one can move within $\Delta(\tilde{N}') = \Delta_p$ (back) or within $\Gamma(\tilde{N}')$ to $\tilde{N}$-blocks of $\Gamma(\tilde{N}')$, from which one can move within $\Gamma(\tilde{N}')$ (back) or within their Δ-classes to $\mathsf{N}(y)$-blocks in Y, which are dead ends as before. From an $\mathsf{N}(y)$-block $E \subseteq Y$ in Δ_p one can move within Δ_p only. So the blocks of c reachable from C_p are: C_p; $\tilde{N}$-blocks (of Γ_p, or of the classes $\Gamma(\tilde{N}')$ for $\tilde{N}' \subseteq \operatorname{supp} x$ in Δ_p); $\mathsf{N}(c)$-blocks inside $\operatorname{supp} x$ lying in Δ_p; and $\mathsf{N}(y)$-blocks in Y. None of these is a $\mathsf{T}(c)$-block other than C_p, which proves (6).

Step 8: counting. Let $l \in \{1, \dots, t\}$ and $I_l = \{p : A_p \subseteq U_l\}$; by Step 3 every A_p lies in some U_l. Consider the class of U_l in $\varepsilon_e \vee \varepsilon_f$ (where $\varepsilon_e = \varepsilon_y \vee \varepsilon_a$, $\varepsilon_f = \varepsilon_y \vee \varepsilon_c$); by $e \bowtie f$ (Step 1(iv)) it contains exactly one $\mathsf{T}(f)$-block, $V_{k(l)}$ say. Every Γ_p is contained in a class of $\varepsilon_e \vee \varepsilon_f$ (its blocks of a and of c are contained in blocks of e and of f), so $C_p \subseteq V_{k(l)}$ for $p \in I_l$, and conversely $C_q \subseteq V_{k(l)}$ implies $A_q \subseteq U_l$; thus $I_l = \{q : C_q \subseteq V_{k(l)}\}$. Now compare the two sides of $y\,ab = y\,acb$ (Step 1(vi)). The upper blocks of ab are the $\mathsf{N}(a)$-blocks and the sets $D \cap \operatorname{supp} a$ for the classes D of $\varepsilon_a \vee \varepsilon_b$ meeting $\operatorname{supp} a$, the latter being transversal iff D meets $\operatorname{supp} b$, with lower part $D \cap \operatorname{supp} b$. By (6), $D \cap \operatorname{supp} a$ is a single $\mathsf{T}(a)$-block; write $D(A_p)$ for the class of A_p. In the product $y \cdot (ab)$ the middle join is $\varepsilon_y \vee \varepsilon_{ab}$, where ε_{ab} is the upper partition of ab; the class of U_l in it is U_l itself, because U_l is a union of $\mathsf{T}(y)$-blocks and of the blocks A_p ($p \in I_l$) of ε_{ab}, and no $\mathsf{N}(a)$-block or $\mathsf{N}(y)$-block meets it. So $y\,ab$ has the upper block U_l, with lower part $\bigcup_{p \in I_l} (D(A_p) \cap \operatorname{supp} b)$ (transversal iff this is nonempty). In $y\,acb = (ya)(cb) = e\,(cb)$ the middle join is $\varepsilon_e \vee \varepsilon_c$, whose class containing U_l contains no other $\mathsf{T}(e)$-block and, being contained in the class of U_l in $\varepsilon_e \vee \varepsilon_f$, contains exactly the $\mathsf{T}(c)$-blocks C_p with $p \in I_l$ (all of which it does contain, as Γ_p is a chain in $\varepsilon_e \vee \varepsilon_c$); so $e\,cb$ has the upper block U_l with lower part $\bigcup_{p \in I_l} B_p$.

Hence

$$\bigcup_{p\in I_l}\big(D(A_p)\cap \operatorname{supp} b\big)=\bigcup_{p\in I_l} B_p.$$

By (6), $D(A_p)\cap\operatorname{supp} b\subseteq B_p$ for each p, and the B_p are pairwise disjoint and nonempty; so $D(A_p)\cap\operatorname{supp} b=B_p$, i.e. $B_p\subseteq D(A_p)$, for every $p\in I_l$ and every l, that is, for every p.

Conclusion. Every class of $\varepsilon_a\vee\varepsilon_b$ meeting $\operatorname{supp} a$ is some $D(A_p)$, which contains exactly one $\mathsf{T}(a)$-block and exactly one $\mathsf{T}(b)$-block (B_p) by (6) and Step 8; and every class meeting $\operatorname{supp} b$ contains some B_p, hence is $D(A_p)$. By Lemma 2.6(c), $a\bowtie b$. □

Remark 5.4. The conclusion obtained is stronger than $a\bowtie b$: by (6) and Step 8 the classes of $\varepsilon_a\vee\varepsilon_b\vee\varepsilon_c$ meeting $\operatorname{supp} a\cup\operatorname{supp} b\cup\operatorname{supp} c$ are exactly the classes containing the triples A_i,B_i,C_i, so that the three matchings are coherent and $ab=acb$. The count of Step 8 alone cannot replace (6): it is blind to the merging of two $\mathsf{T}(a)$-blocks of the same U_l into one class of $\varepsilon_a\vee\varepsilon_b$, which is exactly what happens in the witness $\mathfrak{P}_8^3$.

Corollary 5.5. (R) *holds in every projection algebra satisfying* (R′)*; in particular in every* $\mathsf{P}(\mathcal{P}_X)$.

Proof. Given the hypotheses of (R), put $x'=xc=cx$ and $y'=x$. Then $x'\le c$, so $x'c=cx'$; $\theta_{x'}(a)=cx\,a\,xc=c(xax)c=c(xa)c=x(cac)=xc=\theta_{x'}(c)$, using $xa=ax$, $xc=cx$ and $cac=c$; $y'=x$ commutes with a; and $\theta_{x'}(b)=c\,\theta_x(b)\,c=c\,\theta_x(c)\,c=cxc=xc=\theta_x(c)=\theta_x(b)=\theta_{y'}(b)$. So (R′) gives $a\bowtie b$. (The substitution was found by Prover9.) □

5.4. **The condition (V).**

Theorem 5.6. (V) *holds in every partition monoid: if a,b,c,t are projections of $\mathcal{P}_X$ with $bcb=b$, $ata\le b$ and $ctc\le a$, then $ata=btb$.*

Proof. Write $z=ata$, $w=ctc$, $s=btb$. We first reduce the claim, by computations valid in every projection algebra satisfying (Q1), to a single statement about $\mathcal{P}_X$.

Step 1: three identities. (i) *If $k\le a$ and $ab=ba$ then $bkb\le a$:* $a(bkb)a=(ab)k(ba)=(ba)k(ab)=b(aka)b=bkb$. (ii) *If $ata\le b$ then $ab=ba$ and $ata=ete$, where $e=ab$:* $z=ata$ lies below a and below b, so a and b commute by (Q1); $e=ab$ is a projection with $e\le a$, $e\le b$, and $eze=a(bzb)a=aza=z$, so $z\le e$ and $z=eze=e(ata)e=(ea)t(ae)=ete$. (iii) *If $bcb=b$ then $s=b(csc)b$:* $s=btb=(bcb)t(bcb)=b\,c(btb)c\,b$.

Step 2: reduction. Suppose we know that

$$csc=w,\qquad \text{i.e.}\qquad \theta_c(\theta_b(t))=\theta_c(t). \tag{7}$$

Then $s=b(csc)b=bwb$ by (iii), and $bwb\le a$ by (i), since $w\le a$ and $ab=ba$ (by (ii), as $z\le b$). So $s\le a$ and $s\le b$, hence $s\le e=ab$ and $s=ese=e(btb)e=(eb)t(be)=ete$; and $z=ete$ by (ii). Thus $s=z$. It remains to prove (7) in $\mathcal{P}_X$; in fact we show that $\theta_c(s)$ and $\theta_c(t)$ have the same blocks.

Step 3: the setting in $\mathcal{P}_X$. Put $J_a=\varepsilon_a\vee\varepsilon_t$, $J_b=\varepsilon_b\vee\varepsilon_t$, $J_c=\varepsilon_c\vee\varepsilon_t$. By Lemma 2.6(a) the blocks of z are the $\mathsf{N}(a)$-blocks and the sets $Z_K=K\cap\operatorname{supp} a$ for the classes K of J_a meeting $\operatorname{supp} a$, Z_K being transversal iff K meets $\operatorname{supp} t$; similarly the blocks of w are the $\mathsf{N}(c)$-blocks and the sets $W_L=L\cap\operatorname{supp} c$ ($L\in J_c$), and those of s are the $\mathsf{N}(b)$-blocks and the sets $S_H=H\cap\operatorname{supp} b$ ($H\in J_b$). From $w\le a$ and Lemma 2.7(a): every block of a lies in a block of w; every block of w is an $\mathsf{N}(a)$-block or a union of $\mathsf{T}(a)$-blocks; and every $\mathsf{N}(a)$-block is an $\mathsf{N}(w)$-block, i.e. an $\mathsf{N}(c)$-block or a set W_L with $L\cap\operatorname{supp} t=\emptyset$. From $z\le b$: every block of b lies in a block of z; every block of z is an $\mathsf{N}(b)$-block or a union of $\mathsf{T}(b)$-blocks; and every $\mathsf{N}(b)$-block is a block of z, non-transversal. From $bcb=b$ and Lemma 2.7(b): every block of c meets at most one $\mathsf{T}(b)$-block, and for every $\mathsf{T}(b)$-block B the class $\Delta(B)$ of $\varepsilon_b\vee\varepsilon_c$ containing B meets $\operatorname{supp} c$.

Since blocks of a lie in blocks of w, which lie in classes of J_c, we have $J_a \leq J_c$ (every class of J_a lies in a class of J_c); likewise $J_b \leq J_a$. For a class K of J_a let $\lambda(K)$ be the class of J_c containing it. Call a block of c *inner* if it meets $\operatorname{supp} a$ and *outer* otherwise. An inner $\mathsf{T}(c)$-block lies in a set W_L which is a union of $\mathsf{T}(a)$-blocks; an inner $\mathsf{N}(c)$-block is a union of $\mathsf{T}(a)$-blocks (being a block of w that is not an $\mathsf{N}(a)$-block); an outer block of c inside a class L of J_c with $W_L \neq \emptyset$ is either a $\mathsf{T}(c)$-block inside W_L, W_L being then an $\mathsf{N}(a)$-block, or an $\mathsf{N}(c)$-block which is an $\mathsf{N}(a)$-block.

Step 4: two facts. (1) *No $\mathsf{T}(b)$-block lies in a block M which is both an $\mathsf{N}(a)$-block and an $\mathsf{N}(c)$-block.* Otherwise let $B \subseteq M$ be one. The only block of c meeting B is M, and the blocks of b meeting M lie inside M (M is a block of z, and blocks of b lie in blocks of z); so $\Delta(B) = M$ does not meet $\operatorname{supp} c$, contradicting Lemma 2.7(b). Consequently every block which is both an $\mathsf{N}(a)$-block and an $\mathsf{N}(c)$-block is an $\mathsf{N}(b)$-block (a block of z containing no $\mathsf{T}(b)$-block).

(2) *Let K be a class of J_a and $M \subseteq K$ an $\mathsf{N}(a)$-block containing a $\mathsf{T}(b)$-block. Then M is the only block of z inside K that meets* $\operatorname{supp} b$*; in particular Z_K, if nonempty, is an $\mathsf{N}(b)$-block.* By (1) M is not an $\mathsf{N}(c)$-block, so $M = W_L$ with $L = \lambda(K)$, $L \cap \operatorname{supp} t = \emptyset$; this already shows that no other $\mathsf{N}(a)$-block in K contains a $\mathsf{T}(b)$-block (it would also equal $W_{\lambda(K)}$). Suppose Z_K contains a $\mathsf{T}(b)$-block B. Every $\mathsf{T}(a)$-block in K lies in L but not in $W_L = M$ (which is disjoint from $\operatorname{supp} a$), hence in an $\mathsf{N}(c)$-block, which is then inner, a union of $\mathsf{T}(a)$-blocks, and contained in L. Take a chain $B = B_0, C_1, B_1, C_2, \dots$ from B to a $\mathsf{T}(c)$-block in $\Delta(B)$ (it exists by Lemma 2.7(b)). C_1 meets $B \subseteq Z_K \subseteq \operatorname{supp} a$ in a point of a $\mathsf{T}(a)$-block of K, so C_1 is an inner $\mathsf{N}(c)$-block, $C_1 \subseteq L \cap \operatorname{supp} a$; then B_1 meets $\operatorname{supp} a$ in a point of a $\mathsf{T}(a)$-block $A_1 \subseteq C_1 \subseteq L$, whose class K_1 satisfies $\lambda(K_1) = L$, and $B_1 \subseteq Z_{K_1}$; so C_2 meets a $\mathsf{T}(a)$-block of K_1, which lies in L but not in W_L, and C_2 is again an inner $\mathsf{N}(c)$-block. Inductively no block of the chain is a $\mathsf{T}(c)$-block, a contradiction.

Step 5: the blocks of $\theta_c(s)$ and $\theta_c(t)$ inside $\operatorname{supp} c$. By Lemma 2.6(a) the blocks of $\theta_c(s)$ are the $\mathsf{N}(c)$-blocks and the sets $\Theta \cap \operatorname{supp} c$ for the classes Θ of $\varepsilon_c \vee \varepsilon_s$ meeting $\operatorname{supp} c$, transversal iff Θ meets $\operatorname{supp} s$; the blocks of $\theta_c(t) = w$ are the $\mathsf{N}(c)$-blocks and the W_L, transversal iff L meets $\operatorname{supp} t$. Every block of s lies in a class of J_c: an $\mathsf{N}(b)$-block is a block of z, hence lies in a class of J_a, and $S_H \subseteq H$, a class of $J_b \leq J_a \leq J_c$. Hence $\varepsilon_c \vee \varepsilon_s \leq J_c$: every class Θ lies in a class L of J_c, $\Theta \cap \operatorname{supp} c \subseteq W_L$, and if Θ meets $\operatorname{supp} s$ then L meets $\operatorname{supp} t$ (a block $S_H \subseteq \Theta$ is transversal only if $H \subseteq L$ meets $\operatorname{supp} t$). So (7) follows once we show, for every class L of J_c with $W_L \neq \emptyset$: *(A) any two $\mathsf{T}(c)$-blocks in W_L lie in a common class of $\varepsilon_c \vee \varepsilon_s$, and (B) if L meets* $\operatorname{supp} t$ *then this class meets* $\operatorname{supp} s$. Fix L; then W_L is a union of $\mathsf{T}(a)$-blocks (*type T*) or an $\mathsf{N}(a)$-block M (*type N*, and then $L \cap \operatorname{supp} t = \emptyset$, so (B) is vacuous).

Type T. Let C, C' be $\mathsf{T}(c)$-blocks in W_L and take a chain $C = C_0, T_1, C_1, \dots, T_r, C_r = C'$ in J_c (blocks C_i of c, blocks T_i of t, all inside L). We claim that consecutive inner blocks of c on the chain, say C_i and C_j with $C_{i+1}, \dots, C_{j-1}$ outer, lie in a common class of $\varepsilon_c \vee \varepsilon_s$; since C_0 and C_r are inner ($W_L \subseteq \operatorname{supp} a$), (A) follows. The outer blocks $C_{i+1}, \dots, C_{j-1}$ are $\mathsf{N}(c)$-blocks which are $\mathsf{N}(a)$-blocks (the $\mathsf{T}(c)$-blocks of L lie in $W_L \subseteq \operatorname{supp} a$), hence $\mathsf{N}(b)$-blocks by (1); the blocks $T_{i+1}, C_{i+1}, \dots, T_j$ lie in a single class K'' of J_a (each lies in a class, and consecutive ones meet). Let $x \in T_{i+1} \cap C_i$ and $y \in T_j \cap C_j$; both lie in $\operatorname{supp} a$ (C_i, C_j are inner: a $\mathsf{T}(c)$-block inside W_L, or an $\mathsf{N}(c)$-block which is a union of $\mathsf{T}(a)$-blocks), so $x, y \in Z_{K''}$. If $Z_{K''}$ is an $\mathsf{N}(b)$-block, it is a block of s meeting C_i and C_j. Otherwise $Z_{K''}$ is a union of $\mathsf{T}(b)$-blocks; let $B_x \ni x$, $B_y \ni y$ be the $\mathsf{T}(b)$-blocks; then $B_x, T_{i+1}, C_{i+1}, T_{i+2}, \dots, T_j, B_y$ is a chain in J_b, so $B_x, B_y \subseteq H$ for one class H of J_b, and the block $S_H \supseteq B_x \cup B_y$ of s meets C_i and C_j. This proves the claim. For (B), let L meet $\operatorname{supp} t$, and take a chain $C = C_0, T_1, C_1, \dots, C_q, T_{q+1}$ in L from C to a $\mathsf{T}(t)$-block T_{q+1}. Let C_j ($j \leq q$) be the last inner block of c on it; then $C_{j+1}, \dots, C_q$ are $\mathsf{N}(b)$-blocks as above, and $T_{j+1}, C_{j+1}, \dots, T_{q+1}$ lie in one class K' of J_a, which meets $\operatorname{supp} t$. The point $x \in T_{j+1} \cap C_j$ lies in $Z_{K'}$, a transversal block of z, hence a union of $\mathsf{T}(b)$-blocks (an $\mathsf{N}(b)$-block is a non-transversal

block of z); the $\mathsf{T}(b)$-block $B_x \ni x$ is joined to T_{q+1} by the chain $B_x, T_{j+1}, C_{j+1}, \dots, C_q, T_{q+1}$ in J_b, so its class H meets $\operatorname{supp} t$, S_H is a transversal block of s, and S_H meets C_j, which lies in the class of C by the claim.

Type N. Here $W_L = M$ is an $\mathsf{N}(a)$-block and a block of z. If M is an $\mathsf{N}(b)$-block, it is a block of s containing every $\mathsf{T}(c)$-block of W_L, and (A) holds. Otherwise M is a union of $\mathsf{T}(b)$-blocks $B_1, \dots, B_k$; each $\mathsf{T}(c)$-block $C \subseteq M$ lies inside one B_i (it is covered by them and meets at most one), so (A) follows if we show that $B_1, \dots, B_k$ lie in a single class H of J_b (then $S_H \supseteq M$). Let K_0 be the class of J_a containing M. By (2), M is the only block of z inside K_0 meeting $\operatorname{supp} b$; so every $\mathsf{N}(a)$-block in K_0 other than M is an $\mathsf{N}(b)$-block, and Z_{K_0} is an $\mathsf{N}(b)$-block if nonempty. Hence every block of b inside K_0 is one of: a $\mathsf{T}(b)$-block B_i; an $\mathsf{N}(a)$-block $M' \neq M$ in K_0; or Z_{K_0}; and for a point $p \in K_0$ we write $\beta(p)$ for the block of b containing it, which is Z_{K_0} whenever $p \in \operatorname{supp} a$. Let $C \subseteq B_1$ and $C' \subseteq B_2$ be $\mathsf{T}(c)$-blocks and take a chain $C = C_0, T_1, C_1, \dots, T_r, C_r = C'$ in L. Every block of t meeting M, Z_{K_0} or an $\mathsf{N}(a)$-block in K_0 lies in K_0; in particular $T_1, T_r \subseteq K_0$. Consider a block C_l of the chain with $T_l \subseteq K_0$, and let $u \in T_l \cap C_l$. If C_l is a $\mathsf{T}(c)$-block, then $C_l \subseteq M$ lies inside one B_i, and $\beta(u) = \beta(v) = B_i$ for the point $v \in T_{l+1} \cap C_l$. If C_l is an outer $\mathsf{N}(c)$-block, it is an $\mathsf{N}(a)$-block inside K_0 and an $\mathsf{N}(b)$-block by (1), and $\beta(u) = \beta(v) = C_l$. If C_l is an inner $\mathsf{N}(c)$-block, then $u \in \operatorname{supp} a \cap K_0$ and $\beta(u) = Z_{K_0}$; if also $T_{l+1} \subseteq K_0$ then $\beta(v) = Z_{K_0}$; if not, let $T_{l'}$ ($l' > l+1$) be the next block of t on the chain contained in K_0 (it exists, as $T_r \subseteq K_0$): the block $C_{l'-1}$ meets $T_{l'-1} \not\subseteq K_0$ and $T_{l'} \subseteq K_0$, so it meets two classes of J_a and is therefore an inner $\mathsf{N}(c)$-block, and the point $v' \in T_{l'} \cap C_{l'-1}$ lies in $\operatorname{supp} a \cap K_0$, so $\beta(v') = Z_{K_0} = \beta(u)$. Since consecutive points of the chain on a common block $T_l \subseteq K_0$ have blocks β joined by T_l in J_b, we obtain, walking along the chain and skipping the excursions outside K_0, a chain in J_b from $\beta(x_0) = B_1$ ($x_0 \in T_1 \cap C_0$) to $\beta(y_r) = B_2$ ($y_r \in T_r \cap C_r$). So B_1, B_2 lie in one class of J_b, as required.

Conclusion. By Step 5 the blocks of $\theta_c(s)$ inside $\operatorname{supp} c$ are the sets W_L, transversal iff L meets $\operatorname{supp} t$, and its other blocks are the $\mathsf{N}(c)$-blocks; so $\theta_c(s) = \theta_c(t) = w$, which is (7), and $ata = btb$ by Step 2. □

Remark 5.7. (1) The hypothesis $bcb = b$ enters the proof of (7) only through the facts (1) and (2) of Step 4. (2) The algebraic part shows more generally: in any projection algebra satisfying (Q1), if $bcb = b$, $ata \leq b$, $ctc \leq a$ and $c(btb)c = ctc$, then $ata = btb$. (3) In every projection algebra, (V) is equivalent to the three-atom clause $\theta_a(t) \leq \theta_d(c),\ \theta_c(t) \leq a \Rightarrow \theta_a(t) \leq \theta_{\theta_d(c)}(t)$ (Prover9): the half-matching hypothesis $bcb = b$ is absorbed by writing $b = \theta_d(c)$, since $\theta_{\theta_d(c)}(c) = \theta_d(c)$ is an identity of projection algebras (Lemma 2.3) and every b with $bcb = b$ is of this form. For $d = c$ the clause is a consequence of (Q1) ($\theta_a(t)$ is then a common lower bound of a and c, so $ac = ca$ and $\theta_a(t) = \theta_{ac}(t) = \theta_c(t)$), and the witness $\mathfrak{P}_8^7$ satisfies this special case while violating (V).

5.5. **The condition ($\mathbf{P}_1$).** We write the path of (P_1) as $a \bowtie c \bowtie b \bowtie d$; the hypotheses are $xa = ax$, $xc = cx$, $\theta_a(x) = \theta_b(x)$ and $\theta_x(c) = \theta_x(d)$, and the conclusion is $a \bowtie d$. For a path of length three we extend the notation of Section 2.3: D_i is the $\mathsf{T}(d)$-block in the class Λ_i of $\varepsilon_b \vee \varepsilon_d$ containing B_i, so that Γ_i contains exactly the T-blocks A_i, C_i; Δ_i exactly C_i, B_i; and Λ_i exactly B_i, D_i.

Theorem 5.8. (P_1) *holds in every partition monoid: if a, b, c, d, x are projections of $\mathcal{P}_X$ with $a \bowtie c$, $c \bowtie b$, $b \bowtie d$, $xa = ax$, $xc = cx$, $\theta_a(x) = \theta_b(x)$ and $\theta_x(c) = \theta_x(d)$, then $a \bowtie d$.*

Proof. Write $e = xa = ax$ and $f = xc = cx$, and $Y = X \setminus \operatorname{supp} x$.

Step 1: x commutes with b, and $xb = e$. We have $\theta_b(x) = \theta_a(x) = axa = e$, so $e \leq b$; as also $e \leq x$, (Q1) gives $xb = bx$, and then $\theta_b(x) = bxb = xb$, i.e. $xb = bx = e$.

Step 2: the blocks of a, b, c. Since x commutes with a, b and c, every block of a, b or c lies inside $\operatorname{supp} x$ or is an $\mathsf{N}(x)$-block, and Lemma 5.2, applied to the matched pairs $a \bowtie c$ and $b \bowtie c$, shows that *a, b and c have the same blocks inside Y, with the same labels*: for $i \notin I := \{i : A_i \subseteq \operatorname{supp} x\}$ we have $A_i = B_i = C_i \subseteq Y$, and an $\mathsf{N}(x)$-block is an $\mathsf{N}(a)$-block iff it is an $\mathsf{N}(b)$-block iff it is an $\mathsf{N}(c)$-block. Inside $\operatorname{supp} x$ we compare the elements xa and xb, which are equal by Step 1: by Lemma 2.6(b) the upper blocks of xa are the $\mathsf{N}(x)$-blocks, the $\mathsf{T}(e)$-blocks $U_1, \dots, U_t$ (the classes of type (i) of $\varepsilon_x \vee \varepsilon_a$, each a union of $\mathsf{T}(a)$-blocks and of $\mathsf{T}(x)$-blocks, with lower part U_l') and the $\mathsf{N}(a)$-blocks inside $\operatorname{supp} x$ (type (ii)); likewise for xb. Hence *the $\mathsf{N}(a)$-blocks inside $\operatorname{supp} x$ are exactly the $\mathsf{N}(b)$-blocks inside $\operatorname{supp} x$, and each U_l is a union of $\mathsf{T}(a)$-blocks and also a union of $\mathsf{T}(b)$-blocks.* Altogether: a and b have the same N-blocks, and $\operatorname{supp} a \cap \operatorname{supp} x = \operatorname{supp} b \cap \operatorname{supp} x = \operatorname{supp} e = U_1 \cup \dots \cup U_t$. Finally, for $p \in I$ the block C_p meets $\operatorname{supp} e$ only inside $A_p \cap B_p$: a point of $C_p \cap \operatorname{supp} e$ lies in some A_q and some B_r, and C_p meets no $\mathsf{T}(a)$-block other than A_p and no $\mathsf{T}(b)$-block other than B_p.

Step 3: the blocks of d inside $\operatorname{supp} x$. By Lemma 2.6(a) the blocks of $\theta_x(d)$ inside $\operatorname{supp} x$ are the sets $K \cap \operatorname{supp} x$ for the classes K of $\varepsilon_x \vee \varepsilon_d$ meeting $\operatorname{supp} x$, transversal iff K meets $\operatorname{supp} d$; the blocks of $\theta_x(c) = f$ inside $\operatorname{supp} x$ are the $\mathsf{T}(f)$-blocks $V_1, \dots, V_{t'}$ (unions of $\mathsf{T}(c)$-blocks; transversal) and the $\mathsf{N}(c)$-blocks inside $\operatorname{supp} x$ (non-transversal). Since $\theta_x(d) = \theta_x(c)$: *every block Z of d meets $\operatorname{supp} x$ inside a single V_k, or inside a single $\mathsf{N}(c)$-block $\tilde{N} \subseteq \operatorname{supp} x$, or not at all; and in the second case Z is an $\mathsf{N}(d)$-block*, because the class of $\varepsilon_x \vee \varepsilon_d$ whose trace is $\tilde{N}$ does not meet $\operatorname{supp} d$. We call Z of *V-type*, *$\tilde{N}$-type* or *outer* accordingly.

Now let Z be a block of d lying in Λ_s. A point of Z in $\operatorname{supp} b$ lies in some B_r, and then $r = s$: *a block of Λ_s meets $\operatorname{supp} b$ only inside B_s*; in particular, if Z meets a $\mathsf{T}(a)$-block $A_i \subseteq Y$, then Z meets $B_i = A_i$ and $i = s$, and if Z meets $\operatorname{supp} e$ then $s \in I$ and Z meets no $\mathsf{T}(a)$-block inside Y. Suppose Z meets $\operatorname{supp} e$ at a point q; then $q \in B_s$. If Z is of V-type, $q \in V_k \cap \operatorname{supp} e \subseteq \operatorname{supp} c \cap \operatorname{supp} e$ lies in some C_p, hence in $A_p \cap B_p$ by Step 2, and $p = s$: *a V-type block of Λ_s meets $\operatorname{supp} a$ only inside A_s*. If Z is of $\tilde{N}$-type, $q \in \tilde{N} \cap B_s$, so $\tilde{N} \subseteq \Delta_s$ (a block of c meeting B_s), and $\tilde{N}$ lies in a class $\Gamma(\tilde{N})$ of $\varepsilon_a \vee \varepsilon_c$ which contains at most one $\mathsf{T}(a)$-block, A_r say; then *Z meets $\operatorname{supp} a$ only inside $\tilde{N} \cap A_r$*, and $\tilde{N} \subseteq \Gamma_r \cap \Delta_s$.

Step 4: the key invariant. Call two blocks of d *linked* if they are joined by a chain of blocks of d and $\mathsf{N}(a)$-blocks (consecutive blocks meeting); a linked family lies in a single class of $\varepsilon_b \vee \varepsilon_d$, since $\mathsf{N}(a)$-blocks are $\mathsf{N}(b)$-blocks (Step 2). We claim: *a linked family $\mathcal{F}$ of blocks of d meets at most one $\mathsf{T}(a)$-block; and if $\mathcal{F}$ contains a $\mathsf{T}(d)$-block D_p, it meets $\operatorname{supp} a$ only inside A_p.* If no block of $\mathcal{F}$ meets $\operatorname{supp} b$ there is nothing to prove (a $\mathsf{T}(a)$-block met by a block of $\mathcal{F}$ is then $A_i = B_i \subseteq Y$, and $\mathcal{F} \subseteq \Lambda_i$). Otherwise $\mathcal{F} \subseteq \Lambda_s$ for a unique s, and by Step 3 a block of $\mathcal{F}$ meets $\operatorname{supp} a$ only inside A_s (if it is of V-type, or of $\tilde{N}$-type with $\tilde{N} \subseteq \Gamma_s$), or inside A_r for an $\mathsf{N}(c)$-block $\tilde{N} \subseteq \operatorname{supp} x$ with $\tilde{N} \subseteq \Gamma_r \cap \Delta_s$, $r \neq s$ (if it is of $\tilde{N}$-type and meets $\operatorname{supp} e$), or inside a $\mathsf{T}(a)$-block $A_i \subseteq Y$, which forces $i = s$. So the claim follows if we show: *if some $Z_0 \in \mathcal{F}$ is of $\tilde{N}_0$-type with $\tilde{N}_0 \subseteq \Gamma_r \cap \Delta_s$ and $r \neq s$ (or with $\Gamma(\tilde{N}_0)$ containing no $\mathsf{T}(a)$-block), then every block of $\mathcal{F}$ meeting $\operatorname{supp} x$ is of $\tilde{N}$-type for some $\tilde{N} \subseteq \Gamma_r \cap \Delta_s$ (resp. $\tilde{N} \subseteq \Gamma(\tilde{N}_0) \cap \Delta_s$).* Indeed, such a family meets $\operatorname{supp} a$ only inside A_r (resp. not at all) by Step 3, and it contains no $\mathsf{T}(d)$-block: a $\mathsf{T}(d)$-block is not of $\tilde{N}$-type, and if it were outer, the chain joining it to Z_0 would have a first block meeting $\operatorname{supp} x$, joined to it through $\mathsf{N}(a)$-blocks lying in Y, which are $\mathsf{N}(x)$-blocks (the blocks of d in Y need not be), so that this first block would lie in a class of $\varepsilon_x \vee \varepsilon_d$ meeting $\operatorname{supp} d$ and could not be of $\tilde{N}$-type. We prove the statement by induction along a chain $Z_0, N_1, Z_1, N_2, \dots$ in $\mathcal{F}$. Let Z_j be of $\tilde{N}_j$-type with $\tilde{N}_j \subseteq \Gamma_r \cap \Delta_s$ and let N_{j+1} be an $\mathsf{N}(a)$-block meeting Z_j. If $N_{j+1} \subseteq Y$, it is an $\mathsf{N}(x)$-block, so Z_{j+1} lies in the same class of $\varepsilon_x \vee \varepsilon_d$ as Z_j, whose trace on $\operatorname{supp} x$ is $\tilde{N}_j$: Z_{j+1} is of $\tilde{N}_j$-type or outer, and in the latter case the next

$\mathsf{N}(a)$-block meeting it lies in Y as well (an $\mathsf{N}(a)$-block inside $\operatorname{supp} x$ cannot meet an outer block), so the class of $\varepsilon_x \vee \varepsilon_d$ persists until the next block meeting $\operatorname{supp} x$, which is then of $\tilde{N}_j$-type. If $N_{j+1} \subseteq \operatorname{supp} x$, it meets Z_j inside $\tilde{N}_j$; being a block of a meeting the block $\tilde{N}_j$ of c, it lies in Γ_r, and being a block of b (Step 2) meeting $\tilde{N}_j$, it lies in Δ_s. Let Z_{j+1} meet N_{j+1} at a point $q \in \operatorname{supp} x$, and let E be the block of c containing q. Then E meets N_{j+1}, so $E \subseteq \Gamma_r \cap \Delta_s$; E is not a $\mathsf{T}(c)$-block, since $C_r \subseteq \Delta_r \neq \Delta_s$, $C_s \subseteq \Gamma_s \neq \Gamma_r$ and $C_p \not\subseteq \Gamma_r$ for $p \neq r$ (in the case of a class without $\mathsf{T}(a)$-block, no $\mathsf{T}(c)$-block lies in it); so E is an $\mathsf{N}(c)$-block, inside $\operatorname{supp} x$ as $q \in \operatorname{supp} x$, and by Step 3 $Z_{j+1} \cap \operatorname{supp} x \subseteq E$: Z_{j+1} is of E-type with $E \subseteq \Gamma_r \cap \Delta_s$. This completes the induction and the proof of the claim.

Step 5: the classes of $\varepsilon_a \vee \varepsilon_d$. Let $\rho = \varepsilon_a \vee \varepsilon_d$ and let $D(A_i)$ be the ρ-class of A_i.

(a) *$D(A_i)$ contains no $\mathsf{T}(a)$-block other than A_i.* Let A_i, A_j lie in one class and take a chain in ρ from A_i to A_j; between two consecutive $\mathsf{T}(a)$-blocks on it, the blocks are blocks of d and $\mathsf{N}(a)$-blocks, so the blocks of d there are linked, and by Step 4 the two $\mathsf{T}(a)$-blocks coincide. Hence $A_i = A_j$.

(b) $D_i \subseteq D(A_i)$. Take a chain from B_i to D_i in Λ_i; its blocks of b other than B_i are $\mathsf{N}(b)$-blocks, i.e. $\mathsf{N}(a)$-blocks. If $i \notin I$ then $B_i = A_i$ and the chain is a chain in ρ. If $i \in I$, let Z be the block of d following B_i on the chain; the blocks of d on the chain are linked, and the family contains D_i, so by Step 4 Z meets $\operatorname{supp} a$ only inside A_i; but Z meets $B_i \subseteq \operatorname{supp} e \subseteq \operatorname{supp} a$, so Z meets A_i, and replacing B_i by A_i gives a chain in ρ from A_i to D_i.

(c) *$D(A_i)$ contains no $\mathsf{T}(d)$-block other than D_i.* Let $D_p \subseteq D(A_i)$ and take a chain in ρ from D_p to A_i; let A_r be the first $\mathsf{T}(a)$-block on it. The blocks of d before A_r are linked to D_p, so by Step 4 the one meeting A_r meets $\operatorname{supp} a$ only inside A_p, whence $r = p$; and by (a) applied along the rest of the chain, $A_i = A_r = A_p$.

Conclusion. By (a)–(c) the classes $D(A_1), \dots, D(A_m)$ are distinct and $D(A_i)$ contains exactly the T-blocks A_i and D_i; every class of ρ meeting $\operatorname{supp} a \cup \operatorname{supp} d$ contains some A_i or some D_i, hence is one of them. By Lemma 2.6(c), $a \bowtie d$. □

Remark 5.9. The proof is shorter than that of (R′) because the hypotheses make a and b almost equal: they have the same blocks outside $\operatorname{supp} x$ and the same N-blocks, and differ only in how they partition each $\mathsf{T}(e)$-block into T-blocks. Off the "diagonal" $A_p \cap B_p$, which is where the $\mathsf{T}(c)$-blocks meet $\operatorname{supp} e$, the overlaps $A_r \cap B_s$ lie in $\mathsf{N}(c)$-blocks, and Step 4 says that the blocks of d reaching such an overlap form a closed family of $\mathsf{N}(d)$-blocks. Unlike (R′), the conclusion is not accompanied by a separation statement for a coarser join: in $\mathsf{P}(\mathcal{P}_4)$ the classes of $\varepsilon_a \vee \varepsilon_d \vee \varepsilon_b$ may contain two $\mathsf{T}(a)$-blocks.

5.6. **The condition (Q) and its corollaries ($\mathbf{P}_2$), (S).** We write the path of (Q) as $a \bowtie c \bowtie b \bowtie d$, with the T-blocks A_i, C_i, B_i, D_i and the classes $\Gamma_i, \Delta_i, \Lambda_i$ as in Section 5.5; the hypotheses are $xa = ax$, $\theta_c(x) = \theta_d(x)$, $ya = ay$, $yb = by$, $\theta_y(c) = \theta_y(b)$, and the conclusion is $a \bowtie d$. The proof occupies the rest of this subsection; the statements of Steps 2–7 were found and checked on all instances of the hypotheses in $\mathsf{P}(\mathcal{P}_4)$ and on sampled instances in $\mathsf{P}(\mathcal{P}_5)$ and $\mathsf{P}(\mathcal{P}_6)$ before being proved. Besides (Q) itself we obtain the two conditions

$$(\mathrm{P}_2): \quad a \bowtie c \bowtie b \bowtie d,\ xa = ax,\ xb = bx,\ \theta_x(c) = \theta_x(b),\ \theta_c(x) = \theta_d(x) \implies a \bowtie d,$$

$$(\mathrm{S}): \quad a \bowtie c \bowtie b \bowtie d,\ cd = dc,\ \theta_a(cd) = \theta_b(cd) \implies a \bowtie d,$$

of which (P_2) is the case $y = x$ of (Q) and (S) is a consequence of (P_2) (Corollary 5.13).

Step 1: normalising the auxiliary x. Put $e = xa = ax = \theta_a(x)$. Then $e \leq a$ and $\theta_c(e) = \theta_d(e)$. Indeed, for projections p, q, r one has $\theta_{\theta_r(p)}(\theta_p(q)) = \theta_r(\theta_p(q))$, since $rpr\,pqp\,rpr = (rprp)q(prpr) = rp\,q\,pr$ by the idempotence of rp and pr (Section 2.1); hence $\theta_c(e) = \theta_c(\theta_x(a)) = \theta_{\theta_c(x)}(\theta_x(a)) = \theta_{\theta_d(x)}(\theta_x(a)) = \theta_d(\theta_x(a)) = \theta_d(e)$. So we may and do assume $x = e \leq a$. Since

$z := \theta_c(e) = \theta_d(e)$ is a common lower bound of c and d, (Q1) gives $cd = dc$. By Lemma 2.7(a) the projection $e \le a$ is determined by a partition P of the index set $\{1, \dots, m\}$ into *groups* G and a label for each group: the blocks of e are the $\mathsf{N}(a)$-blocks and the sets $A_G = \bigcup_{i \in G} A_i$, transversal iff G is labelled T. The labels carry no information (changing the label of a group from T to N preserves the hypothesis $\theta_c(e) = \theta_d(e)$, as Step 2 shows), but it is convenient to keep them. We write $C_G = \bigcup_{i \in G} C_i$, B_G, D_G similarly.

Step 2: the structure of c and d. By Lemma 2.8, $z = \theta_c(e)$ has the blocks $\mathsf{N}(c) \cup \{C_G : G \in P\}$, C_G transversal iff G is. Put $\Pi_1 = \varepsilon_e \vee \varepsilon_c$. Its classes are the sets $\Gamma_G = \bigcup_{i \in G} \Gamma_i$ and the *plain* classes of $\varepsilon_a \vee \varepsilon_c$ (those without T-blocks), and since $z \le c$, $z \le d$ and $z = \theta_d(e)$ we have $\Pi_1 = \varepsilon_e \vee \varepsilon_d = \varepsilon_a \vee \varepsilon_z$, so that $\varepsilon_a \vee \varepsilon_d \le \Pi_1$. More precisely, since $\varepsilon_a \le \varepsilon_e$, $\Pi_1 = \varepsilon_e \vee \varepsilon_d = \rho \vee \varepsilon_e$ with $\rho = \varepsilon_a \vee \varepsilon_d$, and $\rho \vee \varepsilon_e$ is obtained from ρ by identifying the classes that meet the same block A_G; so a class of ρ containing no $\mathsf{T}(a)$-block is a class of Π_1, and hence *every class of ρ contained in some Γ_G contains a $\mathsf{T}(a)$-block A_i, $i \in G$*, and every class of ρ not contained in any Γ_G is a plain class. Comparing the two descriptions of z (blocks $\mathsf{N}(c) \cup \{C_G\}$, and $\mathsf{N}(d)$ together with the traces $L \cap \operatorname{supp} d$ of the classes L of Π_1) gives, for each group G, one of two cases:

(nh) $C_G \subseteq \operatorname{supp} d$ and $C_G = \Gamma_G \cap \operatorname{supp} d$ (a *non-hub* group);

(h) C_G is an $\mathsf{N}(d)$-block, and $\Gamma_G \cap \operatorname{supp} d$ is empty or a single $\mathsf{N}(c)$-block E_G (a *hub*);

every other $\mathsf{N}(c)$-block is an $\mathsf{N}(d)$-block and conversely, and a plain class W meets $\operatorname{supp} d$ in an $\mathsf{N}(c)$-block E_W or not at all. The same construction from the side of b, with $w = \theta_b(z) \le b$ (which satisfies $\theta_c(w) = \theta_d(w) = z$ by Lemma 2.8 along $b \bowtie c$ and $b \bowtie d$) and *the same partition* P (because $\theta_c(w)$ has the blocks $\mathsf{N}(c) \cup \{C_{G'}\}$ for the groups G' of w), gives $\Delta_G = \Lambda_G$ for every G, where $\Delta_G = \bigcup_{i \in G} \Delta_i$ and $\Lambda_G = \bigcup_{i \in G} \Lambda_i$, with $\Lambda_G \cap \operatorname{supp} d = D_G$ (a block of $z = \theta_d(w)$, by transport along $b \bowtie d$); hence in case (nh) $C_G = D_G$, and in case (h) D_G is a single $\mathsf{N}(c)$-block $E'_G \subseteq \Lambda_G$. Comparing the blocks $\mathsf{N}(c) \cup \{C_G\}$ of $z = \theta_c(e)$ with the blocks $\mathsf{N}(d) \cup \{D_G\}$ of $z = \theta_d(w)$, and using $C_G = D_G$ for non-hubs, we see that the $\mathsf{N}(c)$-blocks which are not $\mathsf{N}(d)$-blocks are exactly the sets $D_{G'}$, G' a hub. Each nonempty trace E_G or E_W is an $\mathsf{N}(c)$-block contained in $\operatorname{supp} d$, hence one of these; conversely each $D_{G'}$, G' a hub, is an $\mathsf{N}(c)$-block contained in $\operatorname{supp} d$, hence contained in the trace of its class of Π_1, which is a hub class or a plain class (as $D_{G'} \cap \operatorname{supp} c = \emptyset$), and equals that trace. So the nonempty traces E_G of the hubs and E_W of the plain classes are exactly the sets $D_{G'} = E'_{G'}$, G' a hub, each occurring once; write $E_G = D_{\pi(G)}$, $E_W = D_{\pi(W)}$. In particular the $\mathsf{T}(d)$-blocks contained in a non-hub class Γ_G are exactly D_k, $k \in G$, and those contained in a hub class Γ_G form $D_{\pi(G)}$ for a hub $\pi(G)$. Consequently $\rho \le \Pi_1$ has, inside a non-hub class, $|G|$ blocks of each kind, and a proof of $a \bowtie d$ has to show that these are matched by ρ, and that $|\pi(G)| = |G|$ for hubs, again with a matching.

The hub C_G lies in a single class Λ_j, $j \in G$, and E_G in a single class $\Delta_{k'}$, $k' \in \pi(G)$. Hubs have $|G| \le 2$, and a hub G of size 2 is *cross-wired*: writing $G = \{k, l\}$ with $C_G \subseteq \Lambda_l$, we have $D_G \subseteq \Delta_k$, so that $C_k \subseteq \Delta_k \cap \Lambda_l$, $C_l \subseteq \Delta_l \cap \Lambda_l$, $D_k \subseteq \Delta_k \cap \Lambda_k$, $D_l \subseteq \Delta_k \cap \Lambda_l$, and in particular C_k and D_l do not meet $\operatorname{supp} b$ (the $\mathsf{T}(b)$-block B_m lies in $\Delta_m \cap \Lambda_m$). Indeed, let $C_G \subseteq \Lambda_j$ and $D_G \subseteq \Delta_{j'}$ with $j, j' \in G$, and let $i \in G$. A path in Λ_i from D_i to B_i alternates between blocks of d and blocks of b; its interior blocks of d are $\mathsf{N}(d)$-blocks, i.e. common blocks or hubs $C_{G''}$; every block of b, every common block (an $\mathsf{N}(c)$-block) and $D_i \subseteq D_G$ (contained in the $\mathsf{N}(c)$-block D_G) lies in a single class of $\varepsilon_c \vee \varepsilon_b$, while a hub $C_{G''}$ meets exactly the classes Δ_m, $m \in G''$. So the class of $\varepsilon_c \vee \varepsilon_b$ changes along the path only at hubs $C_{G''}$, and only between indices in G''. The path starts in $\Delta_{j'}$ and ends in Δ_i; if $i \ne j'$, the first change happens at a hub $C_{G''}$ with $j' \in G''$, i.e. $G'' = G$, and since $C_G \subseteq \Lambda_j$ lies on a path in Λ_i, we get $i = j$. Hence $G \subseteq \{j, j'\}$, and if $|G| = 2$ then $j \ne j'$; put $l = j$, $k = j'$. The map π need not be the identity: in $\mathsf{P}(\mathcal{P}_4)$ take $a = (01|23)$, $c = (0|\mathsf{N}1|2|\mathsf{N}3)$, $b = (03|12)$, $d = (\mathsf{N}0|1|\mathsf{N}2|3)$ (writing only the upper parts, N

marking the N-blocks), $e = a$ with its blocks relabelled N, and y the projection of rank 0 with the single block X; then there are two singleton hubs, exchanged by π.

Step 3: the structure imposed by y. Since y commutes with a and b, Lemma 2.6(b) describes $\varepsilon_y \vee \varepsilon_a$ and $\varepsilon_y \vee \varepsilon_b$; in particular every $\mathsf{N}(y)$-block is an $\mathsf{N}(a)$-block or a union of $\mathsf{T}(a)$-blocks, and an $\mathsf{N}(b)$-block or a union of $\mathsf{T}(b)$-blocks (four kinds: *ab-common* blocks, the $\mathsf{N}(a)$-blocks that are $\mathsf{N}(b)$-blocks; *bridges*, the $\mathsf{N}(b)$-blocks that are unions of $\mathsf{T}(a)$-blocks; *co-bridges*; and unions of $\mathsf{T}(a)$-blocks that are unions of $\mathsf{T}(b)$-blocks), and every block of a or b lies in $\operatorname{supp} y$ or in an $\mathsf{N}(y)$-block. The hypothesis $\theta_y(c) = \theta_y(b)$ says (Lemma 2.6(a),(b)): the classes K of $\varepsilon_y \vee \varepsilon_c$ meeting $\operatorname{supp} y$ have traces $K \cap \operatorname{supp} y$ equal to the classes of $\varepsilon_y \vee \varepsilon_b$ of type (i) (unions Q of $\mathsf{T}(y)$-blocks which are unions of $\mathsf{T}(b)$-blocks; these K meet $\operatorname{supp} c$; we call them classes *of the first kind*) and of type (ii) (single $\mathsf{N}(b)$-blocks $F \subseteq \operatorname{supp} y$; these K do not meet $\operatorname{supp} c$; *of the second kind*). We shall use the following consequence repeatedly.

($*$) *Let K be a class of $\varepsilon_y \vee \varepsilon_c$ of the second kind, with trace an $\mathsf{N}(b)$-block $F \subseteq \operatorname{supp} y$, and let P be a path in a class Δ_m of $\varepsilon_c \vee \varepsilon_b$ (blocks of c and of b alternately, consecutive blocks meeting, all distinct) or in a class Λ_k of $\varepsilon_b \vee \varepsilon_d$ (blocks of d and of b), whose first block is F or a block of c lying in K. Then every block of c on P or meeting a block of b of P lies in K; in particular P contains no $\mathsf{T}(c)$-block; every block of d on P is a common block (an $\mathsf{N}(c)$-block which is an $\mathsf{N}(d)$-block; it lies in K), except possibly the $\mathsf{T}(d)$-block D_k, which is then not contained in $\operatorname{supp} c$, so that the $\mathsf{N}(c)$-block containing it lies in K.*

Call a set *covered* if each of its points lies in a block of y or of c belonging to K; F is covered (its points lie in $\mathsf{T}(y)$-blocks, which are in K as F is the trace), and so is a block of c in K. We show inductively that every block of P is covered, which gives all assertions: let W be covered and let W' be the next block of P, meeting W at q, so that q lies in a block Y of y or of c with $Y \subseteq K$. If W' is a block of c, then $W' = Y$ or W' meets the y-block Y, so $W' \subseteq K$; the same applies to any block of c meeting W if W is a block of b, and shows that a block of c meeting a covered set lies in K. If W' is an $\mathsf{N}(b)$-block, it is an $\mathsf{N}(y)$-block, and then $W' \subseteq K$ ($W' = Y$, or W' meets the block Y of c), or a union of $\mathsf{T}(y)$-blocks, and then the $\mathsf{T}(y)$-block through q lies in K, hence in F, so $W' = F$; in both cases W' is covered. If W' is the $\mathsf{T}(b)$-block B_m (resp. B_k), it is not contained in $\operatorname{supp} y$ (the $\mathsf{T}(y)$-block through q would lie in K, hence in F, but $F \cap W' = \emptyset$), so it lies in an $\mathsf{N}(y)$-block Q', which meets Y and lies in K, and W' is covered. If W' is a block of d, it is a common block (a block of c, covered by the first case), or a hub, or D_k; a hub is a union of $\mathsf{T}(c)$-blocks, and the $\mathsf{T}(c)$-block through q would lie in K, which does not meet $\operatorname{supp} c$, impossible; and for D_k the same argument shows $D_k \not\subseteq \operatorname{supp} c$, so D_k lies in an $\mathsf{N}(c)$-block (Step 2), which lies in K.

From this: (a) an $\mathsf{N}(b)$-block F lying in the class Δ_m of a $\mathsf{T}(c)$-block is an $\mathsf{N}(y)$-block: otherwise it is a union of $\mathsf{T}(y)$-blocks, the class K with trace F is of the second kind, and a path in Δ_m from F to C_m contradicts ($*$); (a′) an $\mathsf{N}(b)$-block F lying in the class Λ_k of a $\mathsf{T}(d)$-block is an $\mathsf{N}(y)$-block: otherwise, with K as before, a path in Λ_k from F to D_k shows by ($*$) that D_k lies in an $\mathsf{N}(c)$-block $E \subseteq K$, which is $E_{G'}$ or E_W for a hub G' or a plain class W, hence equals $D_{G''}$ for a hub G'' (Step 2) and lies in a class $\Delta_{k''}$, $k'' \in G''$ (as $D_{G''} \subseteq \Lambda_{G''} = \Delta_{G''}$), and a path in $\Delta_{k''}$ from E to $C_{k''}$ contradicts ($*$); (b) a $\mathsf{T}(y)$-block meeting $\operatorname{supp} c$ lies in a class K of the first kind, hence in a set Q as above, hence in $\operatorname{supp} b$; so a $\mathsf{T}(y)$-block meets C_i only inside $B_i \cap C_i$ (the only $\mathsf{T}(b)$-block meeting $C_i \subseteq \Delta_i$ is B_i); (b′) a $\mathsf{T}(y)$-block Y meeting D_k lies in $\operatorname{supp} b$, and meets D_k only inside $B_k \cap D_k$: if $D_k \subseteq \operatorname{supp} c$ this is (b), and otherwise D_k lies in an $\mathsf{N}(c)$-block $E \subseteq \Delta_{k''}$ as in (a′), and if the class K of Y were of the second kind, $E \subseteq K$ and a path in $\Delta_{k''}$ from E to $C_{k''}$ would contradict ($*$); (c) the $\mathsf{T}(c)$-blocks in the class with trace Q are exactly the C_i with $B_i \subseteq Q$; (d) every $\mathsf{N}(b)$-block meeting $\operatorname{supp} c \cup \operatorname{supp} d$ lies in some Δ_m or Λ_k, hence is an $\mathsf{N}(y)$-block by (a), (a′), hence an $\mathsf{N}(a)$-block or a bridge; (e) every block Z of a

satisfies $Z \cap (\operatorname{supp} c \cup \operatorname{supp} d) \subseteq \operatorname{supp} b$, or Z is contained in an $\mathsf{N}(b)$-block: if $Z \subseteq \operatorname{supp} y$ this is (b), (b′), and otherwise Z lies in an $\mathsf{N}(y)$-block, which is an $\mathsf{N}(b)$-block or a union of $\mathsf{T}(b)$-blocks.

Step 4: plain classes contain no $\mathsf{T}(d)$-block. Let W be a plain class; then $W \cap \operatorname{supp} a = W \cap \operatorname{supp} c = \emptyset$, every block of a or of c meeting W lies in W, and so does every block of y meeting W: an $\mathsf{N}(y)$-block meeting W is an $\mathsf{N}(a)$-block (it cannot be a union of $\mathsf{T}(a)$-blocks), and a $\mathsf{T}(y)$-block lies in a union of $\mathsf{T}(a)$-blocks or in an $\mathsf{N}(a)$-block (Lemma 2.6(b), as $ya = ay$), the latter alternative being forced if it meets W. Hence W *is a union of classes of* $\varepsilon_y \vee \varepsilon_c$. Suppose now $D_j \subseteq E_W \subseteq W$. Take a path in Λ_j from D_j to B_j. Its $\mathsf{N}(b)$-blocks lie in Λ_j, so they are $\mathsf{N}(y)$-blocks (Step 3(a′)), hence $\mathsf{N}(a)$-blocks or bridges; a bridge is a union of $\mathsf{T}(a)$-blocks and cannot meet W, and an $\mathsf{N}(a)$-block meeting W lies in W; the interior $\mathsf{N}(d)$-blocks of the path are common blocks (blocks of c, which lie in W if they meet it) or hubs (unions of $\mathsf{T}(c)$-blocks, which cannot meet W). So inductively every block of the path except possibly B_j lies in W, and B_j meets W. If B_j lies in an $\mathsf{N}(y)$-block Q, then Q meets W, so Q is an $\mathsf{N}(a)$-block contained in W and $B_j \subseteq W$; a path in Δ_j from B_j to C_j then lies in W by the same argument (its $\mathsf{N}(b)$-blocks are $\mathsf{N}(y)$-blocks by Step 3(a), and its blocks of c lie in W as soon as they meet it), so $C_j \subseteq W$, contradicting $W \cap \operatorname{supp} c = \emptyset$. If $B_j \subseteq \operatorname{supp} y$, take $q \in B_j \cap W$ and let Y be the $\mathsf{T}(y)$-block of q and K its class in $\varepsilon_y \vee \varepsilon_c$; then $K \subseteq W$. The trace $K \cap \operatorname{supp} y$ is a class of $\varepsilon_y \vee \varepsilon_b$ containing Y, which meets B_j, so it is not a single $\mathsf{N}(b)$-block and contains B_j: K is of the first kind, with trace $Q \supseteq B_j$, and $C_j \subseteq K \subseteq W$ by Step 3(c), again contradicting $W \cap \operatorname{supp} c = \emptyset$. Consequently every hub G has $E_G \neq \emptyset$, and π is a permutation of the hubs.

Step 5: types, index pairs and contacts. By Step 2, $\rho \leq \Pi_1$, every class of ρ contained in a class Γ_G contains some A_i, $i \in G$, and a class of ρ not contained in any Γ_G is a plain class, which contains no T-block of a (Step 2) or of d (Step 4). By Lemma 2.6(c), $a \bowtie d$ is therefore equivalent to: *for every group G, every class of ρ contained in Γ_G contains exactly one $\mathsf{T}(a)$-block and exactly one $\mathsf{T}(d)$-block.* This is proved for non-hub groups in Step 6 and for hubs in Step 7. Inside Γ_G the blocks of a are A_i ($i \in G$) and $\mathsf{N}(a)$-blocks, and the blocks of d are the *common* blocks (the $\mathsf{N}(c)$-blocks which are $\mathsf{N}(d)$-blocks; in a non-hub class these are all the N-blocks of c and of d in Γ_G, and $C_G = D_G$ is the union of the T-blocks of both), and, in a hub class, the hub C_G and the $\mathsf{T}(d)$-blocks D_m, $m \in \pi(G)$, whose union is the $\mathsf{N}(c)$-block E_G.

Types of blocks. By Step 3 every block Z of a either lies in $\operatorname{supp} y$, or lies in an $\mathsf{N}(y)$-block Q, which is then an $\mathsf{N}(b)$-block or a union of $\mathsf{T}(b)$-blocks. Call Z of N-*type* if it lies in an $\mathsf{N}(y)$-block which is an $\mathsf{N}(b)$-block (if Z is an $\mathsf{N}(a)$-block this means Z itself is an $\mathsf{N}(b)$- and $\mathsf{N}(y)$-block; if $Z = A_i$ it means that A_i lies in a bridge Q), and of b-*type* otherwise; a b-type block lies in $\operatorname{supp} y$ or in an $\mathsf{N}(y)$-block contained in $\operatorname{supp} b$, so by Step 3(b), (b′) it meets each C_m only inside $B_m \cap C_m$ and each D_m only inside $B_m \cap D_m$. For an $\mathsf{N}(b)$-block Q write $\delta(Q) = m$ if $Q \subseteq \Delta_m$ and $\lambda(Q) = k$ if $Q \subseteq \Lambda_k$ (undefined if the class of Q in $\varepsilon_c \vee \varepsilon_b$, resp. $\varepsilon_b \vee \varepsilon_d$, contains no T-block); an N-type block $Z \subseteq Q$ meets C_m only if $\delta(Q) = m$ and D_k only if $\lambda(Q) = k$. A block of a meeting $\operatorname{supp} c \cup \operatorname{supp} d$ at a point outside $\operatorname{supp} b$ is of N-type (Step 3(e), (d)).

In a non-hub class every point of $C_G = D_G$ lies in a unique $\mathsf{T}(c)$-block C_m and a unique $\mathsf{T}(d)$-block D_k (its *cell* (m,k)), and a point of C_G lying in $\operatorname{supp} b$ lies in $B_m \cap C_m \cap D_m$ for some m: its cell is *diagonal*. Hence a b-type block of a meets C_G only in diagonal cells, and since it meets at most one $\mathsf{T}(c)$-block (it lies in one class Γ_m), only in the cell (m,m); while an N-type block $Z \subseteq Q$ meets C_G only in points outside $\operatorname{supp} b$, in the cell $(\delta(Q), \lambda(Q))$.

In a hub class, with $C_G \subseteq \Lambda_j$ ($j \in G$) and $E_G \subseteq \Delta_{k'}$ ($k' \in \pi(G)$) as in Step 2: a b-type block meets C_G only in the sets $B_i \cap C_i$, $i \in G$, and E_G only in the sets $B_m \cap D_m$, $m \in \pi(G)$; an N-type block $Z \subseteq Q$ meets C_G only if $\lambda(Q) = j$, and then inside $C_{\delta(Q)}$, and meets E_G only if $\delta(Q) = k'$, and then inside $D_{\lambda(Q)}$.

Index pairs. For a common block E let $\lambda(E)$ be the index k with $E \subseteq \Lambda_k$ (undefined if the class of E in $\varepsilon_b \vee \varepsilon_d$ contains no $\mathsf{T}(d)$-block) and $\delta(E)$ the index m with $E \subseteq \Delta_m$ (undefined likewise). (i) *If E meets a b-type block of a at a point q, then either $q \in B_m$ for some m and $(\lambda(E), \delta(E)) = (m, m)$, or $q \notin \operatorname{supp} b$ and both indices are undefined.* Indeed, if $q \in B_m$ then E meets B_m. Otherwise q lies in $\operatorname{supp} y$ (a-type block not contained in $\operatorname{supp} y$ lies in $\operatorname{supp} b$), in a $\mathsf{T}(y)$-block Y, which lies in an $\mathsf{N}(b)$-block $F \subseteq \operatorname{supp} y$ (Step 3), and the class K of $\varepsilon_y \vee \varepsilon_c$ with trace F is of the second kind and contains E (c block meeting Y). If $\delta(E) = m$ were defined, a path in Δ_m from E to C_m would contradict $(*)$. If $\lambda(E) = k$ were defined, a path in Λ_k from E to D_k shows by $(*)$ that D_k lies in an $\mathsf{N}(c)$-block $E' \subseteq K$, which lies in a class $\Delta_{k''}$ (Step 2, as in Step 3(a′)), and a path in $\Delta_{k''}$ from E' to $C_{k''}$ contradicts $(*)$. (ii) If E meets an N-type block $Z \subseteq Q$ then $(\lambda(E), \delta(E)) = (\lambda(Q), \delta(Q))$ (as indices or as "undefined"), as E meets the b-block Q. (iii) Two common blocks meeting the same N-type $\mathsf{N}(a)$-block have the same pair. Consequently, along a path $E_0, F_1, E_1, \ldots, F_r, E_r$ alternating between common blocks and $\mathsf{N}(a)$-blocks (consecutive blocks meeting; no hub involved), the pair is preserved across N-type $\mathsf{N}(a)$-blocks and, at a b-type $\mathsf{N}(a)$-block, the pairs of the two adjacent common blocks are each either diagonal ((m, m)) or undefined in both coordinates. In particular (*rigidity*): *if some common block of such a path, or some N-type $\mathsf{N}(a)$-block of it, has a defined non-diagonal pair (k, m), $k \neq m$, then every $\mathsf{N}(a)$-block of the path is of N-type and every block of the path has the pair (k, m); and a $\mathsf{T}(a)$-block met by a common block of the path is then of N-type with the pair (k, m) as well* (by (i) and (ii), which apply to $\mathsf{T}(a)$-blocks too).

Step 6: non-hub classes.

Proposition 5.10. *In the situation of Steps* 1–5*, let G be a non-hub group. Then every class of ρ contained in Γ_G contains exactly one $\mathsf{T}(a)$-block and exactly one $\mathsf{T}(d)$-block.*

Proof. Let $\mathcal{K}$ be the set of components of the graph on the $\mathsf{N}(a)$-blocks and the common blocks contained in Γ_G (blocks joined when they meet); each $\kappa \in \mathcal{K}$ is contained in a single class Γ_m, so it touches at most one $\mathsf{T}(a)$-block, A_m, and its $\mathsf{N}(a)$-blocks meet C_G only in C_m. Say that D_k is *attached* to A_i if A_i meets D_k, or if some κ touching A_i (a common block of κ meets A_i) contains an $\mathsf{N}(a)$-block meeting D_k. The classes of ρ inside Γ_G are generated by these adjacencies, so it suffices to prove:

(a) every κ contains $\mathsf{N}(a)$-blocks meeting at most one $\mathsf{T}(d)$-block;
(b) at most one $\mathsf{T}(d)$-block is attached to each A_i;
(c) at least one $\mathsf{T}(d)$-block is attached to each A_i.

Given (a)–(c): by (b), (c) the attachment defines a map $\tau\colon G \to G$; every D_k lies in a class of ρ containing some A_i (Step 5), and a ρ-path from D_k to A_i starts with D_k, a block of a meeting it, and, if that block is an $\mathsf{N}(a)$-block, continues inside its κ until the first T-block met, which by (a) is a $\mathsf{T}(a)$-block; so every D_k is attached to some A_i, τ is surjective, hence bijective; and then the ρ-class of A_i contains the T-blocks A_i and $D_{\tau(i)}$ only (a further $\mathsf{T}(a)$-block A_j in it would be reached through some D_k, which would be attached to both, and a further $\mathsf{T}(d)$-block through some κ or A, contradicting (a) or (b)), and every class of ρ in Γ_G is of this form.

(c): Take a path in Γ_i from A_i to C_i; its blocks of c are common blocks and its blocks of a are $\mathsf{N}(a)$-blocks, so they lie in one κ touching A_i, and the last block of a on it (or A_i itself) meets $C_i \subseteq C_G = D_G$, hence some D_k.

(b): Let D_k be attached to A_i. *Case A_i of b-type.* A direct contact $A_i \cap D_k \neq \emptyset$ is in the cell (i, i), so $k = i$. Otherwise κ touches A_i at a common block E whose pair is diagonal or undefined (i), and contains an $\mathsf{N}(a)$-block F meeting D_k at a point of C_i. If F is of b-type, the contact is in the cell (i, i) and $k = i$. If F is of N-type, F is an $\mathsf{N}(b)$-block in $\Lambda_k \cap \Delta_i$, so the common block of κ adjacent to F on a path from E has the pair (k, i); but along the path from E all pairs are diagonal or undefined, so $k = i$. *Case $A_i \subseteq Q$ of N-type.* A direct contact lies in the cell

$(i, \lambda(Q))$, so $k = \lambda(Q)$ and $\delta(Q) = i$. Otherwise the common block E of κ meeting A_i has the pair $(\lambda(Q), \delta(Q))$, and along the path from E to the contact block F the pair is preserved up to the first b-type $\mathsf{N}(a)$-block, if any. If there is none: F is of N-type with $(\lambda(F), \delta(F)) = (k, i)$ equal to $(\lambda(Q), \delta(Q))$, or of b-type with $k = i$ and the pair $(\lambda(Q), \delta(Q))$ diagonal or undefined; if there is one, then $(\lambda(Q), \delta(Q))$ is diagonal or undefined and the contact gives $k = i$ (for an N-type F the adjacent pair (k, i) is diagonal). So every attached D_k is $D_{\lambda(Q)}$, and then $\delta(Q) = i$, or D_i, and then $\lambda(Q) = \delta(Q)$ or both are undefined; if both kinds occur then $\lambda(Q) = \delta(Q) = i$. Hence at most one $\mathsf{T}(d)$-block is attached.

(a): Let κ contain F meeting D_k and F' meeting D_l, in cells (γ, k) and (γ, l) where $\kappa \subseteq \Gamma_\gamma$. If F is of b-type then $k = \gamma$, and likewise for F'. If F' is of N-type, it is an $\mathsf{N}(b)$-block in $\Lambda_l \cap \Delta_\gamma$, and the common block next to F' on a path from F' to F has the pair (l, γ); following the path, if it contains no b-type block before F then either F is of N-type and $(k, \gamma) = (l, \gamma)$, or F is of b-type and the pair (l, γ) is diagonal, $l = \gamma = k$; if it contains a b-type block, the pair (l, γ) is diagonal, $l = \gamma$, and by symmetry (the case F of N-type gives $k = \gamma$ in the same way) $k = \gamma$. In all cases $k = l$. □

Step 7: hub classes.

Proposition 5.11. *In the situation of Steps* 1–5*, let G be a hub. Then $|\pi(G)| = |G|$, and every class of ρ contained in Γ_G contains exactly one $\mathsf{T}(a)$-block and exactly one $\mathsf{T}(d)$-block.*

Proof. Write $G' = \pi(G)$; by Steps 2 and 4, $E_G = D_{G'}$ is a nonempty $\mathsf{N}(c)$-block contained in Γ_G, hence in a single class Γ_e, $e \in G$, and in a single class $\Delta_{k'}$, $k' \in G'$; and $C_G \subseteq \Lambda_j$, $j \in G$. (In this proof e denotes an index, not the projection of Step 1, which is no longer needed.) If $|G| = 2$ we write $G = \{k, l\}$ with $l = j$, so that (cross wiring, Step 2) $C_k \subseteq \Delta_k \cap \Lambda_l$, $C_l \subseteq \Delta_l \cap \Lambda_l$ and $C_k \cap \operatorname{supp} b = \emptyset$. If $|G'| = 2$ we write $G' = \{k', l'\}$, so that (cross wiring for G') $D_{k'} \subseteq \Delta_{k'} \cap \Lambda_{k'}$, $D_{l'} \subseteq \Delta_{k'} \cap \Lambda_{l'}$ and $D_{l'} \cap \operatorname{supp} b = \emptyset$. Let $\mathcal{N}$ be the set of components ν of the graph on the N-blocks of a and d inside Γ_G (the $\mathsf{N}(a)$-blocks, the common blocks and C_G; joined when they meet), and say that D_m ($m \in G'$) is *attached* to A_i ($i \in G$) if there is a path in the graph of blocks of a and d from A_i to D_m all of whose interior blocks are N-blocks, i.e. A_i meets D_m, or some ν touching A_i contains an $\mathsf{N}(a)$-block meeting D_m. All paths considered below are shortest paths between their endpoints in the graph in question, hence simple, and an attachment path from A_i to D_m is always taken shortest, so that its interior is a shortest path in a component ν. A component ν not containing C_G lies in a single class Γ_i; a shortest path in a component ν between two blocks lying in the same class Γ_i (or a shortest attachment path with both ends in Γ_i) passes through C_G at most once, and if it does, both neighbours of C_G on it are $\mathsf{N}(a)$-blocks in Γ_i meeting C_G inside C_i, and the two parts of the path avoid C_G (each part lies in a single class of $\varepsilon_a \vee \varepsilon_c$, being a chain of blocks of a and of c); a path in ν between blocks in different classes Γ_k, Γ_l passes through C_G. We write $G = \{k, l\}$, $G' = \{k', l'\}$ as above; when $G' = G$ we have $k' = k$ and $l' = l$, because $E_G = D_G$ lies in $\Delta_{k'}$ and in Δ_k. By Step 5, inside Γ_G:

(i) a block of a meeting C_G in C_i is of N-type, contained in an $\mathsf{N}(b)$-block Q with $(\lambda(Q), \delta(Q)) = (j, i)$, or it is of b-type, $i = j$ and the contact lies in $B_j \cap C_j$;
(ii) a block of a meeting $D_{k'}$ is of N-type with pair (k', k'), or of b-type with contact in $B_{k'} \cap D_{k'}$;
(iii) if $|G'| = 2$, a block of a meeting $D_{l'}$ is of N-type with pair (l', k').

Claim 1: the $\mathsf{N}(a)$-blocks of a single component ν meet at most one $\mathsf{T}(d)$-block. Suppose $Z, Z' \in \nu$ meet D_m, $D_{m'}$ with $m \neq m'$, so $|G'| = 2$, say $m' = l'$, $m = k'$. Both lie in Γ_e; take a shortest path in ν from Z' to Z. If it avoids C_G, then by (iii) and rigidity all its blocks have the pair (l', k'), contradicting (ii) for Z. If it passes through C_G, the part from Z' to the neighbour Z_1 of C_G avoids C_G, so Z_1 has the pair (l', k') by rigidity, and Z_1 meets C_G inside C_e,

so $(\lambda, \delta) = (j, e)$ by (i); thus $k' = e \in G \cap G'$, so $G' = G$, $|G| = 2$, $k' = k$, $e = k$, $l' = l = j$. Then the other neighbour Z_2 of C_G meets C_G inside C_k, so it is of N-type with pair (l, k) by (i) (as $C_k \cap \operatorname{supp} b = \emptyset$), and rigidity along the part from Z_2 to Z gives Z the pair (l, k), contradicting (ii).

Claim 2: at most one $\mathsf{T}(d)$-block is attached to each A_i. Suppose $D_{k'}$ and $D_{l'}$ are attached to A_i, so $|G'| = 2$. By (iii) the contact block of $D_{l'}$ (which is A_i itself or an $\mathsf{N}(a)$-block Z') has the pair (l', k'). *Case $G' \neq G$.* An attachment path from A_i to $D_{l'}$ cannot pass through C_G: the neighbour of C_G on the side of $D_{l'}$ would have the pair (l', k') by rigidity and $\delta = i'' \in G$ by (i), but $k' \notin G$. So by rigidity A_i is of N-type, $A_i \subseteq Q$ with $(\lambda(Q), \delta(Q)) = (l', k')$. Now consider an attachment path from A_i to $D_{k'}$. It does not start with a direct contact ($\lambda(Q) = l' \neq k'$), nor with C_G ($\lambda(Q) = l' \neq j$, as $l' \notin G$), so it starts with a common block meeting A_i, of pair (l', k'), and by rigidity all its blocks have this pair until it reaches C_G or the contact block of $D_{k'}$; the former is excluded as before ($k' \notin G$), the latter by (ii). *Case $G' = G$*, so $|G| = 2$, $k' = k$, $l' = l = j$. Suppose first $i = k$. If A_k meets D_l, then A_k is of N-type with pair (l, k) by (iii) and $e = k$. Otherwise take a shortest attachment path from A_k to D_l; its contact block has the pair (l, k). If the path passes through C_G, the neighbour of C_G on the side of the contact block has the pair (l, k) by rigidity, so by (i) it meets C_G inside C_k and lies in Γ_k, and so does the contact block, i.e. $e = k$; and the neighbour on the side of A_k then also lies in Γ_k, meets C_k and has the pair (l, k) by (i) ($C_k \cap \operatorname{supp} b = \emptyset$). If the path avoids C_G, all its blocks lie in Γ_k, so again $e = k$. In both cases rigidity along the part of the path next to A_k (or (i), if that part is empty and A_k meets C_G) shows that A_k is of N-type with pair (l, k). The attachment path from A_k to D_k then starts with the pair (l, k) (a direct contact is excluded by $\lambda = l \neq k$; a common block meeting A_k has the pair (l, k); and C_G, if it comes first or later, has both neighbours in Γ_k, of pair (l, k), as $e = k$), so by rigidity it never reaches a block of pair (k, k) or of b-type, contradicting (ii). Suppose now $i = l$. If the attachment path from A_l to D_l avoids C_G, then $e = l$, A_l is of N-type with pair (l, k) and, as the neighbours of C_G in Γ_l have $\delta = l$ by (i), no path from A_l with pair (l, k) reaches C_G; so the attachment path from A_l to D_k keeps the pair (l, k) by rigidity and cannot reach a contact block as in (ii); a direct contact is excluded by $\lambda(Q) = l \neq k$. If it passes through C_G, then $e = k$ (a shortest path between blocks of Γ_l through C_G would have both neighbours of C_G in Γ_l, of b-type or with $\delta = l$ by (i), whereas rigidity from the contact block gives the neighbour on its side the pair (l, k)); a direct contact $A_l \cap D_k$ is then impossible ($A_l \subseteq \Gamma_l$, $D_k \subseteq \Gamma_k$), and an attachment path from $A_l \subseteq \Gamma_l$ to $D_k \subseteq \Gamma_k$ passes through C_G with its neighbour on the side of D_k in Γ_k, of pair (l, k) by (i), which by rigidity cannot reach a contact block as in (ii).

Consequences. Every D_m, $m \in G'$, is attached to some A_i: its class of ρ contains some A_i (Step 5), and on a shortest ρ-path from D_m to A_i the first T-block after D_m is reached through N-blocks of one component ν, which contains an $\mathsf{N}(a)$-block meeting D_m (unless the block following D_m is already a $\mathsf{T}(a)$-block), so by Claim 1 it is a $\mathsf{T}(a)$-block. Hence, by Claim 2, attachment is a partial map $\tau\colon G \rightharpoonup G'$ which is surjective, and $|\pi(G)| \leq |G|$ for every hub G. Since π is a permutation of the hubs (Step 4), $\sum_G |\pi(G)| = \sum_G |G|$, so $|\pi(G)| = |G|$ for every hub, and τ is a bijection $G \to G'$. In particular $|G'| = 2$ when $|G| = 2$.

Claim 3: every component ν touches at most one $\mathsf{T}(a)$-block. Suppose ν touches A_k and A_l, so $|G| = 2 = |G'|$. Then ν contains C_G, and a shortest path in ν from a block meeting A_k to a block meeting A_l passes through C_G; its neighbour of C_G on the side of A_k lies in Γ_k and meets C_k, so it is of N-type with pair (l, k) by (i), and by rigidity A_k is of N-type with pair (l, k) (this includes the case that A_k meets C_G directly, by (i)). *Case $e = k$.* The block $D_{k'}$ is attached to some A_i, and every attachment path to $D_{k'}$ ends at a contact block as in (ii), of pair (k', k') or of b-type, lying in Γ_k. An attachment path from A_k: a direct contact would need $\lambda = l = k'$, impossible ($k' = k$ if $G' = G$, $k' \notin G$ otherwise); otherwise it starts with the pair (l, k), and if

it passes through C_G both neighbours lie in Γ_k and have the pair (l,k); so by rigidity it cannot reach the contact block. A direct contact $A_l \cap D_{k'}$ is impossible ($A_l \subseteq \Gamma_l$), and an attachment path from $A_l \subseteq \Gamma_l$ to $D_{k'} \subseteq \Gamma_k$ passes through C_G with its neighbour on the side of $D_{k'}$ in Γ_k, of pair (l,k), and rigidity gives the same contradiction. So $D_{k'}$ is attached to no A_i, contradicting the consequences of Claims 1 and 2. *Case $e = l$.* The block $D_{l'}$ is attached to some A_i, through a contact block of pair (l',k') in Γ_l (iii). A direct contact $A_k \cap D_{l'}$ is impossible ($A_k \subseteq \Gamma_k$), and an attachment path from $A_k \subseteq \Gamma_k$ passes through C_G, whose neighbour on the side of $D_{l'}$ lies in Γ_l and meets C_l, so by (i) it is of b-type or has the pair (l,l), whereas rigidity from the contact block gives it the pair (l',k'), $l' \neq k'$: impossible. So $D_{l'}$ is attached to A_l, and by rigidity (the neighbours of C_G in Γ_l have $\delta = l$ by (i), and $k' = l$ is impossible: it would give $l \in G'$, $G' = G$, $k' = k$) the attachment path avoids C_G and A_l is of N-type with pair (l',k'); a direct contact $A_l \cap D_{l'} \neq \emptyset$ gives the same. But then A_l is not touched by ν: a block of ν meeting A_l is C_G (excluded, as $\lambda = l'$ and $\lambda = j = l$ would give $l \in G'$, $G' = G$, and then $\delta = k' = k$, so the contact would lie in $C_k \subseteq \Gamma_k$) or a common block of pair (l',k'), from which, by rigidity, no path in ν reaches a neighbour of C_G.

Conclusion. Let R be a class of ρ contained in Γ_G; it contains some A_i (Step 5). We claim that its T-blocks are exactly A_i and $D_{\tau(i)}$. Otherwise take a shortest path in the graph of blocks of a and d from $\{A_i, D_{\tau(i)}\}$ to another T-block; its interior consists of N-blocks. It cannot lead from A_i to a $\mathsf{T}(d)$-block (this would be attached to A_i, hence equal to $D_{\tau(i)}$), nor from A_i to a $\mathsf{T}(a)$-block $A_{i'}$ (Claim 3), nor from $D_{\tau(i)}$ to a $\mathsf{T}(a)$-block $A_{i'}$ ($D_{\tau(i)}$ would be attached to $A_{i'}$, so $i' = i$ as τ is injective), nor from $D_{\tau(i)}$ to a $\mathsf{T}(d)$-block (Claim 1). So every class of ρ in Γ_G contains exactly one $\mathsf{T}(a)$-block and one $\mathsf{T}(d)$-block. □

Theorem 5.12. (Q) *holds in every partition monoid: if $a \bowtie c \bowtie b \bowtie d$, $xa = ax$, $\theta_c(x) = \theta_d(x)$, $ya = ay$, $yb = by$ and $\theta_y(c) = \theta_y(b)$ in $\mathcal{P}_X$, then $a \bowtie d$.*

Proof. By Step 1 we may assume $x = e \leq a$. By Step 5 it suffices that every class of $\rho = \varepsilon_a \vee \varepsilon_d$ contained in a class Γ_G contains exactly one $\mathsf{T}(a)$-block and one $\mathsf{T}(d)$-block, which is Proposition 5.10 for the non-hub groups and Proposition 5.11 for the hubs. □

Corollary 5.13. *In every projection algebra,* (Q) *implies* (P_2) *and* (P_2) *implies* (S). *In particular* (P_2) *and* (S) *hold in every $\mathcal{P}_X$.*

Proof. Putting $y = x$ in (Q) gives exactly (P_2). For (P_2)⇒(S), let $a \bowtie c \bowtie b \bowtie d$, $cd = dc =: g$ and $\theta_a(g) = \theta_b(g)$ be the hypotheses of (S), and apply (P_2) to the reversed path $d \bowtie b \bowtie c \bowtie a$ with the auxiliary g: $g \leq d$ and $g \leq c$, so g commutes with d and with c; $\theta_g(b) = gbg = g\,cbc\,g = gcg = g = \theta_g(c)$, since $g = gc = cg$ and $cbc = c$; and $\theta_b(g) = \theta_a(g)$. The conclusion is $d \bowtie a$. □

Remark 5.14. (1) Of the hypotheses, the one on y enters only through the classification of the blocks of a and b into $\mathsf{N}(y)$-blocks and unions of $\mathsf{T}(y)$-blocks and through the description of the classes of $\varepsilon_y \vee \varepsilon_c$ meeting $\operatorname{supp} y$ (Step 3); the one on x only through Step 2. The identity $\theta_y(d) = \theta_y(b)$ holds in every instance in $\mathcal{P}_X$ but is not a consequence of the hypotheses in projection algebras, not even together with (S) (Mace4 finds countermodels of order 6); it is not used. (2) The hub-free case ($\operatorname{supp} c = \operatorname{supp} d$, Proposition 5.10 alone) already contains the case of (S) in which c and d have the same support; the hubs, with the bound $|G| \leq 2$, the cross wiring of Step 2 and the rigidity of Step 5, are where the argument is delicate, and an earlier proof of (S) handled them by a finite computer check. (3) The conclusion of (Q) is not accompanied by a coherence statement for the matchings: in $\mathsf{P}(\mathcal{P}_4)$ the identity $ad = acbd$ fails in instances of the hypotheses with distinct path elements, and the example of Step 2 has π exchanging two hubs.

5.7. **The condition (K4).**

Theorem 5.15. (K4) *holds in every partition monoid: if a, b, c, y, z are projections of $\mathcal{P}_X$ with $a \bowtie c$, $c \bowtie b$, $aya = a$, $b \leq z$, $yc = cy$, $\theta_z(y) = \theta_b(a)$ and $\theta_z(a) = \theta_a(b)$, then $a \bowtie b$.*

Proof. We use the notation of Section 2.3: $A_i, C_i, B_i, \Gamma_i, \Delta_i$ $(1 \leq i \leq m)$, $u = \theta_a(b) = \theta_z(a)$ and $v = \theta_b(a) = \theta_z(y)$; put $n = \operatorname{rank} u$.

Step 1: algebraic consequences (valid in every projection algebra with (Q1)). $u \leq a$ and $u \leq z$, so $az = za = u$ by (Q1); $v \leq b \leq z$; $\theta_b(y) = \theta_b(\theta_z(y)) = \theta_b(v) = v$; $u \bowtie v$ (Lemma 2.3); $\theta_u(y) = \theta_u(\theta_a(y)) = \theta_u(a) = u$ and $\theta_u(c) = u$ likewise; and $\theta_z(c) = b$. For the last claim put $w = \theta_z(c)$ and $f = \theta_c(y) = \theta_y(c)$. Then $\theta_w(b) = zcz\,b\,zcz = zc(zbz)cz = z(cbc)z = w$, using $b \leq z$ and $cbc = c$, and $\theta_b(w) = \theta_b(\theta_z(c)) = \theta_b(c) = b$, using $\theta_b\theta_z = \theta_b$ (as $b \leq z$); so $b \bowtie w$. Next, the identity $\theta_{\theta_r(p)}(\theta_p(q)) = \theta_r(\theta_p(q))$ of Section 5.6 (Step 1 of (Q)) gives, with $(p, q, r) = (c, y, z)$, $\theta_w(f) = \theta_z(f)$, and with $(p, q, r) = (y, c, b)$ and (y, c, z), $\theta_b(f) = \theta_{\theta_b(y)}(f) = \theta_{\theta_z(y)}(f) = \theta_z(f)$, since $\theta_b(y) = v = \theta_z(y)$. Hence $g := \theta_w(f) = \theta_b(f)$ is a common lower bound of w and b, so $wb = bw$ by (Q1), and then $b = \theta_b(w) = \theta_w(b) = w$. (This derivation was found by Prover9, 25 steps.) Finally $\theta_y(u) = \theta_y(z)$: the identity $\theta_p(\theta_{\theta_q(p)}(x)) = \theta_p(\theta_q(x))$, valid in every projection algebra (it is the axiom $\theta_p\theta_q\theta_p\theta_q = \theta_p\theta_q$ with the inner $\theta_q\theta_p\theta_q$ rewritten as $\theta_{\theta_q(p)}$), gives, with $(p, q) = (y, b)$ and $(p, q) = (y, z)$, $\theta_y(\theta_b(x)) = \theta_y(\theta_v(x)) = \theta_y(\theta_z(x))$ for every x (both $\theta_b(y)$ and $\theta_z(y)$ equal v); with $x = a$ this is $\theta_y(v) = \theta_y(u)$, and $\theta_y(v) = \theta_y(\theta_z(y)) = \theta_y(z)$ by the axiom $\theta_p(\theta_q(p)) = \theta_p(q)$.

Step 2: the blocks of a, b, z. Since $a \sim z$, the classes of $\varepsilon_a \vee \varepsilon_z$ are of the four types of Lemma 2.6(b) and $\varepsilon_u = \varepsilon_a \vee \varepsilon_z$: the $\mathsf{T}(u)$-blocks are the classes $K_1, \ldots, K_n$ of type (i), each a union of $\mathsf{T}(a)$-blocks and of $\mathsf{T}(z)$-blocks; the $\mathsf{N}(u)$-blocks are the $\mathsf{N}(a)$-blocks inside $\operatorname{supp} z$ (type (iii), unions of $\mathsf{T}(z)$-blocks), the $\mathsf{N}(z)$-blocks $Z^{(1)}, \ldots, Z^{(t)}$ that are unions of $\mathsf{T}(a)$-blocks (type (ii)), and the common N-blocks of a and z (type (iv)). We call $K_1, \ldots, K_n, Z^{(1)}, \ldots, Z^{(t)}$ the *generalised K's*; every $\mathsf{T}(a)$-block lies in exactly one of them. Every block of a lies inside one block of z (the $\mathsf{N}(a)$-blocks outside $\operatorname{supp} z$, which we call *outer*, and the $\mathsf{T}(a)$-blocks inside some $Z^{(k)}$) or inside $\operatorname{supp} z$. Since $b \leq z$, every block of b is an $\mathsf{N}(z)$-block or a union of $\mathsf{T}(z)$-blocks, $\varepsilon_z \leq \varepsilon_b$, and $B_i \subseteq \operatorname{supp} z$. By $u \bowtie v$, every class of $\varepsilon_u \vee \varepsilon_v = \varepsilon_a \vee \varepsilon_b$ meeting $\operatorname{supp} u \cup \operatorname{supp} v$ contains exactly one K_j and exactly one $\mathsf{T}(v)$-block; write D_j for the class containing K_j, $S_j = D_j \cap \operatorname{supp} b$ for its $\mathsf{T}(v)$-block (a union of $\mathsf{T}(b)$-blocks), and $I_j = \{i : A_i \subseteq K_j\}$. By Lemma 2.6(a) for $v = \theta_b(a)$, the blocks of v inside $\operatorname{supp} b$ are the sets $D \cap \operatorname{supp} b$ for the classes D of $\varepsilon_a \vee \varepsilon_b$ meeting $\operatorname{supp} b$: the S_j, and the $\mathsf{N}(v)$-blocks $D \cap \operatorname{supp} b$ for the classes D meeting $\operatorname{supp} b$ but not $\operatorname{supp} a$; the other $\mathsf{N}(v)$-blocks are the $\mathsf{N}(b)$-blocks. In particular every block of v inside $\operatorname{supp} z$ lies in a single class of $\varepsilon_a \vee \varepsilon_b$, which we denote $D(\cdot)$; and a $Z^{(k)}$ is a block of b which is a union of blocks of a, hence a class of $\varepsilon_a \vee \varepsilon_b$ by itself.

Step 3: the blocks of c and y relative to z. By $\theta_z(c) = b$ and Lemma 2.6(a), the blocks of b inside $\operatorname{supp} z$ are the sets $K \cap \operatorname{supp} z$ for the classes K of $\varepsilon_z \vee \varepsilon_c$ meeting $\operatorname{supp} z$; hence $\varepsilon_b \leq \varepsilon_z \vee \varepsilon_c$, so $\varepsilon_z \vee \varepsilon_c = \varepsilon_b \vee \varepsilon_c$ and $\Delta_i \cap \operatorname{supp} z = B_i$: every block of c meets $\operatorname{supp} z$ inside a single block of b, and the only $\mathsf{T}(z)$-blocks in Δ_i are those of B_i. By $\theta_z(y) = v$ and Lemma 2.6(a), every class K of $\varepsilon_z \vee \varepsilon_y$ meeting $\operatorname{supp} z$ meets it in exactly one block of v, which is a $\mathsf{T}(v)$-block S_j if K meets $\operatorname{supp} y$ and an $\mathsf{N}(v)$-block otherwise. We call a class of $\varepsilon_z \vee \varepsilon_y$ *bright* if it contains a $\mathsf{T}(y)$-block and *dark* otherwise; for a dark class K meeting $\operatorname{supp} z$ put $D(K) = D(K \cap \operatorname{supp} z)$. Since every block of b inside $\operatorname{supp} z$ lies in a block of v, $\varepsilon_b \leq \varepsilon_z \vee \varepsilon_y$ and $\varepsilon_z \vee \varepsilon_y = \varepsilon_b \vee \varepsilon_y$. By $yc = cy$ and Lemma 2.6(b), a block of c of type (iv) is a block of y and one of type (iii) lies in an $\mathsf{N}(y)$-block, so these lie in a single class of $\varepsilon_z \vee \varepsilon_y$; a block of c of type (ii) is a union of $\mathsf{T}(y)$-blocks and one of type (i) lies in $\operatorname{supp} y$, so these lie in bright classes. Hence *a block of c meeting a dark class lies inside it.* Also, every block of y meeting an $\mathsf{N}(c)$-block lies inside it (types (ii), (iv)), and a block of y meeting no $\mathsf{N}(c)$-block is contained in $\bigcup_i C_i$ (types (i), (iii)).

Step 4: two counts. Put $\Omega = \varepsilon_u \vee \varepsilon_c = \varepsilon_a \vee \varepsilon_b \vee \varepsilon_c$ (by Step 3 and $\varepsilon_z \leq \varepsilon_b$). Lemma 2.9 for the pair (a, c), which a identifies, and $u \leq a$ gives $\delta'_{ac}(u) = \delta'_{ac}(a) = 0$, i.e. $|\varepsilon_u|_{\operatorname{supp} a} = |\Omega|_{\operatorname{supp} c}$; the left side is $n + t$. By $\theta_u(c) = u$ every class of Ω meeting $\operatorname{supp} u$ contains exactly one K_j, and it meets $\operatorname{supp} c$; so exactly t classes of Ω meet $\operatorname{supp} c$ without meeting $\operatorname{supp} u$, and each of them contains some C_i, hence A_i, hence a $Z^{(k)}$ (as $A_i \not\subseteq \operatorname{supp} u$). Since there are t blocks $Z^{(k)}$, *every class of* Ω *meeting* $\operatorname{supp} c$ *contains exactly one generalised* K*, and a class containing some* $Z^{(k)}$ *contains no* K_j. Next, by $aya = a$ the classes of $\varepsilon_a \vee \varepsilon_y$ meeting $\operatorname{supp} a$ are $E_1, \ldots, E_m$ with $E_i \cap \operatorname{supp} a = A_i$ and each E_i meets $\operatorname{supp} y$. Lemma 2.9 for $u \leq a$ with the pairs (a, c) and (y, c) (both identified by a, as $\theta_a(y) = a = \theta_a(c)$) gives $|\varepsilon_u| - |\Omega| = |\varepsilon_a| - |\varepsilon_a \vee \varepsilon_c|$ and $|M| - |\Omega| = |\varepsilon_a \vee \varepsilon_y| - |\varepsilon_a \vee \varepsilon_c|$, where $M := \varepsilon_u \vee \varepsilon_y = \varepsilon_a \vee \varepsilon_z \vee \varepsilon_y$; hence $|\varepsilon_a \vee \varepsilon_y| - |M| = |\varepsilon_a| - |\varepsilon_u| = m - n - t$. The classes E_i whose $\mathsf{T}(a)$-block lies in a common generalised K lie in a common class of M (a generalised K is a block of z or a union of blocks of a and z in one class of $\varepsilon_a \vee \varepsilon_z$), so the m classes E_i fall into at most $n + t$ classes of M, which already accounts for a drop of at least $m - n - t$; hence they fall into exactly $n + t$ classes and no other class is merged: writing M_j for the class of M containing K_j and $M(Z^{(k)})$ for the one containing $Z^{(k)}$, these are $n + t$ distinct classes, $M_j = \bigcup_{i \in I_j} E_i$ and $M(Z^{(k)}) = \bigcup_{A_i \subseteq Z^{(k)}} E_i$. Since $\varepsilon_b \leq \varepsilon_z \vee \varepsilon_y$, $D_j \subseteq M_j$; in particular a block of a inside $\operatorname{supp} z$ meeting some S_j lies in $D_j \subseteq M_j$.

Step 5: $t = 0$*, and every* B_i *lies in some* S_j. Suppose that $A_1 \subseteq Z := Z^{(1)}$ (case I), or that $t = 0$ and B_1 lies in an $\mathsf{N}(v)$-block (case II; then $A_1 \subseteq K_j$ for some j). Let O be the class of Ω containing C_1; it contains Γ_1 and Δ_1, and by Step 4 it contains Z and no $K_{j'}$ in case I, and K_j and no other generalised K in case II.

(A) $D(B_1)$ *does not meet* $\operatorname{supp} a$*, the class* K' *of* $\varepsilon_z \vee \varepsilon_y$ *containing* B_1 *is dark,* $D(K') = D(B_1)$*, and* $\Delta_1 \subseteq K'$. In case II the first assertion is the definition of an $\mathsf{N}(v)$-block. In case I, if $D(B_1)$ contained some A_i, then either $A_i \subseteq K_{j'}$, so $D(B_1) = D_{j'}$, $D_{j'} \subseteq O$ and $K_{j'} \subseteq O$, which is impossible, or $A_i \subseteq Z^{(k)}$, so $D(B_1) = Z^{(k)}$ (Step 2), which does not contain the $\mathsf{T}(b)$-block B_1. So B_1 lies in the $\mathsf{N}(v)$-block $D(B_1) \cap \operatorname{supp} b$, the class K' meets $\operatorname{supp} z$ in that $\mathsf{N}(v)$-block, hence is dark (Step 3), and $D(K') = D(B_1)$. Finally $\Delta_1 \subseteq K'$: every block of b lies in one class of $\varepsilon_z \vee \varepsilon_y$ and every block of c meeting the dark class K' lies inside it (Step 3), so a chain from B_1 in Δ_1 never leaves K'. In particular $C_1 \subseteq K'$.

(B) *In case I, the class of* $\varepsilon_z \vee \varepsilon_y$ *containing* Z *is bright.* E_1 contains a $\mathsf{T}(y)$-block H. Follow a chain $A_1 = P_0, Y_1, P_1, \ldots, Y_r = H$ in E_1 (P's blocks of a, Y's blocks of y) point by point and track the class of $\varepsilon_z \vee \varepsilon_y$ of the current point. A step inside a block of y, or inside a block of a lying in one block of z, keeps the class. The remaining blocks of a on the chain are $\mathsf{N}(a)$-blocks $P \subseteq \operatorname{supp} z$ ($E_1 \cap \operatorname{supp} a = A_1 \not\subseteq \operatorname{supp} z$); a step inside such a P joins two points of $\operatorname{supp} z$, neither of which lies in a bright class: a bright class meets $\operatorname{supp} z$ in some $S_{j'}$, and then $P \subseteq D_{j'} \subseteq M_{j'}$ (Step 4), while $P \subseteq E_1 \subseteq M(Z) \neq M_{j'}$. So the class changes only from a dark class to a dark class; since the chain ends in a bright class (at H) it starts in one, and the start is the class of $Z \supseteq A_1$.

(C) *The contradiction.* Take a chain $A_1 = P_0, F_1, P_1, \ldots, F_\ell, P_\ell$ in Γ_1 with $F_\ell = C_1$ (P's blocks of a, F's blocks of c, all inside O) and follow it point by point, tracking the class $K(x)$ of $\varepsilon_z \vee \varepsilon_y$ of the current point x. We claim that throughout, $K(x)$ *is bright, or dark and disjoint from* $\operatorname{supp} z$*, or dark, meeting* $\operatorname{supp} z$*, with* $D(K(x)) = D_j$ (the third alternative is void in case I). At the start this holds by (B) in case I, and in case II because $x \in A_1 \subseteq K_j \subseteq D_j \cap \operatorname{supp} z$, so if $K(x)$ is dark then $K(x) \cap \operatorname{supp} z$ is the block of v containing x and $D(K(x)) = D_j$. A step inside a block of c preserves the claim by Step 3 (the class does not change, or both points lie in bright classes). A step inside a block of a lying in one block of z keeps the class. For a step inside a block of a, $P \subseteq \operatorname{supp} z$: the current point x lies in $\operatorname{supp} z$, so $K(x)$ is not of the second kind; if $K(x)$ is bright, x lies in some $S_{j'}$, so $P \subseteq D_{j'} \subseteq O$ and $K_{j'} \subseteq O$, impossible in case I

and forcing $j' = j$ in case II; if $K(x)$ is dark, $x \in K(x) \cap \operatorname{supp} z \subseteq D(K(x)) = D_j$. In both cases $P \subseteq D_j$, so the new point x' lies in $D_j \cap \operatorname{supp} z$ and $K(x')$ is bright or satisfies $D(K(x')) = D_j$. This proves the claim. But the chain ends in $C_1 \subseteq K'$, a dark class meeting $\operatorname{supp} z$ (in B_1) with $D(K') = D(B_1)$, which does not meet $\operatorname{supp} a$ (by (A)) while D_j does; a contradiction in both cases.

Consequences. $t = 0$, so $\operatorname{supp} a \subseteq \operatorname{supp} z$ and $\operatorname{supp} u = \operatorname{supp} a$; the $\mathsf{N}(z)$-blocks are exactly the common N-blocks of a, b and z (the outer blocks); every class of $\varepsilon_a \vee \varepsilon_b$ meeting $\operatorname{supp} a$ or $\operatorname{supp} b$ is some D_j; an outer block is a class of $\varepsilon_a \vee \varepsilon_b$ by itself, so $D_j \subseteq \operatorname{supp} z$; every class of $\varepsilon_z \vee \varepsilon_y$ meeting $\operatorname{supp} z$ meets it in some S_j (bright; write K'_j for this class) or in an $\mathsf{N}(b)$-block inside $\operatorname{supp} z$ (dark); $A_i \subseteq K_j$ iff $B_i \subseteq S_j$ (both $D(A_i)$ and $D(B_i)$ lie in the class of Ω containing C_i, which contains one K_j); the classes of M meeting $\operatorname{supp} a$ are $M_1, \dots, M_n$ with $M_j = \bigcup_{i \in I_j} E_i$ and $M_j \cap \operatorname{supp} a = K_j$; and $K'_j \subseteq M_j$. Finally $\theta_y(u) = \theta_y(z)$ (Step 1) says, by Lemma 2.6(a), that the sets $M_j \cap \operatorname{supp} y$ and the sets $K'_j \cap \operatorname{supp} y$ are the same family of blocks; as $K'_j \subseteq M_j$, we get $K'_j \cap \operatorname{supp} y = M_j \cap \operatorname{supp} y$: *every $\mathsf{T}(y)$-block inside M_j lies in K'_j.*

Step 6: labels. Suppose, for a contradiction, that $a \not\bowtie b$. As $u \leq a$ has rank n, this means $n < m$, so some I_j has $|I_j| \geq 2$; fix such a j. Every point x of K'_j lies in M_j, so its block of a lies in a unique E_i, $i \in I_j$; put $\sigma(x) = i$ (the *side* of x; the block of y containing x lies in the same E_i). Let $\gamma(x) \in [m] \cup \{\emptyset\}$ be the index i such that the block of c containing x lies in Γ_i ($\emptyset$ if it lies in a class of $\varepsilon_a \vee \varepsilon_c$ without T-blocks); the block of a containing x lies in the same class, so γ is constant on blocks of a and of c. Let $\lambda(x)$ be the index k such that the block of c containing x lies in Δ_k ($\emptyset$ if in a T-less class of $\varepsilon_b \vee \varepsilon_c$); the block of b containing x lies in the same class, so λ is constant on blocks of b and of c, and $\lambda(x) = k$ for $x \in B_k$. The class K'_j is connected by chains alternating blocks of z and blocks of y; we record how the label $(\sigma, \gamma, \lambda)$ can vary inside a block of z or of y lying in K'_j. An outer block is a block of a and of b, so all three labels are constant on it. A $\mathsf{T}(z)$-block $Z \subseteq S_j$ lies in one block of b, hence λ is constant; it lies in a class of $\varepsilon_a \vee \varepsilon_z$ of type (i) or (iii); in type (iii) it lies in one $\mathsf{N}(a)$-block and σ, γ are constant; in type (i) it lies in K_j, its points lie in the $\mathsf{T}(a)$-blocks A_i, $i \in I_j$, and a point of A_i has $\sigma = \gamma = i$: so the labels on Z are of the form (i, i, k) with k fixed. A block Y of y has constant σ; if Y lies in an $\mathsf{N}(c)$-block then γ and λ are constant on Y; otherwise $Y \subseteq \bigcup C_i$, and a point of C_i has $\gamma = \lambda = i$: the labels on Y are of the form (s, i, i) with s fixed. Hence the labels occurring in K'_j form a connected set in the graph $\mathcal{L}$ on $[m] \times ([m] \cup \{\emptyset\})^2$ whose edges join (i, i, k) to (i', i', k) and (s, i, i) to (s, i', i'). The connected components of $\mathcal{L}$ are the sets $\mathcal{C}_s = \{(s, i, i) : i \in [m]\} \cup \{(i, i, s) : i \in [m]\}$, $s \in [m]$, the set $\{(i, i, \emptyset) : i \in [m]\}$, and single vertices. Since K'_j contains B_i and $B_{i'}$ for distinct $i, i' \in I_j$, with labels $(\cdot, \cdot, i)$ and $(\cdot, \cdot, i')$, its labels lie in one component $\mathcal{C}_s$, for a unique s; as s is the side of some point, $s \in I_j$; and no point of K'_j has $\gamma = \emptyset$ or $\lambda = \emptyset$.

Step 7: the contradiction. Take $i \in I_j$, $i \neq s$. A point $x \in B_i$ has $\lambda(x) = i \neq s$, so its label in $\mathcal{C}_s$ is (s, i, i): *every point of B_i has side s and its block of c lies in Γ_i.* Choose $x \in B_i$ and let P be its block of a; then $P \subseteq \Gamma_i$, $P \subseteq \operatorname{supp} z$ and $P \subseteq E_s$. For a block Q of a write $\hat{\sigma}(Q)$ for its class of $\varepsilon_a \vee \varepsilon_y$ (an E_k or a class without $\mathsf{T}(a)$-blocks). Take a chain $A_i = P_0, F_1, P_1, \dots, F_\ell, P_\ell = P$ in Γ_i (P_k blocks of a, F_k blocks of c, consecutive ones meeting). We show by induction on k that either $\hat{\sigma}(P_k) = E_i$, or P_k is an outer block contained in Δ_s; since $P_\ell = P$ lies in $\operatorname{supp} z$ and in $E_s \neq E_i$, this is the desired contradiction. First let $\hat{\sigma}(P_{k-1}) = E_i$ and $x' \in P_{k-1} \cap F_k$, $x'' \in P_k \cap F_k$. If F_k is a single block of y (type (iv)), $\hat{\sigma}(P_k) = E_i$. If F_k is an $\mathsf{N}(c)$-block of type (ii), x' lies in a $\mathsf{T}(y)$-block $H \subseteq F_k$ with $H \subseteq E_i \subseteq M_j$, so $H \subseteq K'_j$ (Step 5) and x' has label $(i, i, \lambda(F_k)) \in \mathcal{C}_s$; as $i \neq s$ this forces $\lambda(F_k) = s$. If $\hat{\sigma}(P_k) = E_{i''}$ with $i'' \in I_j$, the same argument at x'' gives the label $(i'', i, s) \in \mathcal{C}_s$, which forces $i'' = i$. Otherwise $P_k \not\subseteq M_j$; but $F_k \cap \operatorname{supp} z \subseteq B_s \subseteq K'_j \subseteq M_j$,

so $P_k \cap \operatorname{supp} z = \emptyset$, i.e. P_k is an outer block (Step 5), hence a block of b meeting $F_k \subseteq \Delta_s$, so $P_k \subseteq \Delta_s$. If $F_k = C_i$ then, as $C_i \subseteq \Delta_i$ and $i \neq s$, the point x' cannot lie in a $\mathsf{T}(y)$-block of K'_j (its label would be $(i,i,i) \notin \mathcal{C}_s$), so C_i lies in an $\mathsf{N}(y)$-block (type (iii)), which is a single block of y, and $\hat{\sigma}(P_k) = E_i$. Now let P_{k-1} be an outer block contained in Δ_s; then $F_k \subseteq \Delta_s$. If P_k is outer, it lies in Δ_s. Otherwise $P_k \subseteq \operatorname{supp} z$ meets $F_k \cap \operatorname{supp} z \subseteq B_s \subseteq K'_j$ at a point with label $(\hat{\sigma}(P_k), i, s) \in \mathcal{C}_s$ (its γ-label is i because $F_k \subseteq \Gamma_i$), which forces $\hat{\sigma}(P_k) = E_i$. This completes the induction. Hence $|I_j| = 1$ for every j, $n = m$, $u = a$, $v = b$ and $a \bowtie b$. □

Remark 5.16. Each hypothesis on y is used where a computation says it is needed (dropping any one hypothesis produces counterexamples in $\mathsf{P}(\mathcal{P}_4)$): $yc = cy$ confines the blocks of y to single blocks of c or to $\bigcup C_i$ (Step 3), $aya = a$ gives the sides, $\theta_z(y) = \theta_b(a)$ forces the connection of B_i with $B_{i'}$ inside K'_j and the bright/dark dichotomy of Step 3, and $\theta_z(a) = \theta_a(b)$ gives $\theta_y(u) = \theta_y(z)$, i.e. that the connection carries all $\mathsf{T}(y)$-blocks of M_j. The proofs of this section were checked by the *dropped-hypothesis test*: for each intermediate statement of a proof and each hypothesis h, the statement is tested on instances in $\mathsf{P}(\mathcal{P}_m)$ satisfying all hypotheses but h; a statement that fails when h is dropped cannot be proved by an argument that does not use h. The test exposed a gap in a first version of Step 5. The special role of the side s is curious: the connection of the B_i through y and z can only be made from one side, and that side cannot see its own C's.

5.8. **The condition (K0).**

Theorem 5.17. *Let a, b, c, y, z be projections of $\mathcal{P}_X$ with $a \bowtie c$, $c \bowtie b$, $yc = cy$, $\theta_y(b) = \theta_b(a)$, $\theta_b(z) = b$, $\theta_z(a) = \theta_z(b)$ and $\theta_y(z) = \theta_a(b)$. Then $a \bowtie b$. The same holds with the hypothesis $\theta_z(a) = \theta_z(b)$ replaced by $\theta_a(z) = a$.*

The second statement is a different quasi-identity, which we denote (K0)″: it is not a consequence of (K0) and the other conditions of this paper in projection algebras (a projection algebra of order 9 separates them, Section 8), while conversely (K0)″ implies (K0) in every finite projection algebra, by finite descent (Corollary 5.18); the block proof below gives both at once.

Proof. We use the notation of Section 2.3: A_i, C_i, B_i are the T-blocks of a, c, b indexed through c, $\Gamma_i \ni A_i, C_i$ and $\Delta_i \ni C_i, B_i$ are the corresponding classes, $u = \theta_a(b)$, $v = \theta_b(a)$, and $m = \operatorname{rank} a$, $n = \operatorname{rank} u$. Every block of c meeting A_i lies in Γ_i, every block of b meeting C_i lies in Δ_i, and no T-block of one of a, b, c is an N-block of a matched partner (Lemma 2.6(c)). By Lemma 2.6(a), $a \bowtie b$ iff $n = m$ and every class of $\varepsilon_a \vee \varepsilon_b$ meeting $\operatorname{supp} a$ meets $\operatorname{supp} b$.

Step 1: algebra, and the cells. From $\theta_y(u) = \theta_y(\theta_y(z)) = \theta_y(z) = u$ and $\theta_y(v) = \theta_y(\theta_y(b)) = v$ we get $u \leq y$ and $v \leq y$; as $u \leq a$ and $v \leq b$, (Q1) gives $ay = ya$ and $by = yb$, so $\theta_y(a) = ya$ and $\theta_y(b) = yb = v$. Put $p = \theta_z(a) = \theta_z(b)$. Then $b = \theta_b(z) \bowtie \theta_z(b) = p$ (Lemma 2.3), so $\operatorname{rank} p = m$. The T-blocks of $p = \theta_z(a)$ are the sets $L \cap \operatorname{supp} z$ for the classes L of $\varepsilon_a \vee \varepsilon_z$ meeting both $\operatorname{supp} a$ and $\operatorname{supp} z$; there are m of them, each contains a $\mathsf{T}(a)$-block, and there are m $\mathsf{T}(a)$-blocks; so every class of $\varepsilon_a \vee \varepsilon_z$ meeting $\operatorname{supp} a$ meets $\operatorname{supp} z$ and contains exactly one $\mathsf{T}(a)$-block, which by Lemma 2.6(a) says $\theta_a(z) = a$. (Under the hypotheses of (K0)″ this is given, and from here on the two conditions are treated together; only $\theta_a(z) = a$ and $\theta_b(z) = b$ will be used.) Now $(ya)z(ya) = y(aza)y = yay = ya$ and $(ya)z(ya) = a(yzy)a = aua = u$, so $\theta_y(a) = ya = u$.

Let L_i be the class of $\varepsilon_a \vee \varepsilon_z$ containing A_i and L'_i the class of $\varepsilon_b \vee \varepsilon_z$ containing B_i; the L_i are distinct ($\theta_a(z) = a$) and so are the L'_i ($\theta_b(z) = b$). A *cell* is a nonempty set $L \cap L'$ with L a class of $\varepsilon_a \vee \varepsilon_z$ and L' a class of $\varepsilon_b \vee \varepsilon_z$; the cells partition X, every block of z lies in a cell, every block of a lies in one class of $\varepsilon_a \vee \varepsilon_z$, and every block of b in one class of $\varepsilon_b \vee \varepsilon_z$. The *diagonal* cells are $D_i = L_i \cap L'_i$ (possibly empty).

Step 2: y against a, b, c. Lemma 2.6(b) applies to the commuting pairs (y,a), (y,b), (y,c). We claim that for (y,a) and (y,b) there is no class of type (iii): *every $\mathsf{T}(a)$-block and every $\mathsf{T}(b)$-block meets* $\operatorname{supp} y$. Let N be an $\mathsf{N}(y)$-block which is a union of $\mathsf{T}(a)$-blocks, $N \supseteq A_i$. As an $\mathsf{N}(y)$-block, N is a nontransversal block of $\theta_y(b) = v = \theta_b(a)$, i.e. an $\mathsf{N}(b)$-block or a set $M \cap \operatorname{supp} b$ for a class M of $\varepsilon_a \vee \varepsilon_b$ missing $\operatorname{supp} a$; the latter is impossible as $N \supseteq A_i$, so N is an $\mathsf{N}(b)$-block. For (y,c), N is of type (iii) or (iv). If N is an $\mathsf{N}(c)$-block, it meets A_i, so $N \subseteq \Gamma_i$, and $N \subseteq \operatorname{supp} a$ gives $N = A_i$, a $\mathsf{T}(a)$-block that is an $\mathsf{N}(c)$-block, impossible. So N is a union of $\mathsf{T}(c)$-blocks; being a block of b it lies in one class Δ_l, which contains the single $\mathsf{T}(c)$-block C_l, so $N = C_l$ is a $\mathsf{T}(c)$-block that is an $\mathsf{N}(b)$-block, impossible. Next let N be an $\mathsf{N}(y)$-block which is a union of $\mathsf{T}(b)$-blocks, $N \supseteq B_k$. It is a nontransversal block of $v = \theta_b(a)$ which is not an $\mathsf{N}(b)$-block, so $N = M \cap \operatorname{supp} b$ for a class M of $\varepsilon_a \vee \varepsilon_b$ missing $\operatorname{supp} a$; thus N contains no $\mathsf{T}(a)$-block, and by what was just proved N is an $\mathsf{N}(a)$-block. For (y,c), if N were an $\mathsf{N}(c)$-block then $N \subseteq \Delta_k$ and $N = B_k$, impossible; so N is a union of $\mathsf{T}(c)$-blocks. Every $\mathsf{T}(b)$- and every $\mathsf{T}(c)$-block lies in some Δ_l, so $N = \bigsqcup_l (N \cap \Delta_l)$, where $N \cap \Delta_l$ is both a union of $\mathsf{T}(b)$-blocks (hence B_l or empty) and a union of $\mathsf{T}(c)$-blocks (hence C_l or empty): N is a disjoint union of blocks $B_l = C_l$. Being a block of a, N lies in one Γ_l, so $N = B_k = C_k$ is an $\mathsf{N}(a)$-block which is a $\mathsf{T}(c)$-block, impossible.

Consequences. Every $\mathsf{N}(y)$-block is an $\mathsf{N}(a)$-block and an $\mathsf{N}(b)$-block (type (iv) for (y,a) and (y,b)); every $\mathsf{T}(y)$-block Y lies in a class of type (i) or (ii), so Y lies in a union of $\mathsf{T}(a)$-blocks or in an $\mathsf{N}(a)$-block, in a union of $\mathsf{T}(b)$-blocks or in an $\mathsf{N}(b)$-block, and in a union of $\mathsf{T}(c)$-blocks or in an $\mathsf{N}(c)$-block; and every N-block of a, b or c meeting a $\mathsf{T}(y)$-block contains it (types (ii) and (iv)). The T-blocks of $u = ya$ are the classes $U_1, \dots, U_n$ of type (i) of $\varepsilon_y \vee \varepsilon_a$, each a union of $\mathsf{T}(a)$-blocks and of $\mathsf{T}(y)$-blocks, and its N-blocks are the $\mathsf{N}(a)$-blocks; comparing with the description of $u = \theta_a(b)$ (Section 2.3), the U_j are the sets $M_j \cap \operatorname{supp} a$ for the classes $M_1, \dots, M_n$ of $\varepsilon_a \vee \varepsilon_b$ meeting $\operatorname{supp} a$, and every M_j meets $\operatorname{supp} b$. Put $I_j = \{i : A_i \subseteq U_j\}$; the I_j partition $[m]$, and $a \bowtie b$ iff $n = m$ iff every $|I_j| = 1$.

Step 3: the classes of $\varepsilon_y \vee \varepsilon_z$. By Lemma 2.6(a) the blocks of $\theta_y(z)$ inside $\operatorname{supp} y$ are the sets $K \cap \operatorname{supp} y$ for the classes K of $\varepsilon_y \vee \varepsilon_z$ meeting $\operatorname{supp} y$, transversal iff K meets $\operatorname{supp} z$; the blocks of $u = ya$ inside $\operatorname{supp} y$ are the U_j (transversal) and the $\mathsf{N}(a)$-blocks contained in $\operatorname{supp} y$ (nontransversal; type (ii)). As $\theta_y(z) = u$, the classes of $\varepsilon_y \vee \varepsilon_z$ meeting $\operatorname{supp} y$ are $K_1, \dots, K_n$ with $K_j \cap \operatorname{supp} y = U_j$ and $K_j \cap \operatorname{supp} z \neq \emptyset$, and, for each $\mathsf{N}(a)$-block $F \subseteq \operatorname{supp} y$, a class K_F with $K_F \cap \operatorname{supp} y = F$. In particular each A_i lies in K_j for $i \in I_j$, and each $\mathsf{N}(a)$-block is an $\mathsf{N}(y)$-block or lies in its K_F.

Step 4: blocks of y and cells. Call a $\mathsf{T}(y)$-block *crossing* if it meets two $\mathsf{T}(a)$-blocks.

(a) *A crossing block Y satisfies $Y \subseteq \bigcup_i (A_i \cap B_i \cap C_i)$, the union over the i with $Y \cap A_i \neq \emptyset$; in particular Y lies in the union of the diagonal cells.* Y meets $\operatorname{supp} a$, so it lies in a type (i) class U_j and $Y \subseteq \operatorname{supp} a$. Y is not inside an $\mathsf{N}(c)$-block F': such an F' would meet A_i and $A_{i'}$ for two $i \neq i'$, so $F' \subseteq \Gamma_i \cap \Gamma_{i'}$. So $Y \subseteq \bigcup_k C_k$; a point of $Y \cap A_i$ lies in some $C_k \subseteq \Gamma_k$ and in $A_i \subseteq \Gamma_i$, so $k = i$: $Y \cap A_i \subseteq C_i$. A point $x \in Y \cap C_i$ lies in a block of b inside Δ_i, namely B_i or an $\mathsf{N}(b)$-block $G \subseteq \Delta_i$; in the second case $G \supseteq Y$, but $Y \cap A_{i'} \subseteq C_{i'} \subseteq \Delta_{i'}$ for a second index i'. So $Y \cap A_i \subseteq A_i \cap B_i \cap C_i$, and a point of $A_i \cap B_i$ lies in $L_i \cap L_i' = D_i$.

(b) *A non-crossing $\mathsf{T}(y)$-block Y lies in one cell.* If Y meets no $\mathsf{T}(a)$-block then Y lies in an $\mathsf{N}(a)$-block, and if it meets exactly one, A_i, then it lies in $U_j \subseteq \operatorname{supp} a$ and $Y \subseteq A_i$; either way Y lies in one block of a, hence in one class of $\varepsilon_a \vee \varepsilon_z$ and in one class of $\varepsilon_a \vee \varepsilon_c$. For $\varepsilon_b \vee \varepsilon_z$: if Y lies in an $\mathsf{N}(b)$-block we are done; otherwise $Y \subseteq \bigcup_k B_k$, and we claim that Y meets only one B_k. Suppose Y meets B_k and B_l, $k \neq l$. Y is not inside an $\mathsf{N}(c)$-block (it would lie in $\Delta_k \cap \Delta_l$), so $Y \subseteq \bigcup C$'s, and as in (a) $Y \cap B_k \subseteq C_k \subseteq \Gamma_k$, $Y \cap B_l \subseteq C_l \subseteq \Gamma_l$, contradicting the fact that Y lies in one class of $\varepsilon_a \vee \varepsilon_c$. So $Y \subseteq B_k \subseteq L_k'$.

(c) *Every* $\mathsf{N}(y)$*-block and every block of* z *lies in one cell*: an $\mathsf{N}(y)$-block is a block of a and of b (Step 2).

(d) *Every class* K *of* $\varepsilon_y \vee \varepsilon_z$ *containing no crossing block lies in one cell, and every class containing a crossing block lies in* $\bigcup_i D_i$. K is connected by chains of blocks of y and z. By (b), (c) a non-crossing block lies in one cell, so it does not leave the cell of the block before it; by (a) a crossing block lies in $\bigcup D_i$, and a non-crossing block meeting it lies in some D_i. Starting the chains at a crossing block, resp. noting that without crossing blocks no chain changes cell, gives the claim.

Step 5: $\varepsilon_a \le \varepsilon_b \vee \varepsilon_z$. Let $A_i \subseteq U_j \subseteq K_j$. If K_j contains no crossing block, A_i lies in one cell. Otherwise $K_j \subseteq \bigcup_{i'} D_{i'}$, and $A_i \subseteq L_i$ with $D_{i'} \subseteq L_{i'}$ and the L's distinct, so $A_i \subseteq D_i \subseteq L'_i$. An $\mathsf{N}(a)$-block $F \subseteq \operatorname{supp} y$ lies in K_F, which does not meet $\operatorname{supp} a$ and so contains no crossing block; so F lies in one cell. An $\mathsf{N}(a)$-block not inside $\operatorname{supp} y$ is an $\mathsf{N}(y)$-block, in one cell by Step 4(c). So every block of a lies in one class of $\varepsilon_b \vee \varepsilon_z$, and therefore $\varepsilon_a \vee \varepsilon_b \le \varepsilon_b \vee \varepsilon_z$.

Step 6: the contradiction. Suppose $n < m$, so that some I_j contains two indices, say $1, 2 \in I_j$. The class U_j of type (i) is connected by chains alternating $\mathsf{T}(a)$- and $\mathsf{T}(y)$-blocks, so some $\mathsf{T}(y)$-block of U_j meets two $\mathsf{T}(a)$-blocks: K_j contains a crossing block, and by Step 5, $A_1 \subseteq L'_1$ and $A_2 \subseteq L'_2$, two distinct classes of $\varepsilon_b \vee \varepsilon_z$. But $A_1, A_2 \subseteq M_j$, a single class of $\varepsilon_a \vee \varepsilon_b$, which by Step 5 lies in one class of $\varepsilon_b \vee \varepsilon_z$. Hence $n = m$, every class of $\varepsilon_a \vee \varepsilon_b$ meeting $\operatorname{supp} a$ meets $\operatorname{supp} b$ (Step 2), and $a \bowtie b$. □

Corollary 5.18. *In every finite projection algebra,* (K0)″ *implies* (K0).

Proof. Assume the hypotheses of (K0) and put $p = \theta_z(a) = \theta_z(b)$. Then $b = \theta_b(z) \bowtie \theta_z(b) = p$ by Lemma 2.3, and $\theta_p(a) = \theta_{\theta_z(a)}(a) = \theta_z(a) = p$; so $a \bowtie c \bowtie b \bowtie p$ and Lemma 2.10(c) gives $\theta_a(p) = a$, i.e. $\theta_a(\theta_z(a)) = \theta_a(z) = a$. Now (K0)″ applies. □

Remark 5.19. Corollary 5.18 explains an experiment. Mace4 finds a projection algebra of order 9 satisfying (Q1) and (K0) but not (K0)″ (so (K0) does not imply (K0)″), but no finite model of (Q1)+(K0)″ + ¬(K0) at any size, while Prover9 does not derive (K0) from (Q1) and (K0)″: the implication holds in every finite projection algebra by descent, and we do not know whether it holds in general (Question 9.3). The derivation of (E_1) from (K0) is discussed in Remark 5.23.

5.9. **The witnesses and the classification up to order 8.** The witnesses of the wiring conditions were found by enumerating small projection algebras and searching for embeddings into $\mathsf{P}(\mathcal{P}_m)$ (Section 8). The enumeration rests on a simple observation.

Lemma 5.20. *Let* P *be a finite projection algebra with* $|P| \ge 2$ *and let* e *be a maximal element of* $(P, \le)$. *Then* $P \setminus \{e\}$ *is a subalgebra. Consequently, for a class* $\mathcal{K}$ *of projection algebras defined by quasi-identities, the members of* $\mathcal{K}$ *of order* $n+1$ *are exactly the one-point extensions, by a new maximal element, of the members of* $\mathcal{K}$ *of order* n.

Proof. If $\theta_q(p) = e$ then $e \le q$, so $q = e$. Subalgebras of members of $\mathcal{K}$ are in $\mathcal{K}$. □

Starting from the one-element algebra, Mace4 computes all one-point extensions of each algebra of order n by a maximal element subject to the axioms and the chosen quasi-identities, and a canonical form removes duplicates. The numbers of projection algebras satisfying (Q1) alone are $1, 2, 5, 16, 62, 296, 1\,671, 10\,932$ for the orders $1, \dots, 8$; those satisfying (Q1), (S) and (R) number $1, 2, 5, 16, 62, 295, 1\,662, 10\,828, 80\,039$ for the orders $1, \dots, 9$. Every algebra of order ≤ 8 in these lists satisfies (Tri); indeed no projection algebra of order ≤ 10 satisfies (Q1) and violates (Tri) (Section 4.3). The embedding searches give: every algebra of order ≤ 5 satisfying (Q1) embeds in $\mathsf{P}(\mathcal{P}_4)$; of the 296 of order 6 all but $\mathfrak{P}$ embed in $\mathsf{P}(\mathcal{P}_5)$; of the 1 662 of order 7

satisfying (Q1), (S), (R) all embed in $\mathsf{P}(\mathcal{P}_6)$; and of the 10 828 of order 8 all but seven embed in $\mathsf{P}(\mathcal{P}_7)$. The seven of order 8 embed in no $\mathsf{P}(\mathcal{P}_8)$ either, and they are the following algebras (rows p, columns q, entries $\theta_p(q)$).

$\mathfrak{P}_8^1$:

	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	0	1	0	0	0	0	6	6
2	0	0	2	0	0	2	0	2
3	0	0	0	3	0	3	0	3
4	4	4	4	4	4	4	4	4
5	4	4	5	5	4	5	4	5
6	6	6	6	6	6	6	6	6
7	6	6	7	7	6	7	6	7

$\mathfrak{P}_8^2$:

	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	0	1	0	0	6	6	6	6
2	0	0	2	0	0	0	0	2
3	0	0	0	3	0	3	0	3
4	4	4	4	4	4	4	4	4
5	4	4	4	5	4	5	4	4
6	6	6	6	6	6	6	6	6
7	6	6	7	7	6	6	6	7

$\mathfrak{P}_8^3$:

	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	0	1	0	0	0	1	0	1
2	0	0	2	2	0	0	0	2
3	0	0	2	3	4	4	0	2
4	4	4	4	4	4	4	4	4
5	4	5	4	4	4	5	4	4
6	6	6	6	6	6	6	6	6
7	6	7	7	7	6	6	6	7

$\mathfrak{P}_8^4$:

	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	0	1	0	6	6	6	6	6
2	0	0	2	0	0	2	0	0
3	3	3	3	3	3	3	3	3
4	3	3	3	3	4	3	3	4
5	3	3	5	3	3	5	3	5
6	6	6	6	6	6	6	6	6
7	6	6	6	6	7	7	6	7

$\mathfrak{P}_8^5$:

	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	0	1	0	0	0	1	0	0
2	0	0	2	6	7	7	6	7
3	3	3	3	3	3	3	3	3
4	3	3	4	3	4	3	3	4
5	3	5	5	3	3	5	3	5
6	6	6	6	6	6	6	6	6
7	6	6	7	6	7	7	6	7

$\mathfrak{P}_8^6$:

	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	0	1	0	0	0	0	0	1
2	0	0	2	0	2	2	0	2
3	3	3	3	3	3	3	3	3
4	3	3	4	3	4	4	3	3
5	3	3	4	3	4	5	6	6
6	6	6	6	6	6	6	6	6
7	6	7	7	6	6	6	6	7

$\mathfrak{P}_8^7$:

	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	0	1	0	3	3	0	3	0
2	0	0	2	0	2	5	5	5
3	3	3	3	3	3	3	3	3
4	3	3	4	3	4	6	6	3
5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7

$\mathfrak{P}_7$:

	a	c	b	x	u	v	w
a	a	a	u	u	u	u	u
c	c	c	c	v	v	v	v
b	w	b	b	w	w	w	w
x	u	v	v	x	u	v	v
u	u	u	u	u	u	u	u
v	v	v	v	v	v	v	v
w	w	w	w	w	w	w	w

The algebra $\mathfrak{P}_7$, also listed, is the witness of (R): $a \bowtie c \bowtie b$ is a path, u, v, w form a bottom rectangular band with $u \leq a$, $v \leq c$, $w \leq b$ (so far the projection algebra of the graph semigroup $Q(a - c - b, \perp)$ with $\perp$ injective, which embeds in $\mathcal{P}_4$), and the new element x commutes with a and c ($xa = u$, $xc = v$) and sees b as it sees c ($\theta_x(b) = v$), while $\theta_b(x) = w$; there is a 19-element regular $*$-semigroup with projection algebra $\mathfrak{P}_7$. Six of the seven algebras of order 8 are likewise one-point extensions of projection algebras of graph semigroups $Q(G, \perp)$ with $|V| = 4$ and three colours; for instance $\mathfrak{P}_8^3$ is $Q(5 - 1 - 7 - 2, \perp)$ with $\perp\colon 1, 2 \mapsto 0$, $5 \mapsto 4$, $7 \mapsto 6$, extended by the element 3, which lies above 0, 4 and the vertex 2, commutes with 1 and with 5, and has $\theta_3(7) = 2$, $\theta_7(3) = 7$. The seventh, $\mathfrak{P}_8^7$, has a bottom band of five elements $0, 3, 5, 6, 7$ above

which lie $1 \geq 0, 3$, $2 \geq 0, 5$, $4 \geq 3, 6$ with $2 \bowtie 4$, 1 commuting with 2 and with 4, and an element 7 below no other element.

Proposition 5.21. *Each of the nine algebras $\mathfrak{P}$, $\mathfrak{P}_7$, $\mathfrak{P}_8^1, \ldots, \mathfrak{P}_8^7$ satisfies* (Q1), (Tri) *and all chain conditions* (W_n) *except as stated, and violates exactly the following of the conditions* (S), (R), (R′), (V), (P_1), (P_2), (Q), (K4), (K0) *(with a violating instance):*

- $\mathfrak{P}$*:* (S), (P_2), (Q), (K4) *((a, c, b, d) as named in Definition 7.2);*
- $\mathfrak{P}_7$*:* (R), (R′), *and every* (W_n) *((a, c, b, x) as named;* (R′) *with $(x, y) = (v, x)$);*
- $\mathfrak{P}_8^1$*:* (K0) *($(a, c, b, y, z) = (2, 7, 3, 1, 5)$);* $\mathfrak{P}_8^2$*:* (P_1) *($(a, c, b, d; x) = (2, 7, 3, 5; 1)$);* $\mathfrak{P}_8^3$*:* (R′) *($(a, c, b, x, y) = (5, 1, 7, 2, 3)$);*
- $\mathfrak{P}_8^4$*:* (P_2), (Q) *($(a, c, b, d; x) = (2, 5, 7, 4; 1)$);* $\mathfrak{P}_8^5$*:* (Q) *($(a, c, b, d; x, y) = (1, 5, 7, 4; 0, 2)$);*
- $\mathfrak{P}_8^6$*:* (K4) *($(a, c, b, y, z) = (2, 7, 1, 5, 1)$);* $\mathfrak{P}_8^7$*:* (V) *($(a, b, c, t) = (1, 2, 4, 7)$: $ata = 0$, $btb = 5$, $ctc = 3$).*

Consequently the nine algebras embed in no $\mathsf{P}(\mathcal{P}_X)$; the six conditions (R′), (V), (P_1), (Q), (K4), (K0) *are independent, each failing in an algebra of order* 8 *satisfying* (Q1), (Tri), *all* (W_n) *and the other five; and* (S) *and* (R), *which are consequences of* (Q) *and* (R′) *(Corollaries 5.13, 5.5), have the witnesses $\mathfrak{P}$ and $\mathfrak{P}_7$ of orders* 6 *and* 7, *of which $\mathfrak{P}$ satisfies* (R), (R′), (V), (P_1), (K0) *and all* (W_n), *and $\mathfrak{P}_7$ satisfies everything except* (R), (R′) *and the chain conditions.*

Proof. A direct computation (the conditions are quasi-identities with at most seven variables, and (W_n) for all n is decided by the component formulation (W_∞) of Section 6.1); the first statement for (Tri) also follows from the Mace4 result quoted above. Non-embeddability follows from the theorems of this section and Section 6. □

Theorem 5.22. *Let P be a projection algebra of order at most* 8 *satisfying* (Q1). *Then P embeds in some $\mathsf{P}(\mathcal{P}_X)$ if and only if it satisfies the eight wiring conditions, and then it embeds in $\mathsf{P}(\mathcal{P}_7)$. The minimal non-embeddable projection algebras of order at most* 8 *satisfying* (Q1) *are exactly $\mathfrak{P}$, $\mathfrak{P}_7$ and $\mathfrak{P}_8^1, \ldots, \mathfrak{P}_8^7$.*

Proof. If P satisfies the eight conditions, in particular (S) and (R), it belongs to the census of algebras with (Q1), (S), (R), and it embeds in $\mathsf{P}(\mathcal{P}_7)$ unless it is one of $\mathfrak{P}_8^1, \ldots, \mathfrak{P}_8^7$, which violate one of the conditions. The converse is Theorem 1.3. For the second statement, a minimal non-embeddable algebra P of order $n \leq 8$ satisfying (Q1) has every proper subalgebra embeddable; deleting a maximal element (Lemma 5.20) gives an embeddable algebra of order $n - 1$, so P is a one-point extension, by a new maximal element, of an embeddable algebra of order $n - 1$, subject to the axioms and (Q1) only. These extensions were enumerated (296, 1 671 and 10 932 algebras of orders $6, 7, 8$). Those satisfying (S) and (R) belong to the census above, where the non-embeddable ones are known; those violating (S) or (R) are non-embeddable, and all but $\mathfrak{P}$ (order 6) and $\mathfrak{P}_7$ (order 7) contain $\mathfrak{P}$ or $\mathfrak{P}_7$ as a proper subalgebra, hence are not minimal (8 of the 9 such algebras of order 7, and all 104 of order 8). Minimality of the nine: every proper subalgebra of $\mathfrak{P}_8^k$ is an algebra of order ≤ 7 satisfying (Q1), (S), (R), hence embeddable; similarly for $\mathfrak{P}_7$ and $\mathfrak{P}$. □

Remark 5.23. The census produced, for $\mathfrak{P}_8^1$ and $\mathfrak{P}_8^6$, quasi-identities in seven and five variables (greedy-minimal subsets of the diagrams of the algebras, valid in $\mathsf{P}(\mathcal{P}_7)$ by SAT), which we called (E_1) and (E_6):

$$(\mathrm{E}_1): \quad z \leq b,\ \theta_a(b) = z,\ dbd = d,\ \theta_x(c) = z,\ aca = a,\ bcb = b,\ e \leq x,\ e \leq d,\ ada = a \implies aba = a;$$

$$(\mathrm{E}_6): \quad dbd = d,\ aca = a,\ ada = a,\ u \leq c,\ u \leq d,\ bcb = aua \implies aua = a.$$

Prover9 (M. Kinyon) derives (E_1) from (Q1), (W_2) and (K0) (227 steps, with five auxiliary lemmas) and (E_6) from (Q1) and (K4) (72 steps); the chain condition is needed for the first

((Q1)+(K0) alone has a model of order 10 violating (E_1)). So (K0) and (K4), the five-variable path-closing forms of the same obstructions, are the conditions of this paper, and (E_1), (E_6) need no separate proof. The proofs are in the repository accompanying the paper (`embed/pa8/p9/k0_implies_e1.pf`, `k0_lemmas.pf`, `k4_implies_e6.pf`; the 25-step derivation of $\theta_z(c) = b$ in Theorem 5.15 is `k4.pf1`).

6. The chain conditions and the counting invariant

6.1. **The chain conditions.** The hypotheses of (R) say that a single auxiliary x commutes with a and c and identifies b with c. Exchanging the roles of a and b and allowing a chain of commuting auxiliaries, each of which identifies a with c, gives for every $n \geq 1$ the quasi-identity

$$(W_n): \qquad a \bowtie c,\ b \bowtie c, \quad c \sim x_1 \sim x_2 \sim \cdots \sim x_n \sim b, \quad \theta_{x_i}(a) = \theta_{x_i}(c)\ (1 \leq i \leq n) \implies a \bowtie b,$$

in $n+3$ variables. (W_1) is (R) with a and b exchanged, and (W_2) is the condition that settles one of the candidates of order 9 (Section 9). One direction between consecutive members is trivial: (W_{n+1}) implies (W_n) in every projection algebra, since an instance of (W_n) becomes an instance of (W_{n+1}) by repeating its last auxiliary. The family is nevertheless strictly increasing (Theorem 6.11).

In $\mathcal{P}_X$ two projections commute iff they have a common lower bound (Proposition 2.4(4)), and the set $\mathrm{Id}(a,c) = \{y : \theta_y(a) = \theta_y(c)\}$ of *identifiers* of a and c is a down-set, since $\theta_z(a) = \theta_z(\theta_y(a))$ for $z \leq y$ (Section 2.1). So a chain $c \sim x_1 \sim \cdots \sim x_n$ of identifiers is a zigzag $c \geq m_0 \leq x_1 \geq m_1 \leq x_2 \geq \ldots$ in the poset $\mathrm{Id}(a,c)$, and the possible chain ends are exactly the vertices of the connected component $\mathrm{Comp}(c)$ of c in the comparability graph of $\mathrm{Id}(a,c)$. The validity of all (W_n) in $\mathcal{P}_X$ is therefore the single statement

$$(W_\infty) \qquad \text{for every } y \in \mathrm{Comp}(c) \text{ and every } b \bowtie c \text{ with } b \sim y:\ a \bowtie b.$$

No bounded number of links suffices: the component of c has unbounded radius.

Lemma 6.1 (radius)**.** *Let $k \geq 1$, $X = \{p_1, \ldots, p_k, q_1, \ldots, q_k, t\}$, and let a, c, z be the projections of $\mathcal{P}_X$ with the single transversal block $\{t\}$ and the nontransversal blocks $\{p_1, \ldots, p_k\}, \{q_1\}, \ldots, \{q_k\}$ for a, all singletons for c, and $\{p_1, q_1\}, \ldots, \{p_k, q_k\}$ for z. Then $a \bowtie c$, $z \in \mathrm{Id}(a,c)$, and z is at distance exactly $k+1$ from c in the comparability graph of $\mathrm{Id}(a,c)$.*

Proof. $a \bowtie c$ since both have the single transversal block $\{t\}$. A projection y identifies a with c iff the partitions of $\operatorname{supp} y$ induced by $\varepsilon_y \vee \varepsilon_a$ and by $\varepsilon_y \vee \varepsilon_c$ coincide and a part is bright for a iff it is bright for c; here a part is bright iff it contains t (for both), so only the partitions matter. For z both partitions of $\operatorname{supp} z = \{t\}$ are trivial, so z identifies. Let $B = \{p_1, \ldots, p_k\}$. Let y_i ($1 \leq i \leq k$) have the nontransversal blocks $\{p_j, q_j\}$ for $j < i$, the transversal blocks $\{p_i\}$, $\{q_i\}$, $\{t\}$, and singletons elsewhere. Then y_i identifies: the part of p_i is $\{p_i\}$ for both joins (via a its class is $B \cup \{q_j : j < i\}$, which meets $\operatorname{supp} y_i$ only in p_i; via c it is $\{p_i\}$), and likewise for q_i and t. Consecutive members commute by Lemma 2.6(b): from y_i to y_{i+1} the two transversal blocks $\{p_i\}, \{q_i\}$ merge into the nontransversal block $\{p_i, q_i\}$ (type (ii)), the singletons $\{p_{i+1}\}, \{q_{i+1}\}$ become transversal (type (iii)) and all other blocks are common; the same for $c \sim y_1$ and $y_k \sim z$. So $\mathrm{dist}(c,z) \leq k+1$.

For the lower bound, note first that an identifier y has at most one transversal block meeting B: two such blocks lie in one class of $\varepsilon_y \vee \varepsilon_a$ (through the block B of a), but in different classes of $\varepsilon_y \vee \varepsilon_c$, since every block of c is a singleton. Now let $c = y_0 \sim y_1 \sim \cdots \sim y_n = z$ be any chain of identifiers. Fix i and let j be the largest index such that $\{p_i, q_i\}$ is not a block of y_j (j exists, since it is not a block of c, and $j < n$). Then $\{p_i, q_i\}$ is a nontransversal block of y_{j+1} which is not a block of y_j, so by Lemma 2.6(b) it is a union of transversal blocks of y_j; hence y_j has a transversal block containing p_i and contained in $\{p_i, q_i\}$. Call $j = j(i)$. Since $y_{j(i)}$ has

exactly one transversal block meeting B, and that block determines i, the indices $j(1), \dots, j(k)$ are distinct; they lie in $\{1, \dots, n-1\}$ ($y_0 = c$ has no transversal block meeting B). Hence $n - 1 \geq k$. □

So a proof of the family has to handle arbitrary n at once. The tool is the counting invariant of Lemma 2.9, which is constant on $\mathrm{Comp}(c)$ and whose value at c carries enough information to close the chain.

6.2. **Separation and the meaning of the count.**

Corollary 6.2 (separation)**.** *Let $a \bowtie c$ in $\mathsf{P}(\mathcal{P}_X)$ and $\kappa = |\varepsilon_a \vee \varepsilon_c|$. Then $\delta_{ac}(c) = \kappa - |\varepsilon_c| \leq 0 \leq |\varepsilon_a| - \kappa = \delta_{ac}(a)$ and $\delta'_{ac}(c) = \delta'_{ac}(a) = 0$; the two values of δ_{ac} coincide iff $a = c$. Hence for $a \neq c$ the projections a and c lie in different components of the comparability graph of $\mathrm{Id}(a, c)$, and every $y \in \mathrm{Comp}(c)$ has $\delta_{ac}(y) = \kappa - |\varepsilon_c|$ and $\delta'_{ac}(y) = 0$.*

Proof. c identifies a with c ($\theta_c(a) = c = \theta_c(c)$) and so does a. $\varepsilon_c \vee \varepsilon_a$ has κ classes and $\varepsilon_c \vee \varepsilon_c = \varepsilon_c$ has $|\varepsilon_c| \geq \kappa$; symmetrically for a. Both inequalities are equalities iff $\varepsilon_a \leq \varepsilon_c \leq \varepsilon_a$, and if $\varepsilon_a = \varepsilon_c$ then each block is a class of $\varepsilon_a \vee \varepsilon_c$ by itself and $a \bowtie c$ forces it to be transversal in a iff in c (Lemma 2.6(c)), so $a = c$. For δ': the classes of $\varepsilon_a \vee \varepsilon_c$ meeting $\operatorname{supp} a$ are the classes Γ_i, as many as the $\mathsf{T}(c)$-blocks. The rest is Lemma 2.9 along a chain in $\mathrm{Comp}(c)$. □

The value of the count has structural content, which we extract with a lemma on contracting partitions.

Lemma 6.3 (contraction)**.** *Let $\pi \leq \pi'$ and ρ be equivalence relations on X. Then $|\pi| - |\pi \vee \rho| \geq |\pi'| - |\pi' \vee \rho|$, and the inequality is strict as soon as two π-classes lying in a common π'-class lie in a common class of $\pi \vee \rho$.*

Proof. Let H be the graph on the π-classes in which two classes are adjacent iff they meet a common ρ-class; its components are the classes of $\pi \vee \rho$, so $|\pi| - |\pi \vee \rho| = |V(H)| - c(H)$. Identify, one pair at a time, the π-classes lying in a common π'-class. Each identification lowers $|V|$ by one and c by at most one, by exactly one iff the two vertices lie in different components at that moment; at the end the vertices are the π'-classes and the components are the classes of $\pi' \vee \rho$. So $|V| - c$ never increases, and if some identification joins two vertices of one component we may perform that identification first, so that $|V| - c$ drops by one at that step and never increases afterwards. □

Lemma 6.4 (the value of the count)**.** *Let $a \bowtie c$, $\kappa = |\varepsilon_a \vee \varepsilon_c|$, and let ε be an equivalence relation on X with $|\varepsilon \vee \varepsilon_a| - |\varepsilon \vee \varepsilon_c| = \kappa - |\varepsilon_c|$ (for instance $\varepsilon = \varepsilon_y$ with $y \in \mathrm{Comp}(c)$). Then*

(A) $\varepsilon_c \leq \varepsilon \vee \varepsilon_a$: every block of c lies in one class of $\varepsilon \vee \varepsilon_a$;

(M) $(\varepsilon \vee \varepsilon_c) \wedge (\varepsilon_a \vee \varepsilon_c) = \varepsilon_c$: two blocks of c in one class of $\varepsilon_a \vee \varepsilon_c$ lie in different classes of $\varepsilon \vee \varepsilon_c$.

If moreover $|\varepsilon \vee \varepsilon_a|_{\operatorname{supp} a} = |\varepsilon \vee \varepsilon_c|_{\operatorname{supp} c}$, then

(G2′) two $\mathsf{T}(c)$-blocks lying in one class of $\varepsilon \vee \varepsilon_a \vee \varepsilon_c$ lie in one class of $\varepsilon \vee \varepsilon_c$.

Proof. Put $g_a = |\varepsilon \vee \varepsilon_a| - |\varepsilon \vee \varepsilon_a \vee \varepsilon_c| \geq 0$ and $g_c = |\varepsilon \vee \varepsilon_c| - |\varepsilon \vee \varepsilon_a \vee \varepsilon_c| \geq 0$; by Lemma 6.3 with $\pi = \varepsilon_c \leq \pi' = \varepsilon \vee \varepsilon_c$ and $\rho = \varepsilon_a$, $g_c \leq |\varepsilon_c| - \kappa$. The hypothesis reads $g_a - g_c = \kappa - |\varepsilon_c|$, so $g_a = g_c - (|\varepsilon_c| - \kappa) \leq 0$, whence $g_a = 0$ and $g_c = |\varepsilon_c| - \kappa$. The first equation is (A); the second is the equality case of Lemma 6.3, which is (M). For (G2′): by (A), $\varepsilon \vee \varepsilon_a = \varepsilon \vee \varepsilon_a \vee \varepsilon_c$, whose classes are unions of classes of $\varepsilon_a \vee \varepsilon_c$; such a class meets $\operatorname{supp} a$ iff it contains some Γ_i iff it meets $\operatorname{supp} c$. So $|\varepsilon \vee \varepsilon_a \vee \varepsilon_c|_{\operatorname{supp} c} = |\varepsilon \vee \varepsilon_a|_{\operatorname{supp} a} = |\varepsilon \vee \varepsilon_c|_{\operatorname{supp} c}$. Every class of $\varepsilon \vee \varepsilon_a \vee \varepsilon_c$ meeting $\operatorname{supp} c$ contains at least one class of the finer relation $\varepsilon \vee \varepsilon_c$ meeting $\operatorname{supp} c$; equal counts force exactly one. □

6.3. **The end step and the chain theorem.**

Theorem 6.5 (end step)**.** *Let $a \bowtie c$ and $b \bowtie c$ in $\mathsf{P}(\mathcal{P}_X)$, let μ be an equivalence relation on X whose nontrivial classes are unions of $\mathsf{T}(b)$-blocks, and put $\varepsilon = \varepsilon_b \vee \mu$. If*

$$|\varepsilon \vee \varepsilon_a| - |\varepsilon \vee \varepsilon_c| = |\varepsilon_a \vee \varepsilon_c| - |\varepsilon_c| \qquad \text{and} \qquad |\varepsilon \vee \varepsilon_a|_{\operatorname{supp} a} = |\varepsilon \vee \varepsilon_c|_{\operatorname{supp} c},$$

then (G1) *and* (G2) *of Section 5.2 hold, and hence $a \bowtie b$.*

Proof. We use the notation B_i, C_i, Δ_i of Section 2.3 for the pair $b \bowtie c$ (so $\Delta_i \cap \operatorname{supp} b = B_i$, $\Delta_i \cap \operatorname{supp} c = C_i$, and the other classes of $\varepsilon_b \vee \varepsilon_c$ meet neither support). Write $\kappa = |\varepsilon_a \vee \varepsilon_c|$, $\kappa_{bc} = |\varepsilon_b \vee \varepsilon_c|$, $\pi = \varepsilon_a \vee \varepsilon_b \vee \varepsilon_c$ and $N = |\pi \vee \mu|$. Call the nontrivial classes of μ the *groups*; let there be g groups and let s' be the number of $\mathsf{T}(b)$-blocks lying in groups. Choose one point p_i in each B_i and a set Λ of $s' - g$ pairs (p_i, p_j) forming a spanning tree on the $\mathsf{T}(b)$-blocks of each group; then $\varepsilon_b \vee \mu = \varepsilon_b \vee \nu$ for the equivalence ν generated by Λ, and for every equivalence relation $\tau \geq \varepsilon_b$ the relation $\tau \vee \mu = \tau \vee \nu$ is obtained from τ by adding the pairs of Λ one at a time, each addition lowering the number of classes by at most one. Hence

(i) $|\tau| - |\tau \vee \mu| \leq s' - g$ for every $\tau \geq \varepsilon_b$, with equality only if no pair of Λ joins two points already in one τ-class; since Λ may be chosen to contain any given pair (p_i, p_j) with B_i, B_j in one group, equality implies that two $\mathsf{T}(b)$-blocks of one group lie in different τ-classes;

(ii) $|\varepsilon \vee \varepsilon_c| = |\varepsilon_b \vee \varepsilon_c \vee \nu| = \kappa_{bc} - (s' - g)$: the pairs of Λ join the distinct classes Δ_i, and a tree edge always joins two different pieces.

By Lemma 6.4(A), $\varepsilon \vee \varepsilon_a = \varepsilon \vee \varepsilon_a \vee \varepsilon_c = \pi \vee \mu$, so the first hypothesis and (ii) give

$$N = |\varepsilon \vee \varepsilon_a| = |\varepsilon \vee \varepsilon_c| + \kappa - |\varepsilon_c| = \kappa_{bc} - (s' - g) + \kappa - |\varepsilon_c|. \tag{8}$$

Put $r_1 = \kappa - |\pi| \geq 0$ and $r_2 = |\pi| - N \geq 0$, so that $r_1 + r_2 = \kappa - N = (|\varepsilon_c| - \kappa_{bc}) + (s' - g)$. Lemma 6.3 with $\varepsilon_c \leq \varepsilon_a \vee \varepsilon_c$ and $\rho = \varepsilon_b$ gives $|\varepsilon_c| - \kappa_{bc} \geq \kappa - |\pi| = r_1$, and (i) with $\tau = \pi$ gives $r_2 \leq s' - g$. Hence both are equalities, $r_1 = |\varepsilon_c| - \kappa_{bc}$ and $r_2 = s' - g$, and the equality clause of (i) yields

$$\text{two } \mathsf{T}(b)\text{-blocks of one group lie in different classes of } \pi = \varepsilon_a \vee \varepsilon_b \vee \varepsilon_c. \tag{9}$$

(G1). Let $r_c = |\varepsilon_a \vee \varepsilon_b| - |\pi| \geq 0$. By (8) and (i) with $\tau = \varepsilon_a \vee \varepsilon_b$, $N = |\varepsilon_a \vee \varepsilon_b \vee \mu| \geq |\varepsilon_a \vee \varepsilon_b| - (s' - g)$, while $N = |\pi| - r_2 = |\varepsilon_a \vee \varepsilon_b| - r_c - (s' - g)$. So $r_c \leq 0$, i.e. $|\varepsilon_a \vee \varepsilon_b| = |\varepsilon_a \vee \varepsilon_b \vee \varepsilon_c|$, which says $\varepsilon_c \leq \varepsilon_a \vee \varepsilon_b$.

(G2). Suppose C_i and C_j, $i \neq j$, lie in one class of π. Since Δ_i and Δ_j lie inside classes of π, so do B_i and B_j, in that same class. If B_i, B_j lie in one group this contradicts (9). Otherwise Δ_i and Δ_j lie in different classes of $\varepsilon \vee \varepsilon_c = \varepsilon_b \vee \varepsilon_c \vee \mu$ (the classes of this relation meeting $\operatorname{supp} c$ are the unions of the Δ_i over the $\mathsf{T}(b)$-blocks of a group, together with the Δ_i of the $\mathsf{T}(b)$-blocks in no group, and these are distinct because $\Delta_i \cap \operatorname{supp} b = B_i$), but in one class of $\pi \leq \pi \vee \mu = \varepsilon \vee \varepsilon_a \vee \varepsilon_c$; this contradicts (G2′) of Lemma 6.4, which applies by the second hypothesis. The conclusion $a \bowtie b$ is Lemma 5.1. □

Theorem 6.6 (the chain family)**.** (W_n) *holds in every partition monoid $\mathcal{P}_X$ for every $n \geq 1$. More generally* (W_∞) *holds: if $a \bowtie c$, $b \bowtie c$, $y \in \operatorname{Comp}(c)$, or merely $y \in \operatorname{Id}(a, c)$ with $\delta_{ac}(y) = \delta_{ac}(c)$ and $\delta'_{ac}(y) = 0$, and $yb = by$, then $a \bowtie b$.*

Proof. By Corollary 6.2 every $y \in \operatorname{Comp}(c)$ has $\delta_{ac}(y) = \kappa - |\varepsilon_c|$ and $\delta'_{ac}(y) = 0$; the end x_n of a chain $c \sim x_1 \sim \cdots \sim x_n$ of identifiers is such a y. Let $m = yb = by$. Then $m \leq y$, so by Lemma 2.9 $\delta_{ac}(m) = \delta_{ac}(y)$ and $\delta'_{ac}(m) = \delta'_{ac}(y)$. By Lemma 2.6(b) the classes of $\varepsilon_m = \varepsilon_y \vee \varepsilon_b$ are common N-blocks, $\mathsf{N}(y)$-blocks which are unions of $\mathsf{T}(b)$-blocks, $\mathsf{N}(b)$-blocks which are unions of $\mathsf{T}(y)$-blocks, and classes inside $\operatorname{supp} y \cap \operatorname{supp} b$, which are unions of $\mathsf{T}(b)$-blocks; so $\varepsilon_m = \varepsilon_b \vee \mu$,

where μ identifies the $\mathsf{T}(b)$-blocks lying in a common class. Theorem 6.5 applies and gives $a \bowtie b$. □

Remark 6.7. (1) Both counts are needed: in $\mathsf{P}(\mathcal{P}_4)$ and $\mathsf{P}(\mathcal{P}_5)$ there are instances with $y \in \mathrm{Id}(a,c)$, $\delta_{ac}(y) = \delta_{ac}(c)$, $b \bowtie c$, $b \sim y$ and $a \not\bowtie b$, in none of which $\delta'_{ac}(y) = 0$. The pair $(\delta_{ac}, \delta'_{ac})$ separates a from c but does not characterise $\mathrm{Comp}(c)$: in $\mathsf{P}(\mathcal{P}_4)$ there are matched pairs for which another component of the comparability graph of $\mathrm{Id}(a,c)$ has the same values; this does not matter for the chain conditions. (2) The invariant is global. The search for a proof of (W_2) had looked for a *local* property of the chain end y, closed under the two kinds of steps of the zigzag (down: $z \le y$; up: $z \le w \in \mathrm{Id}(a,c)$), and several natural candidates, such as $\varepsilon_c \le \varepsilon_y \vee \varepsilon_a$ (statement (A) above) or (M), are indeed true at every chain end but are not closed under the up step by themselves; the pair of counts is.

6.4. The witnesses and the non-finite basis theorem.

Definition 6.8. For $n \ge 1$ let $\mathfrak{P}_n^{\mathrm{ch}}$ be the set of $2n+5$ elements $a, c, b, x_1, \dots, x_n$ (*top*) and $d, e_0, e_1, \dots, e_n$ (*bottom*) with the operation θ defined by: $\theta_e \equiv e$ for every bottom element e; $\theta_a(c) = a$, $\theta_a(y) = d$ for $y \ne a, c$; $\theta_c(a) = \theta_c(b) = c$, $\theta_c(y) = e_0$ for $y \ne a, b, c$; $\theta_b(c) = b$, $\theta_b(y) = e_n$ for $y \ne b, c$; and, for $1 \le i \le n$, $\theta_{x_i}(y) = e_{i-1}$ if $y \in \{a, c, d, e_0, \dots, e_{i-1}, x_1, \dots, x_{i-1}\}$ (the elements to the *left* of x_i) and $\theta_{x_i}(y) = e_i$ if $y \in \{e_i, \dots, e_n, x_{i+1}, \dots, x_n, b\}$ (to the *right*); and $\theta_{x_i}(x_i) = x_i$.

So the bottom elements form a rectangular band of pairwise matched elements, d is the bottom of a, and $e_0 = \theta_c(x_1) = \theta_{x_1}(c)$, $e_i = \theta_{x_i}(x_{i+1}) = \theta_{x_{i+1}}(x_i)$, $e_n = \theta_{x_n}(b) = \theta_b(x_n)$ are the shared lower bounds along the chain $c \sim x_1 \sim \cdots \sim x_n \sim b$. We have $a \bowtie c \bowtie b$, every x_i identifies a with c ($\theta_{x_i}(a) = \theta_{x_i}(c) = e_{i-1}$), and $a \not\bowtie b$ ($\theta_b(a) = e_n$). Hence $\mathfrak{P}_n^{\mathrm{ch}}$ violates (W_n), and (W_k) for every $k \ge n$. For $n = 1$ it is the witness $\mathfrak{P}_7$ of (R) with a and b exchanged, and for $n = 2$ it is one of the candidates of order 9 (Section 9); for $n = 3$ it was found by Mace4 as a projection algebra satisfying (Q1) and (W_2) but not (W_3).

Proposition 6.9. *For every $n \ge 1$, $\mathfrak{P}_n^{\mathrm{ch}}$ is a projection algebra satisfying* (Q1) *and* (W_{n-1}) *and violating* (W_n).

Proof. Write B for the bottoms and T for the tops, and for a top t let $M_t = \{y : \theta_t(y) = t\}$ ($M_a = \{a, c\}$, $M_c = \{a, b, c\}$, $M_b = \{b, c\}$, $M_{x_i} = \{x_i\}$; note $y \in M_t \Leftrightarrow t \in M_y$) and $\beta_t(y) = \theta_t(y) \in B$ for $y \notin M_t$ ($\beta_a \equiv d$, $\beta_c \equiv e_0$, $\beta_b \equiv e_n$, $\beta_{x_i} = e_{i-1}$ on the left of x_i and e_i on the right). The one fact used repeatedly is $(*)$: for a top $t \ne x_i$, β_t maps every element to the same side of x_i as t itself (a, c, d, e_0 are left of every x_i, b, e_n right, and for x_j the value e_j is left of x_i when $j < i$ while e_{j-1} is right when $j > i$), and β_{x_i} maps left to e_{i-1} (left) and right to e_i (right).

The axioms in θ-notation (Section 2.1). $\theta_x(x) = x$ holds by definition. $\theta_x(\theta_x(y)) = \theta_x(y)$ is trivial for $x \in B$ or $\theta_x(y) = x$, and otherwise $\theta_x(\beta) = \beta$ for $\beta = \beta_x(y)$ since $d \notin M_a$, $e_0 \notin M_c$, $e_n \notin M_b$ and e_{i-1}, e_i lie on the correct sides of x_i. $\theta_x(\theta_y(x)) = \theta_x(y)$ is trivial unless x, y are distinct tops with $y \notin M_x$, and then $\theta_x(\beta_y(x)) = \beta_x(y)$ by $(*)$. For $\theta_x(\theta_y(\theta_x(z))) = \theta_{\theta_x(y)}(z)$: if $x \in B$ both sides are x; if $y \in B$ both are $\beta_x(y)$; if $y \in M_x$ the right side is $\theta_x(z)$, and so is the left side (for $z \in M_x$ use $y \in M_x \Leftrightarrow x \in M_y$, otherwise $\beta_x(\beta_y(\beta_x(z))) = \beta_x(z)$ by $(*)$); if $y \notin M_x$ the right side is $\beta_x(y)$ and the left side is $\theta_x(\beta_y(x)) = \beta_x(y)$ for $z \in M_x$ and $\beta_x(\beta_y(\beta_x(z))) = \beta_x(y)$ otherwise, by $(*)$. For $\theta_y\theta_x\theta_y\theta_x(z) = \theta_y\theta_x(z)$ put $w = \theta_y(\theta_x(z))$: if $w = y$ then $\theta_x(z) \in M_y$ forces $\theta_x(z) = x \in M_y$ and $\theta_y\theta_x(y) = \theta_y(x) = y$; if $w \in B$ then $\beta_y(\beta_x(w)) = w$ by $(*)$, since $w = \beta_y(t)$ with $t \in \{x\} \cup B$ on the same side of y as x.

(Q1): $\theta_x(z) = \theta_y(z) = z$ with $x \ne y$ forces $z \in B$ below both; the only tops sharing a bottom are consecutive chain members ($e_0 \le c, x_1$; $e_i \le x_i, x_{i+1}$; $e_n \le x_n, b$), and these commute.

(W_{n-1}): the matched pairs of $\mathfrak{P}_n^{\mathrm{ch}}$ are $a \bowtie c$, $c \bowtie b$, the pairs of bottoms, and the trivial pairs $x \bowtie x$; so the only instances of (W_{n-1}) whose conclusion is not automatic are the paths

$a \bowtie c \bowtie b$ and $b \bowtie c \bowtie a$. For the first, the identifiers of (a, c) are c, the x_i and the bottoms, and the commuting graph on them is the path $c - x_1 - \cdots - x_n$ with e_0 adjacent to c, x_1, e_i to x_i, x_{i+1} and e_n to x_n; the only elements commuting with b are b, x_n, e_n, at distance n or $n+1$ from c, so no chain of $n-1$ identifiers joins c to a neighbour of b. For the second, the identifiers of (b, c) are b, c and the bottoms, the only one commuting with a is d, and d commutes with no identifier other than itself while $c \not\sim d$; so there is no chain at all. Hence (W_{n-1}) holds, and (W_n) fails as noted. □

Proposition 6.10. *For every $n \geq 1$, every maximal proper subalgebra of $\mathfrak{P}_n^{\mathrm{ch}}$ embeds in $\mathsf{P}(\mathcal{P}_m)$ for some $m \leq 2n+3$.*

Proof. The top elements are irreducible ($\theta_x(y)$ is a top element only when it equals x), so the maximal proper subalgebras are obtained by omitting one top element; there are $n+3$ of them. We give explicit embeddings. A projection of rank 0 is a partition ε_e of X with no transversal block; by Lemma 2.6(a), $\theta_e(y) = e$ for every y, and $\theta_p(e)$ is the rank-0 projection whose blocks are the $\mathsf{N}(p)$-blocks and the unions of the $\mathsf{T}(p)$-blocks lying in a common class of $\varepsilon_p \vee \varepsilon_e$. So all rank-0 projections are pairwise matched, as the bottom elements are, and the bottoms will be realised by rank-0 projections. Each chain element x_l present is realised with exactly two T-blocks, the singletons $\{p_l\}$, $\{q_l\}$ of a *pair* $X_l = \{p_l, q_l\}$ of points reserved for it; every other element has X_l either as one N-block ("X_l merged") or as two singleton N-blocks ("X_l separate"), and no other block meets X_l. Then for every element $y \neq x_l$, $\theta_{x_l}(y)$ is $\mathsf{N}(x_l)$ together with X_l merged if X_l is merged in y, and with X_l separate otherwise; this realises the dichotomy left/right of Definition 6.8, with "left" = "X_l merged". Accordingly, the bottom element e_j has X_l separate for $l \leq j$ and merged for $l > j$; d, a, c have all pairs merged, b all pairs separate, and x_l has X_k separate for $k < l$ and merged for $k > l$. Writing $[\,\cdot\,]$ for a T-block and $(\,\cdot\,)$ for an N-block, and listing only the blocks on the remaining points R, the embeddings are:

- *omit x_i* ($R = \{\alpha, \beta, \gamma\}$, $2n+1$ points): $a = [\alpha](\beta\gamma)$, $c = [\alpha\beta](\gamma)$, $b = [\beta](\alpha)(\gamma)$, $d = (\alpha)(\beta\gamma)$; every x_l with $l < i$ and every e_j with $j < i$ is $(\alpha\beta)(\gamma)$ on R, every x_l with $l > i$ and every e_j with $j \geq i$ is $(\alpha)(\beta)(\gamma)$ on R;
- *omit b* ($2n+3$ points): as before, with all x_l, e_j equal to $(\alpha\beta)(\gamma)$ on R;
- *omit c*: $a = [\alpha](\beta\gamma)$, $b = [\beta](\alpha)(\gamma)$, $d = (\alpha)(\beta\gamma)$, all x_l, e_j equal to $(\alpha)(\beta)(\gamma)$ on R;
- *omit a*: $c = [\alpha][\beta](\gamma)$ with all pairs merged, $b = [\alpha][\beta](\gamma)$ with all pairs separate, $d = (\alpha)(\beta\gamma)$, all x_l, e_j equal to $(\alpha)(\beta)(\gamma)$ on R.

Each map is injective (distinct elements differ in a block of R or in a pair), and each value $\theta_y(x)$ is computed by Lemma 2.6(a) from the classes of $\varepsilon_y \vee \varepsilon_x$. The point set is $R \sqcup X_1 \sqcup \cdots \sqcup X_n$ and no block of any listed projection meets two of these parts; hence $\varepsilon_y \vee \varepsilon_x$ is the disjoint union of the joins of the restrictions, $\theta_y(x)$ is computed part by part, and the maps are embeddings into $\mathcal{P}_R \times \prod_l \mathcal{P}_{X_l} \leq \mathcal{P}_X$. On a pair X_l the restriction of every element is one of $[p_l][q_l]$, $(p_l\, q_l)$, $(p_l)(q_l)$, and the pair rule above is immediate: for $y \neq x_l$ no class of $\varepsilon_{x_l} \vee \varepsilon_y$ contains a $\mathsf{T}(y)$-block together with $[p_l]$ or $[q_l]$ (the $\mathsf{T}(y)$-blocks lie in R, where the blocks of x_l are N-blocks), so $\theta_{x_l}(y)$ is $\mathsf{N}(x_l)$ with X_l merged or separate, i.e. e_{l-1} or e_l, as y is to the left or right of x_l. On R the restriction of every element takes one of finitely many values independent of n, and the restriction to R of $\theta_y(x)$ is a function of the restrictions of x and y; the finitely many combinations that occur are checked directly (for instance $a \bowtie c$ and $c \bowtie b$ through $[\alpha\beta]$, $\theta_a(b) = d$ since the class $\{\alpha\}$ of $[\alpha]$ in $\varepsilon_a \vee \varepsilon_b$ does not meet $\operatorname{supp} b$, and the rank-0 values of $\theta_a, \theta_b, \theta_c$ on bottoms and chain elements). All combinations that occur for any n already occur for $n = 3$, and the tables were verified by computer for $n \leq 7$. □

Theorem 6.11 (non-finite basis)**.** *For every $n \geq 1$, $\mathfrak{P}_n^{\mathrm{ch}}$ is a minimal non-embeddable projection algebra of order $2n+5$ which satisfies every quasi-identity in at most $n+2$ variables that is valid in all $\mathsf{P}(\mathcal{P}_X)$. Consequently the quasivariety generated by the projection algebras $\mathsf{P}(\mathcal{P}_X)$*

is not finitely based, the class of finite projection algebras embeddable in some $\mathsf{P}(\mathcal{P}_X)$ *is not the class of finite models of any finite set of quasi-identities, and* (W_{n+1}) *is not a consequence of* $(\mathrm{W}_1),\dots,(\mathrm{W}_n)$ *together with any valid quasi-identities in at most* $n+3$ *variables.*

Proof. A subalgebra of $\mathfrak{P}_n^{\mathrm{ch}}$ generated by k elements contains at most k top elements, the tops being irreducible; for $k \le n+2$ it omits one of the $n+3$ tops, hence lies in a maximal proper subalgebra, which embeds in some $\mathsf{P}(\mathcal{P}_X)$ by Proposition 6.10. Let σ be a quasi-identity in $k \le n+2$ variables valid in all $\mathsf{P}(\mathcal{P}_X)$, and suppose an assignment violates σ in $\mathfrak{P}_n^{\mathrm{ch}}$; it violates σ in the subalgebra generated by its values, which embeds in some $\mathsf{P}(\mathcal{P}_X)$, a contradiction. So $\mathfrak{P}_n^{\mathrm{ch}}$ satisfies σ. On the other hand it violates (W_n), which is valid in all $\mathsf{P}(\mathcal{P}_X)$ by Theorem 6.6; so it embeds in none, and it is minimal since every proper subalgebra lies in a maximal one. If a finite set Σ of quasi-identities valid in all $\mathsf{P}(\mathcal{P}_X)$ axiomatised the quasivariety, or had the embeddable finite algebras as its finite models, then $\mathfrak{P}_n^{\mathrm{ch}}$ with $n+2$ at least the number of variables of every member of Σ would be a model of Σ, hence embeddable: a contradiction. The last assertion is the case $\Sigma = \{(\mathrm{W}_1),\dots,(\mathrm{W}_n)\} \cup \Sigma'$ applied to $\mathfrak{P}_{n+1}^{\mathrm{ch}}$, which has $n+4$ top elements. □

In particular $\mathfrak{P}_n^{\mathrm{ch}}$ satisfies (Tri) and the eight wiring conditions of Section 5 (which have at most seven variables) for $n \ge 5$, and by direct computation also for $n \le 4$; this is the statement of Theorem 1.4. The theorem also explains why the family was found late: (W_n) has $n+3$ variables and its smallest witness has order $2n+5$, so the first member beyond (R) needs order 9 and five variables, the next order 11 and six.

Remark 6.12. In every projection algebra satisfying (Q1), (W_2) implies the two-auxiliary condition $a \bowtie c \bowtie b,\ xc = cx,\ \theta_x(a) = \theta_x(c),\ yb = by,\ \theta_x(b) = \theta_y(a) \Rightarrow a \bowtie b$, a relative of (R') which the census of order 9 produced first (Prover9, 99 steps: the work is to show that y identifies a with c, after which $c \sim x \sim y \sim b$ is an instance of (W_2)). The chain conditions also enter the derivation of (E_1) from (K0) (Remark 5.23).

7. Graph semigroups: sufficiency

The smallest non-embeddable example, and the one that first showed that (Q1) and (Tri) are not sufficient, belongs to a family of regular $*$-semigroups built from a graph and a colouring. For this family, and for a layered generalisation of it, (Q1) and (S) are also sufficient, and both conditions have a transparent graph-theoretic meaning.

7.1. **Graph semigroups.** Let $G = (V, E)$ be a graph, B a set and $\perp\colon V \to B$ a map; extend $\perp$ to $V \sqcup B$ by $\perp(m) = m$ for $m \in B$, and write $v \bowtie v'$ for $v = v'$ or $vv' \in E$. On the set $Q(G, \perp) = (V \times V) \sqcup (B \times B)$ define $(x, y)^* = (y, x)$ and

$$(p, q)(r, s) = \begin{cases} (p, s) & \text{if } q, r \in V \text{ and } q \bowtie r, \\ (\perp p, \perp s) & \text{otherwise.} \end{cases}$$

Lemma 7.1. *$Q(G, \perp)$ is a regular $*$-semigroup with $|V|^2 + |B|^2$ elements. Its projections are the pairs (v, v), $v \in V$, and (m, m), $m \in B$; identifying them with $V \sqcup B$, the projection algebra is*

$$\theta_v(v') = \begin{cases} v & v \bowtie v' \\ \perp v & \textit{otherwise,} \end{cases} \qquad \theta_v(m) = \perp v, \qquad \theta_m(x) = m \qquad (v, v' \in V,\ m \in B,\ x \in V \sqcup B).$$

Its matched pairs of distinct elements of V are the edges of G, and it satisfies (Q1) iff $\perp$ is a proper colouring of G.

Proof. For $x=(p,q)$, $y=(r,s)$, $z=(t,u)$ one checks the four cases according to whether $q \bowtie r$ and $s \bowtie t$: if both hold, $(xy)z=(p,u)=x(yz)$; in the other three cases both products equal $(\bot p, \bot u)$, because a pair with a coordinate in B is never matched. The identities $(xy)^*=y^*x^*$ and $xx^*x=x$ are immediate. The projections are the symmetric pairs, and $\theta_{(v,v)}(v',v') = (v,v')(v,v) = (v,v)$ if $v \bowtie v'$, while otherwise it is $(\bot v, \bot v')(v,v) = (\bot v, \bot v)$; the other values are computed in the same way. The projections below (v,v) are (v,v) and $(\bot v, \bot v)$, and (m,m) has no other projection below it; so two distinct elements $v, v' \in V$ have a common lower bound iff $\bot v = \bot v'$, and then $(v,v)(v',v') = (v,v') \neq (v',v) = (v',v')(v,v)$ if $vv' \in E$, while the two products are both $(\bot v, \bot v)$ if $vv' \notin E$. □

Thus $Q(G,\bot)$ is the Rees matrix semigroup over the trivial group with sandwich matrix "adjacency plus identity" of G, in which the zero has been replaced by the rectangular band $B \times B$ via $\bot$.

Definition 7.2. Let $\mathfrak{Q} = Q(G,\bot)$ where G is the path $a-c-b-d$ on $V=\{a,b,c,d\}$, $B=\{w,z\}$, $\bot(a)=\bot(b)=w$ and $\bot(c)=\bot(d)=z$; so $|\mathfrak{Q}|=20$. Its projection algebra $\mathfrak{P}=\{a,b,c,d,w,z\}$ has the matched pairs $a \bowtie c$, $b \bowtie c$, $b \bowtie d$, $w \bowtie z$, and every other value of θ is the bottom of the subscript: $\theta_a(b)=\theta_b(a)=w$, $\theta_c(d)=\theta_d(c)=z$, $\theta_a(d)=w$, $\theta_d(a)=z$, $\theta_a(z)=\theta_b(z)=w$, $\theta_c(w)=\theta_d(w)=z$, and $\theta_w \equiv w$, $\theta_z \equiv z$.

The colouring is proper, so $\mathfrak{Q}$ satisfies (Q1), and $\mathfrak{P}$ satisfies (Tri) (a direct search, or Lemma 7.6(d) below). But $\mathfrak{P}$ violates (S): $a \bowtie c \bowtie b \bowtie d$, $cd=dc=z$, $\theta_a(z)=w=\theta_b(z)$, and $a \not\bowtie d$. So $\mathfrak{Q}$ embeds in no partition monoid (Corollary 5.13), and $\mathfrak{P}$ is a minimal non-embeddable projection algebra of order 6 (its proper subalgebras embed, Section 5.9). The mechanism is the one named in Section 5.1: in $\mathcal{P}_X$ the commuting pair c,d shares its N-blocks (Lemma 2.6(b)), and the matchings $a \bowtie c$, $b \bowtie c$, $b \bowtie d$ transport the pairing of T-blocks across them to a pairing between a and d.

For a colouring $\bot\colon V \to B$ write $V_m = \bot^{-1}(m)$, and for $m \neq n$ let $G[V_m, V_n]$ be the bipartite graph of edges between V_m and V_n.

Theorem 7.3. *Let $G=(V,E)$ be a graph, B a set and $\bot\colon V \to B$ a map. The following are equivalent:*

(i) *$Q(G,\bot)$ $*$-embeds in some partition monoid;*
(ii) *the projection algebra $V \sqcup B$ of $Q(G,\bot)$ embeds in some $\mathsf{P}(\mathcal{P}_X)$;*
(iii) *$V \sqcup B$ satisfies (Q1) and (S);*
(iv) *$\bot$ is a proper colouring, and for any two colours $m \neq n$ the bipartite graph $G[V_m,V_n]$ contains no induced path $a-c-b-d$ with $\bot a = \bot b = m$ and $\bot c = \bot d = n$; equivalently, every $G[V_m,V_n]$ is a disjoint union of complete bipartite graphs and isolated vertices.*

When they hold, $Q(G,\bot)$ embeds in $\mathcal{P}_X$ with $|X| = |V| + \beta + 2|B|$, where β is the number of connected components with at least one edge of the graphs $G[V_m,V_n]$, $m \neq n$, and the image consists of elements of rank ≤ 1.

Proof. (i)⇒(ii) is clear and (ii)⇒(iii) is (Q1) together with Corollary 5.13.

(iii)⇒(iv). By Lemma 7.1, (Q1) says that $\bot$ is proper. We work out what (S) says in $V \sqcup B$. A vertex and a bottom element are never matched ($\theta_v(m) = \bot v \neq v$), so in an instance of (S) either all of a,b,c,d lie in B, where everything is matched and the conclusion holds, or all lie in V, with $a \bowtie c$, $b \bowtie c$, $b \bowtie d$ meaning equality or adjacency. Two distinct vertices c,d commute iff $\theta_c(d) = \theta_d(c)$, i.e. iff they are non-adjacent with $\bot c = \bot d$ (if $c \bowtie d$ the two values are $c \neq d$). If $c=d$ then $cd=c$, and $\theta_a(c)=a$, $\theta_b(c)=b$, so the last hypothesis forces $a=b$ and the conclusion $a \bowtie d = b \bowtie d$ holds. If $c \neq d$ then $cd = \bot c \in B$, $\theta_a(cd) = \bot a$ and $\theta_b(cd) = \bot b$. Hence, for a proper colouring, (S) is exactly the statement

$$a \bowtie c, \quad b \bowtie c, \quad b \bowtie d, \quad c \neq d, \quad \bot c = \bot d, \quad \bot a = \bot b \implies a \bowtie d \qquad (a,b,c,d \in V).$$

An induced path $a-c-b-d$ in some $G[V_m, V_n]$ with $a, b \in V_m$, $c, d \in V_n$ satisfies the hypotheses (c, d are distinct and, having the same colour, non-adjacent), so (S) gives $a \bowtie d$, i.e. $ad \in E$, contradicting "induced". Finally, a bipartite graph without induced 4-vertex paths is a disjoint union of complete bipartite graphs: if a connected component is not complete bipartite, choose non-adjacent $p \in V_m$, $q \in V_n$ in it at minimal distance; the distance is odd and ≥ 3, and the first four vertices of a shortest path from p to q form an induced path $p-c-b-d$ with $b \in V_m$, $c, d \in V_n$ ($p \not\sim d$ by minimality). The converse is obvious.

(iv)⇒(i). Let $\mathcal{C}$ be the set of connected components with at least one edge of the graphs $G[V_m, V_n]$, $m \neq n$; by (iv) each $C \in \mathcal{C}$ is complete bipartite between its two nonempty sides $C \cap V_m$ and $C \cap V_n$. Let

$$X = \{x_v : v \in V\} \sqcup \{y_C : C \in \mathcal{C}\} \sqcup \{z_m, z'_m : m \in B\},$$

and for $m \in B$ let ε_m be the partition of X whose non-singleton blocks are $\{z_m, z'_m\}$ and, for every $C \in \mathcal{C}$ one of whose colours is m, the block $\{y_C\} \cup \{x_u : u \in C \setminus V_m\}$ (the point y_C together with the points of the side of C *not* coloured m). These blocks are pairwise disjoint: a vertex $u \in V_n$ lies in at most one component of $G[V_m, V_n]$, and components with different second colours involve points x_u of different colours. Note that x_v is a singleton of $\varepsilon_{\perp v}$.

Claim. Let $v \in V_m$ and $v' \in V_n$. Then x_v and $x_{v'}$ lie in the same class of $\varepsilon_m \vee \varepsilon_n$ iff $v \bowtie v'$. Indeed, if $m = n$ the join is ε_m, in which both points are singletons, and $v \bowtie v'$ iff $v = v'$ since the colouring is proper. If $m \neq n$ and v lies in a component $C \in \mathcal{C}$ of $G[V_m, V_n]$, then the set $\{y_C\} \cup \{x_u : u \in C\}$ is a union of blocks of ε_m (the block of y_C, which is $\{y_C\} \cup \{x_u : u \in C \cap V_n\}$, and the singletons x_u, $u \in C \cap V_m$) and likewise of ε_n, so it is a class of the join; if v lies in no such component, $\{x_v\}$ is a class. So $x_v \sim x_{v'}$ iff v, v' lie in a common component of $G[V_m, V_n]$ iff $vv' \in E$, by completeness of the components.

Now define $\varphi\colon Q(G, \perp) \to \mathcal{P}_X$: $\varphi(v, v')$ is the partition of $X \sqcup X'$ whose blocks are the blocks of $\varepsilon_{\perp v}$ on X and of $\varepsilon_{\perp v'}$ on X', except that the two singletons $\{x_v\}$ and $\{x'_{v'}\}$ are merged into one transversal block; and $\varphi(m, n)$ consists of the blocks of ε_m on X and of ε_n on X' (rank 0). Clearly $\varphi(x^*) = \varphi(x)^*$, and φ is injective: $\varphi(x)$ determines its rank, its upper and lower partitions (which determine the colours, since $\{z_m, z'_m\}$ is a block of ε_m only) and, in rank 1, the transversal pair $(x_v, x'_{v'})$. For the products, consider $\varphi(v, q)\varphi(r, s)$ with $v, q, r, s \in V$. The only point of the upper row of $\varphi(v, q)$ connected to the middle row is x_v, and its partner there is x_q; the middle row is glued by $\varepsilon_{\perp q} \vee \varepsilon_{\perp r}$, so by the Claim x_q reaches the point x_r, and through it the lower point x'_s, iff $q \bowtie r$. Hence the product has the upper blocks of $\varepsilon_{\perp v}$, the lower blocks of $\varepsilon_{\perp s}$, and the transversal block $\{x_v, x'_s\}$ if $q \bowtie r$ and no transversal block otherwise: it is $\varphi(v, s)$ or $\varphi(\perp v, \perp s)$, as required. If one of the factors has rank 0 there is no transversal block in the product, whose upper and lower partitions are those of the first and the last factor: $\varphi(m, n)\varphi(r, s) = \varphi(m, \perp s)$, $\varphi(v, q)\varphi(m, n) = \varphi(\perp v, n)$, $\varphi(m, n)\varphi(m', n') = \varphi(m, n')$. So φ is an injective $*$-homomorphism. □

Remark 7.4. (1) For an induced subgraph $G' = G[V']$ the semigroup $Q(G', \perp|_{V'})$ is a $*$-subsemigroup of $Q(G, \perp)$. So a non-embeddable $Q(G, \perp)$ with $\perp$ proper contains a copy of $\mathfrak{Q}$, and $\mathfrak{Q}$ is the smallest non-embeddable graph semigroup satisfying (Q1); the list of forbidden coloured subgraphs consists of the monochromatic edge and the bichromatic path $a-c-b-d$. (2) The construction is not economical: for $\mathfrak{Q}$ with $\perp$ injective it uses $|X| = 15$, while $\mathcal{P}_4$ suffices. (3) The triangle condition holds in every $Q(G, \perp)$ with $\perp$ proper, for a trivial reason: three pairwise commuting projections with a common pairwise product w are either all equal or three pairwise non-adjacent vertices of one colour $w \in B$, so a triangle configuration with $w \neq z$ has $a_i \in V_w$ and $b_i \in V_z$ with $w \neq z$, and then a_i and b_i cannot commute. Thus the family $Q(G, \perp)$ sees only (Q1) and (S), and by the theorem it can witness no further obstruction: an embedding, when it exists, uses only projections of rank ≤ 1, whose T-blocks are single points.

7.2. **Layered graph semigroups.** The following generalisation replaces the bottom band of $Q(G,\perp)$ by another graph semigroup, and so on, so that the product of two unmatched projections may lie several levels below them. It turns out that this family, too, sees only (Q1) and (S).

Definition 7.5. A *layered graph* is a finite set $V = V_1 \sqcup \cdots \sqcup V_L$ ($L \geq 1$) with a reflexive symmetric relation $\bowtie$ on each level V_i, complete on the bottom level V_L, and maps $\perp\colon V_i \to V_{i+1}$ ($1 \leq i < L$) which preserve $\bowtie$. For $p \in V_i$ and $\ell \geq i$ write $p_\ell = \perp^{\ell-i} p$ (p descended to level ℓ), call p_L the *root* of p and $T_r = \{p : p_L = r\}$ the *tree* of the root $r \in V_L$; the trees partition V. For $q, r \in V$ let

$$f(q,r) = \min\{\ell \geq \max(\operatorname{lv} q, \operatorname{lv} r) : q_\ell \bowtie r_\ell\},$$

which exists because the bottom level is complete. The *layered graph semigroup* $Q(V)$ is the set of pairs (p,q) with p,q on a common level, with $(p,q)^* = (q,p)$ and $(p,q)(r,s) = (p_\ell, s_\ell)$, $\ell = f(q,r)$.

For $L = 2$ this is $Q(G,\perp)$ ($V_1 = V$, $V_2 = B$); every map into a complete graph preserves $\bowtie$.

Lemma 7.6. *$Q(V)$ is a regular $*$-semigroup; its projections are the pairs (p,p), identified with V, and*

$$\theta_p(q) = p_{f(p,q)}, \qquad q \leq p \iff q = p_\ell \text{ for some } \ell.$$

Moreover:

(a) *(Q1) holds in $Q(V)$ iff distinct matched elements have different roots;*
(b) *two elements p,q commute iff $p_f = q_f$ for $f = f(p,q)$, and then $pq = p_f$; under (Q1) this says that p,q lie in a common tree, and p_f is their least common ancestor;*
(c) *assuming (Q1), (S) holds in $Q(V)$ iff for every level i and every two roots $r \neq r'$ the bipartite graph of matched pairs between $T_r \cap V_i$ and $T_{r'} \cap V_i$ is a disjoint union of complete bipartite graphs and isolated vertices;*
(d) *assuming (Q1), the triangle condition holds in $Q(V)$.*

Proof. Preservation of $\bowtie$ gives $q_\ell \bowtie r_\ell \Rightarrow q_{\ell'} \bowtie r_{\ell'}$ for $\ell' \geq \ell$, hence $f(q_k, r) = \max(k, f(q,r))$ for $k \geq \operatorname{lv} q$. So $((p,q)(r,s))(t,u) = (p_{\ell_2}, u_{\ell_2})$ with $\ell_2 = f(s_{\ell_1}, t) = \max(f(q,r), f(s,t))$, $\ell_1 = f(q,r)$, and symmetrically $(p,q)((r,s)(t,u)) = (p_{\ell_4}, u_{\ell_4})$ with $\ell_4 = \max(f(s,t), f(q,r))$. The identities $(xy)^* = y^*x^*$ and $xx^*x = x$ are clear (f is symmetric and $f(q,q) = \operatorname{lv} q$). The projections are the symmetric pairs, and $\theta_p(q) = (p,p)(q,q)(p,p) = (p_f, q_f)(p,p) = (p_f, p_f)$ with $f = f(p,q)$, since $f(q_f, p) = f$; so $\theta_p(q) = q$ iff $q = p_f$.

(b) is the formula for θ together with Lemma 2.1. Within a tree, $p_f \bowtie q_f$ with $p_f \neq q_f$ would be two distinct matched elements with a common descendant, so under (Q1) "matched" means "equal" inside a tree and $f(p,q)$ is the level of the least common ancestor of p and q.

(a) Matched elements lie on a common level, so "no common descendant" means "different roots". If $p \bowtie q$ are distinct with a common root then $r = p_L = q_L$ satisfies $r \leq p,q$ but $\theta_p(q) = p \neq q = \theta_q(p)$. Conversely let $r \leq p$, $r \leq q$ with $p \neq q$, say $r = p_\ell = q_\ell$; at the level $f = f(p,q) \leq \ell$ the matched elements p_f, q_f have the common descendant r, so $p_f = q_f$ and $\theta_p(q) = p_f = q_f = \theta_q(p)$.

(c) Matched elements lie on a common level, so in an instance of (S) all of a,b,c,d lie on one level i. If $c = d$ the instance is trivial as in the proof of Theorem 7.3. If $c \neq d$ then $cd = dc$ says by (b) that c,d lie in a common tree, with $z := c_f = d_f$, $f = f(c,d)$; since $a_f \bowtie c_f = z$ we get $f(a,z) = f$ and $\theta_a(z) = a_f$, similarly $\theta_b(z) = b_f$, so the last hypothesis says that a,b lie in a common tree and their least common ancestor is at level $\leq f$. By (Q1) the tree of a,b differs from that of c,d. Thus, writing $\mu(p,q)$ for the level of the least common ancestor of two elements of a tree, (S) says: for every path $a - c - b - d$ of matchings with $a,b \in T_r$, $c,d \in T_{r'}$, $r \neq r'$, if $\mu(a,b) \leq \mu(c,d)$ then $a \bowtie d$. Applied to the reversed path $d - b - c - a$ it says the

same under $\mu(c,d) \leq \mu(a,b)$; one of the two always holds. So (S) is equivalent to: every path $a - c - b - d$ with $a, b \in T_r \cap V_i$ and $c, d \in T_{r'} \cap V_i$ closes up; that is, the bipartite graph in (c) has no induced 4-vertex path, which (as in the proof of Theorem 7.3) means that it is a disjoint union of complete bipartite graphs.

(d) Let $a_1, a_2, a_3, b_1, b_2, b_3$ be a triangle configuration. By (b) the a_i pairwise commute, so they lie in one tree; likewise the b_i; and a_i commutes with b_i, so all six lie in one tree. Then $a_1 \bowtie b_2$ forces $a_1 = b_2$ by (a), and similarly $a_2 = b_3$, $a_3 = b_1$. Hence $\{a_1, a_2, a_3\} = \{b_1, b_2, b_3\}$ and $w = a_1 a_2 = b_2 b_3 = z$. □

Theorem 7.7. *For a layered graph V the following are equivalent:*

(i) *$Q(V)$ $*$-embeds in some partition monoid;*
(ii) *the projection algebra V of $Q(V)$ embeds in some $\mathsf{P}(\mathcal{P}_X)$;*
(iii) *V satisfies (Q1) and (S);*
(iv) *distinct matched elements have different roots, and for every level i and every two roots $r \neq r'$ the bipartite graph of matched pairs between $T_r \cap V_i$ and $T_{r'} \cap V_i$ is a disjoint union of complete bipartite graphs and isolated vertices.*

When they hold, $Q(V)$ embeds in $\mathcal{P}_X$ with $|X| = |V| + |V_L| + \beta$, where β is the number of complete bipartite components in (iv)*, and every T-block of the image is a single point.*

Proof. (i)⇒(ii) is clear, (ii)⇒(iii) is (Q1) and Corollary 5.13, and (iii)⇔(iv) is Lemma 7.6(a),(c). For (iv)⇒(i) we generalise the construction of Theorem 7.3, with trees in the role of colours and the non-root elements in the role of vertices. Let $\mathcal{C}$ be the set of complete bipartite components in (iv), over all levels $i < L$ and pairs of roots; each $C \in \mathcal{C}$ lies on one level and meets exactly two trees. Let

$$X = \{x_v : v \in V \setminus V_L\} \sqcup \{y_C : C \in \mathcal{C}\} \sqcup \{z_r, z'_r : r \in V_L\},$$

and for a root r let ε_r be the partition of X whose non-singleton blocks are $\{z_r, z'_r\}$ and, for each $C \in \mathcal{C}$ meeting T_r, the block $\{y_C\} \cup \{x_u : u \in C \setminus T_r\}$. These blocks are disjoint (an element lies in at most one component per level and pair of trees, and for a fixed r the elements x_u in the blocks belong to the other tree of C). As in the Claim in the proof of Theorem 7.3, for $u \in T_r$ and $v \in T_{r'}$, $r \neq r'$, the points x_u, x_v lie in a common class of $\varepsilon_r \vee \varepsilon_{r'}$ iff u, v lie in a common component $C \in \mathcal{C}$, iff $u \bowtie v$; the class of x_u is $\{y_C\} \cup \{x_w : w \in C\}$ or $\{x_u\}$, and in particular contains no other point $x_{u'}$ with $u' \in T_r$ on a different level. Every x_v, $v \in T_r$, is a singleton of ε_r.

For $p \in T_r$ let $\hat{p}$ be the projection with partition ε_r whose T-blocks are the singletons $\{x_{p_j}\}$, $\operatorname{lv} p \leq j < L$ (the chain from p down to, but excluding, the root); so $\hat{r}$ has rank 0, $\operatorname{rank} \hat{p} = L - \operatorname{lv} p$, and $\operatorname{supp} \hat{p}$ determines p. We check $\theta_{\hat{p}}(\hat{q}) = \widehat{p_f}$, $f = f(p,q)$, using Lemma 2.6(a). If $p, q \in T_r$, the join is ε_r itself, whose classes meeting $\operatorname{supp} \hat{p}$ are the singletons $\{x_{p_j}\}$, transversal in $\theta_{\hat{p}}(\hat{q})$ iff $x_{p_j} \in \operatorname{supp} \hat{q}$, i.e. iff $p_j = q_k$ for some $k \geq \operatorname{lv} q$, i.e. iff $j \geq f$; so $\theta_{\hat{p}}(\hat{q})$ is ε_r with the T-blocks $\{x_{p_j}\}$, $j \geq f$, which is $\widehat{p_f}$. If $p \in T_r$, $q \in T_{r'}$ with $r \neq r'$, the classes of $\varepsilon_r \vee \varepsilon_{r'}$ meeting $\operatorname{supp} \hat{p}$ are the classes K_j of the points x_{p_j}, each containing only this one T-block of $\hat{p}$; K_j meets $\operatorname{supp} \hat{q} = \{x_{q_k} : k \geq \operatorname{lv} q\}$ iff $p_j \bowtie q_k$ for some $k \geq \operatorname{lv} q$, which forces $k = j$, iff $j \geq \operatorname{lv} q$ and $p_j \bowtie q_j$, iff $j \geq f$. So again $\theta_{\hat{p}}(\hat{q}) = \widehat{p_f}$. (For $q = r'$ a root, $\operatorname{supp} \hat{q} = \emptyset$ and $\theta_{\hat{p}}(\hat{q}) = \hat{r} = \widehat{p_L}$; and $\theta_{\hat{r}}(\hat{q}) = \hat{r}$ for every q since $\hat{r}$ has rank 0.) Injectivity: ε_r determines r (the block $\{z_r, z'_r\}$), and the T-blocks determine p. So $p \mapsto \hat{p}$ embeds the projection algebra.

For the semigroup, send (p,q) (with p, q on level i) to the element $\varphi(p,q)$ whose upper and lower partitions are those of $\hat{p}$ and $\hat{q}$, with the transversal blocks $\{x_{p_j}, x'_{q_j}\}$, $i \leq j < L$. This is injective and $*$-compatible, and $\varphi(p,q)\varphi(r,s) = \varphi(p_\ell, s_\ell)$, $\ell = f(q,r)$: in the product, the upper point x_{p_j} is joined to the middle point x_{q_j}, whose class in the middle join contains a point x_{r_k}, $k \geq \operatorname{lv} r$ (necessarily $k = j$), iff $j \geq f(q,r)$, by the two computations above, and then x_{p_j} is

joined to x'_{s_j}; for $j < \ell$ the block of x_{p_j} in the product is the nontransversal singleton $\{x_{p_j}\}$, and the other blocks are unchanged. This is $\varphi(p_\ell, s_\ell)$. □

Remark 7.8. In a layered graph every principal down-set is a chain, every value $\theta_p(q)$ lies in that chain, and under (Q1) the commuting pairs are the pairs in a common tree. This is why the triangle condition is invisible here: it needs two commuting triples with a common meet, which in a tree forces the triples to coincide. The three-level layered graph with $V_1 = \{a, b, c, d\}$ ($a \bowtie c$, $b \bowtie c$, $b \bowtie d$), $V_2 = \{m, n, k\}$ ($m \bowtie n$, $m \bowtie k$), $V_3 = \{w, z\}$ and $\bot\colon a, b \mapsto m$, $c \mapsto n$, $d \mapsto k$, $m \mapsto w$, $n, k \mapsto z$ gives a 29-element regular $*$-semigroup whose projection algebra $\mathfrak{P}_9$ (order 9) satisfies (Q1), (Tri) and the weaker form "$a \bowtie c \bowtie b \bowtie d$, $cd = dc$, $\theta_a(cd) = \theta_b(cd) = \theta_a(d) \Rightarrow a = b$" of (S) in which the obstruction $\mathfrak{P}$ was first found, but violates (S) itself: the pair a, d is only half wired, $\theta_a(d) = m$ lying strictly between w and a. It is a minimal non-embeddable projection algebra, the only minimal one of order 9 that violates (S) (Section 8). Any semigroup separating (Q1)+(S) from embeddability must therefore have projections with non-chain down-sets, or must force T-blocks of size > 1; the constructions of this section use only singleton T-blocks.

8. Computations

We record the computations behind the census, with the three levels of evidence of the introduction kept apart: what is proved, what is verified in $\mathsf{P}(\mathcal{P}_m)$ for stated m, and what is a failure to find an embedding. The programs are in the repository accompanying the paper, https://github.com/mkkinyon/starsemigroup.

Why quasi-identities suffice. For finite semigroups the search for quasi-identities is complete in principle:

Proposition 8.1. *A finite regular $*$-semigroup embeds in some partition monoid if and only if it satisfies every quasi-identity in the language $\{\cdot, ^*\}$ that holds in all partition monoids; and a finite projection algebra embeds in some $\mathsf{P}(\mathcal{P}_X)$ iff it satisfies every quasi-identity valid in all $\mathsf{P}(\mathcal{P}_X)$.*

Proof. Necessity is clear. Conversely, a finite S satisfying all quasi-identities of the class $\{\mathcal{P}_X\}$ lies in the quasivariety it generates, so S embeds in a product $\prod_j A_j$ of ultraproducts A_j of partition monoids $\mathcal{P}_{X_{ij}}$, $i \in I_j$, over ultrafilters $\mathcal{U}_j$. As S is finite, finitely many factors suffice, and by Łoś's theorem each $*$-homomorphism $S \to A_j$ is, for $\mathcal{U}_j$-almost all i, induced by a $*$-homomorphism $S \to \mathcal{P}_{X_{ij}}$ with the same kernel. Taking one such i for each of the finitely many j and using $\mathcal{P}_X \times \mathcal{P}_Y \leq \mathcal{P}_{X \sqcup Y}$ gives an embedding of S into a partition monoid. The same argument applies to projection algebras. □

Regular $*$-semigroups of small order. All regular $*$-semigroups of order $n \leq 8$ were enumerated up to $*$-isomorphism by a backtracking search on multiplication tables with propagation of associativity, $(xy)^* = y^*x^*$ and $xx^*x = x$; for $n = 7, 8$ only the semigroups with a nontrivial involution were generated, since $x^* = x$ for all x forces commutativity and hence an inverse semigroup, which embeds by the Wagner–Preston representation. For each non-inverse semigroup satisfying (Q1), a $*$-embedding into $\mathcal{P}_m$, $m \leq 5$, was searched for by assigning images to the projections first and forcing the images of all products.

order	classes	inverse	non-inverse, (Q1)	non-inverse, not (Q1)	of these, in $\mathcal{P}_5$
4	17	16	1	0	1
5	55	52	2	1	2
6	222	208	9	5	9
7	259†	196	35	28	35
8	1310†	970	178	162	178

† nontrivial involution only. The last column counts the non-inverse semigroups satisfying (Q1) that embed in $\mathcal{P}_5$.

Thus, up to order 8, a regular $*$-semigroup embeds in a partition monoid iff it satisfies (Q1), and then it embeds in $\mathcal{P}_5$. The smallest known regular $*$-semigroup satisfying (Q1) that embeds in no partition monoid has order 19 (a semigroup with projection algebra $\mathfrak{P}_7$); the order-9 enumeration was not completed, so the smallest such semigroup has order between 9 and 19. A cheap bound: if $p \bowtie q$ with $p \neq q$ then pq and qp are distinct non-projections determined by $\{p, q\}$, so $|S| \geq |\mathsf{P}(S)| + 2 \cdot \#\{\{p, q\} : p \bowtie q,\ p \neq q\}$; for the nine minimal non-embeddable projection algebras of order ≤ 8 this gives $|S| \geq 14$, so a counterexample of order ≤ 13 would have an embeddable projection algebra (Question 9.5 below).

Quotients of $\mathcal{P}_3$ and $\mathcal{P}_4$. The ten $*$-congruences of $\mathcal{P}_3$ have quotients of orders 203, 179, 103, 79, 67, 43, 7, 3, 2, 1. The quotients of orders $179, 79, 43$ have a zero and are not inverse, so they fail (Q1); those of orders $7, 3, 2, 1$ are inverse; the two remaining proper quotients, of orders 103 and 67, satisfy (Q1) and are the semigroups of Theorem 4.10 ($\mathcal{P}_3/\mu$ and its quotient collapsing the rank-2 $\mathcal{H}$-classes). For $\mathcal{P}_4$ the classification of [4] leaves two proper $*$-congruences whose quotients have no zero, both satisfying (Q1) and failing (Tri) because they contain the corresponding quotients of $\mathcal{P}_3$; so the congruences of $\mathcal{P}_4$ give nothing new.

Projection algebras. The projection algebras of order ≤ 9 were enumerated as described in Section 5.9: Mace4 with isofilter for the orders ≤ 7 (directly, from the axioms and the clauses for (Q1) and (S)), and the one-point extension method of Lemma 5.20 from order 7 on (with (Q1), (S), (R), and separately with (Q1) alone up to order 9); the two methods agree where both apply. Embeddings into $\mathsf{P}(\mathcal{P}_m)$ were searched for by backtracking for $m \leq 4$ and by a SAT encoding for $5 \leq m \leq 8$ (presence variables for the blocks of the images, the joins computed by iterated squaring, solved with CaDiCaL or kissat); for the seven algebras of order 8 the unsatisfiability at $m = 8$ takes between 25 seconds and 2.6 hours. The validity of the quasi-identities in $\mathsf{P}(\mathcal{P}_m)$ was verified by a SAT search for a violating instance, variables not assumed distinct, for $m \leq 7$ ((Q), the hardest, takes 11 hours at $m = 7$); the proofs of Sections 5 and 6 were found afterwards, and the intermediate statements of the proofs were tested on all instances of the hypotheses in $\mathsf{P}(\mathcal{P}_4)$ and on samples in $\mathsf{P}(\mathcal{P}_5)$, $\mathsf{P}(\mathcal{P}_6)$, by the dropped-hypothesis test of Remark 5.16.

Order 9. Of the 80 039 projection algebras of order 9 satisfying (Q1), (S), (R), exactly 163 contain one of $\mathfrak{P}_8^1, \ldots, \mathfrak{P}_8^7$ as a subalgebra, and one further algebra, $\mathfrak{R}_9$, violates (R′) without containing any of them. Of the remaining 79 875, all but seven embed in $\mathsf{P}(\mathcal{P}_8)$ (79 783 already in $\mathsf{P}(\mathcal{P}_7)$; the 39 semilattices with 8 join-irreducible elements and 46 others need $m = 8$). The seven embed in no $\mathsf{P}(\mathcal{P}_m)$ with $m \leq 8$. Four of them are settled: one violates (W_2) (it is $\mathfrak{P}_2^{\mathrm{ch}}$), one violates (K0), one (K4), and one the variant (K0)″ of (K0) (Corollary 5.18); three remain open (Section 9). The (Q1)-only extension census of order 9 (from the 10 821 embeddable algebras of order 8: 213 084 labelled extensions, 81 259 up to isomorphism, of which 1 227 violate (S) or (R)) shows that every projection algebra of order 9 satisfying (Q1) but violating (S) or (R) contains one of the nine minimal non-embeddable algebras of order ≤ 8, with a single exception: the algebra $\mathfrak{P}_9$ of Remark 7.8, a second witness of (S). So the minimal non-embeddable projection algebras of order 9 satisfying (Q1) are exactly $\mathfrak{P}_9$, $\mathfrak{R}_9$, $\mathfrak{P}_2^{\mathrm{ch}}$, the three algebras settled by (K0),

(K4), (K0)″, and those of the three open candidates that embed in no $\mathsf{P}(\mathcal{P}_X)$: between six and nine algebras; their tables are in the repository (`paper/order9-tables.md`). In particular the number of points needed to embed an embeddable projection algebra of order n reaches $n-1$ at $n=9$.

Prover9 and Mace4. The algebraic parts of the proofs (the identities of Step 1 of (R′), the substitution of Corollary 5.5, $\theta_z(c)=b$ in (K4), the equivalence of (V) with a three-atom clause) were found or confirmed with Prover9, and the implications (K0)+(W$_2$)⇒(E$_1$), (K4)⇒(E$_6$), (W$_2$)⇒(R″) rest on Prover9 proofs (Remarks 5.23, 6.12). Mace4 provided the separating models: $\mathfrak{P}_8^1$ and $\mathfrak{P}_8^6$ as the unique models of order ≤ 8 of all proved conditions violating (K0), resp. (K4); the order-9 model separating (K0)″ from (K0) (Section 5.8); the model of order 11 separating (W$_3$) from (W$_2$); and the nonexistence of a projection algebra of order ≤ 10 satisfying (Q1) and violating (Tri).

9. Open problems

Question 9.1 (order 9)**.** Three projection algebras of order 9 satisfying (Q1), (Tri), the eight wiring conditions and all chain conditions embed in no $\mathsf{P}(\mathcal{P}_m)$ with $m\le 8$, and all their proper subalgebras embed. Each violates a five-variable path-closing condition verified in $\mathsf{P}(\mathcal{P}_m)$ for $m\le 6$: for the first,

$$a\bowtie c\bowtie b,\quad bxb=b,\quad \theta_y(a)=\theta_z(y),\quad \theta_c(z)=\theta_x(a)=\theta_y(x),\quad \theta_z(x)=\theta_b(a)\implies a\bowtie b,$$

and the structurally simpler $a\bowtie c\bowtie b,\ c\sim x,\ x\sim y,\ z\le y,\ \theta_y(a)=\theta_y(z),\ z\sim b,\ \theta_a(x)=\theta_a(b),\ bxb=b\Rightarrow a\bowtie b$; for the third, $a\bowtie c\bowtie b,\ \theta_x(c)=a,\ byb=b,\ bzb=b,\ \theta_y(c)=\theta_y(b),\ \theta_z(c)=\theta_z(b),\ \theta_y(z)=\theta_z(y)=\theta_a(b),\ \theta_x(b)=\theta_b(a)\Rightarrow a\bowtie b$; the second has a five-element bottom band like $\mathfrak{P}_8^7$. Are these conditions valid in every $\mathcal{P}_X$? If so, Theorem 5.22 extends to order 9, with the additional conditions (R′), (W$_2$), (K0), (K4), (K0)″ and these three. Related: is the condition

$$\text{K5}:\qquad a\bowtie c\bowtie b,\quad \theta_x(c)=a,\quad \theta_b(x)=b,\quad zc=cz,\quad \theta_z(x)=\theta_a(b),\quad \theta_z(b)=\theta_b(a)\implies a\bowtie b,$$

verified for $m\le 6$ and the first condition found for the (K0)″-witness, valid in every $\mathcal{P}_X$?

Question 9.2 (a uniform criterion)**.** All the proofs of Section 5 run by transport along the matchings and the propagation of an invariant along a join, and the hypotheses are all of five kinds (commutation, order, half-matching, seeing alike, being seen alike). Is there a combinatorial criterion deciding, for a configuration of auxiliaries attached to a path $a\bowtie c\bowtie b$, whether it forces $a\bowtie b$ in every $\mathcal{P}_X$, with one uniform proof? The chain conditions show that the list of such conditions is infinite, so a parametrised description is the best possible. Is there a hybrid chain family for every one of the eight conditions, as there is for (R) (the family (W$_n$)) and for (R′) (the family $a\bowtie c\bowtie b,\ c\sim x_1\sim\cdots\sim x_n,\ \theta_{x_i}(a)=\theta_{x_i}(c),\ ya=ay,\ \theta_{x_n}(b)=\theta_y(b)\Rightarrow a\bowtie b$, which has explicit witnesses of order $2n+6$ with embeddable maximal subalgebras, but is verified only for $m\le 5$ and $n\le 3$)?

Question 9.3 (finite versus infinite)**.** (K0)″ implies (K0) in every finite projection algebra (Corollary 5.18), by finite descent; does it in every projection algebra? More generally, the proofs in $\mathcal{P}_X$ use rank counts, and Lemma 2.10 replaces them in finite algebras; is there a general mechanism turning a proof in $\mathcal{P}_X$ into a derivation valid in all finite projection algebras satisfying the other conditions? On the digraph side, the hexagon condition does not imply (Tri) in general (Remark 4.9); does it imply (D$_E$) for the non-star digraphs on $k\ge 4$ vertices?

Question 9.4 (compression and decidability)**.** Is embeddability of a finite projection algebra in some $\mathsf{P}(\mathcal{P}_X)$ decidable? A recursive basis of quasi-identities would make non-embeddability semi-decidable, and embeddability is semi-decidable provided a compression lemma holds: if a

finite P embeds in some $\mathsf{P}(\mathcal{P}_X)$, then it embeds in $\mathsf{P}(\mathcal{P}_m)$ for some finite m, say $m \leq f(|P|)$. This is not obvious, since restriction to a finite subset of X does not preserve products. The census data ($m \leq n-1$ suffices for every embeddable algebra of order $n \leq 9$) suggest that it is true, and a bound f would turn "verified for $m \leq 7$" into proofs.

Question 9.5 (the semigroup)**.** Is embeddability of a finite regular $*$-semigroup in a partition monoid determined by its projection algebra? All known obstructions are projection-algebra quasi-identities, and Theorem 3.1 shows that groups and sandwich matrices impose nothing; a regular $*$-semigroup is, however, far from determined by its projection algebra. What is the smallest regular $*$-semigroup satisfying (Q1) that embeds in no partition monoid (its order is between 9 and 19)? By the bound of Section 8, an example of order ≤ 13 would have an embeddable projection algebra and answer the first question negatively.

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